Information and Ordered Mean-Square Estimation in Bosonic Displacement Channels

Abstract

We establish an information–minimum mean-square error (I–MMSE) relation for a classical signal encoded by bosonic displacements in thermal noise. For every input distribution with finite second moment and every positive thermal mean photon number, the derivative of the Holevo information with respect to the signal-to-noise ratio (SNR) parameter equals a thermally weighted integral of ordered minimum mean-square errors. The ordering integral accounts for the noncommutative logarithmic derivative of the output state. The identity implies concavity of the Holevo information in SNR for each fixed input distribution. We also express the known quantum de Bruijn identity as an estimation formula for Gaussian displacement noise. This formula and estimation data processing give an alternative derivation of the known linear and exponential quantum entropy power inequalities, together with exact integral representations of the linear and logarithmic deficits as squared differences between optimal estimators.

Index terms: Quantum information, minimum mean-square error, bosonic channel, Holevo information, quantum conditional expectation, entropy power inequality.

1 Introduction

The I–MMSE relation describes how the information carried by a noisy signal changes as the signal-to-noise ratio increases. For an additive Gaussian channel, this change is determined by an estimation error. If X is a real random vector with finite second moment and Z_0 is an independent standard Gaussian vector, the Guo–Shamai–Verdú identity states that

$$\frac{d}{ds} I(X; \sqrt{s}X + Z_0) = \frac{1}{2} \mathbb{E} \big\| X - \mathbb{E}[X \mid \sqrt{s}X + Z_0] \big\|^2. \tag{1.1}$$

Information is measured in nats. The identity holds for a fixed arbitrary input distribution, including discrete distributions, and provides a direct connection between an information quantity and the accuracy of estimating the transmitted signal [1]. It also converts estimation comparisons into information inequalities. In particular, the fact that discarding part of an observation cannot improve mean-square estimation leads to proofs of the classical entropy power inequality (EPI) [2, 3, 4].

We ask how this connection changes when the channel output is a quantum system. Consider a transmitter that encodes a classical information-bearing signal $z \in \mathbb{C}^m$ in the displacement of m bosonic modes. The receiver obtains the displaced quantum state. For isotropic thermal noise τ_ν with mean photon number $N > 0$ per mode and quadrature covariance νI_{2m}, where $\nu = N + 1/2$, the conditional and average output states are

$$\rho_{s,z} = D_{\sqrt{s}z} \tau_\nu D_{\sqrt{s}z}^\dagger, \qquad \rho_s = \mathbb{E} \rho_{s,Z}. \tag{1.2}$$

Here D_z is the displacement operator, and s controls the SNR while the input distribution and noise state remain fixed. The state τ_ν is the output when no signal is sent. Its modes are independent

and have the same quadrature variance, giving white thermal noise across the modes. The classical–quantum (cq) mutual information is the Holevo quantity $I(Z;A_s) = S(\rho_s) - S(\tau_\nu)$, where A_s denotes the quantum output at SNR parameter s (formally, the thermal signal experiment of Subsection 2.1) and $S(\rho) = -\operatorname{Tr}(\rho \log \rho)$ is the von Neumann entropy. It describes the information in the quantum output before the receiver chooses a measurement. Although the noise is Gaussian, the transmitted signal and the average output need not be Gaussian. We seek an identity that retains this arbitrary-input feature of (1.1).

To identify the appropriate estimation error, we must address two features of quantum outputs. First, a quantum system can be used to estimate a real vector either by choosing one observable for each coordinate or by performing a single measurement whose outcome is a classical vector. These optimizations differ when the optimal coordinate observables are incompatible. Second, even for one real parameter, ordinary quadratic estimation and von Neumann entropy differentiation select different operator geometries. The former averages left and right multiplication by the state, giving the Jordan map $B \mapsto \frac{1}{2}(\rho B + B\rho)$ and the symmetric logarithmic derivative (SLD). The latter differentiates the operator logarithm, giving the Kubo–Mori–Bogoliubov (KMB) map.[1] The SLD and KMB logarithmic derivatives agree for a state tangent $C = \frac{d}{d\theta}\rho_\theta$ that commutes with ρ_θ, but generally differ. The estimation error in the desired identity must account for this difference as well as for measurement incompatibility.

1.1 Main results

Our main result expresses the Holevo-information derivative for an arbitrary classical prior as a weighted integral of optimal quadratic estimation errors. We first define the errors appearing in the formula. For a cq ensemble $z \mapsto \omega_z$ and $0 < u < 1$, let

$$\mathsf{M}_u(Z \mid A) = \inf_{B_1,\dots,B_m} \mathbb{E} \sum_{j=1}^{m} \operatorname{Tr} \omega_Z \big[(1-u)E_j E_j^\dagger + u E_j^\dagger E_j\big], \qquad E_j = Z_j I - B_j. \tag{1.3}$$

The parameter u assigns weights to the two possible orders of the error operator and its adjoint. Both terms are positive. The infimum is over bounded operators on the output Hilbert space; its minimum is attained in the weighted completion described in Subsection 2.3. At $u = 1/2$, the risk is half the sum of the ordinary operator mean-square errors for the real coordinates $X_{2j-1} + iX_{2j} = \sqrt{2}Z_j$. Even at this midpoint, the optimal real coordinate observables need not commute and hence need not be jointly measurable.

Theorem 1 (Thermal signal I–MMSE)**.** *Let τ_ν be the m-mode isotropic thermal state with mean photon number $N > 0$ per mode, where $\nu = N + 1/2$, and let $Z \in \mathbb{C}^m$ satisfy $\mathbb{E}\|Z\|^2 < \infty$. For the ensemble* (1.2)*, set $I(s) = S(\rho_s) - S(\tau_\nu)$. Then I is continuously differentiable on $(0,\infty)$ and*

$$I'(s) = \int_0^1 \frac{\mathsf{M}_u(Z \mid A_s)}{N + 1 - u}\, du, \qquad s > 0. \tag{1.4}$$

For every $s \geq 0$, the integrated identity is

$$I(s) = \int_0^s \int_0^1 \frac{\mathsf{M}_u(Z \mid A_r)}{N + 1 - u}\, du\, dr. \tag{1.5}$$

The proof is in Section 3. The input distribution may be discrete, continuous, or mixed. We call s the SNR parameter, keeping the input distribution and thermal noise fixed; the normalization is

[1] The associated metric is also called the Bogoliubov–Kubo–Mori (BKM) metric. We follow the naming of [5, 6].

specified in Subsection 2.1. In words: the rate at which the quantum output acquires information about Z is a weighted integral, over the operator orderings u, of the minimum ordered mean-square error attainable by operator estimators; the weight of each ordering is the inverse second moment of the thermal noise in that same ordering. The denominator has a direct interpretation: for each mode,

$$n_u := N + 1 - u = (1-u)\,\mathrm{Tr}(\tau_\nu a_j a_j^\dagger) + u\,\mathrm{Tr}(\tau_\nu a_j^\dagger a_j). \tag{1.6}$$

Here a_j and $a_j^\dagger$ are the annihilation and creation operators of mode j (Subsection 2.1). It is the noise state's second moment in the same operator ordering as the loss. The entropy derivative averages over these orderings because the logarithmic derivative of a state is an integral of the corresponding inverse multiplication maps. These are operator estimation quantities, not generally the errors of one joint receiver measurement. Their usefulness here is that trial operator estimators bound the information derivative, while estimation data processing compares such derivatives without computing the spectrum of the average output state.

For a first example, let $Z \sim \mathcal{CN}(0, \sigma^2 I_m)$, the circularly symmetric complex Gaussian law with density $(\pi\sigma^2)^{-m}\exp(-\|z\|^2/\sigma^2)$ on $\mathbb{C}^m \simeq \mathbb{R}^{2m}$, where $\sigma > 0$. Thus its real and imaginary coordinates are independent real Gaussians of variance $\sigma^2/2$, and $\mathbb{E}|Z_j|^2 = \sigma^2$.

The average output is thermal with mean photon number $N + s\sigma^2$, and

$$\mathsf{M}_u(Z \mid A_s) = \frac{m\sigma^2 n_u}{n_u + s\sigma^2}, \qquad I'(s) = m\sigma^2 \log\frac{N + s\sigma^2 + 1}{N + s\sigma^2}. \tag{1.7}$$

Thus the ordering average in (1.4) is exactly the derivative of $m[g(N + s\sigma^2) - g(N)]$, where $g(x) = (x+1)\log(x+1) - x\log x$ is the one-mode thermal entropy.

Each ordered risk obeys data processing. A thermal attenuator (a beam splitter of transmissivity $s_1/s_2 < 1$ whose environment is in the state τ_ν; it maps the output at SNR parameter s_2 to that at s_1, see Subsection 2.2) therefore makes it nonincreasing in s, which proves that $I(s)$ is nondecreasing and concave for every fixed prior. The exact initial slope is

$$I'(0+) = \log\frac{N+1}{N}\ \mathrm{Var}(Z), \qquad \mathrm{Var}(Z) = \mathbb{E}\|Z - \mathbb{E}Z\|^2. \tag{1.8}$$

Corollary 10 proves both statements. For comparison with ordinary operator squared error, Proposition 12 also gives

$$I'(s) = \frac{1}{2\nu}\,\mathrm{mmse}_J(X \mid A_s) + \mathcal{Q}(s), \qquad X_{2j-1} + iX_{2j} = \sqrt{2}Z_j. \tag{1.9}$$

Here $\mathrm{mmse}_J(X \mid A_s)$ is the ordinary operator MMSE of the real coordinates X, obtained with one self-adjoint estimator per coordinate and the Jordan-product loss (Subsection 3.3), and $\mathcal{Q}(s)$ is an explicit KMB–SLD score correction. This geometric correction is distinct from the additional error caused by requiring one joint measurement; for a Gaussian prior, the latter error is recorded in Appendix C.

The second identity concerns estimating Gaussian displacement noise added to an arbitrary finite-energy state. Write

$$\mathcal{N}_t(\rho) = \mathbb{E}D_{\sqrt{t}G}\rho D_{\sqrt{t}G}^\dagger, \qquad G \sim \mathcal{CN}(0, I_m),$$

where G has density $\pi^{-m}e^{-\|g\|^2}$. Let $\mathsf{M}_{\rho,u}(t)$ denote the ordered error (1.3) for estimating G from this ensemble.

Theorem 2 (Gaussian noise I–MMSE)**.** *Let ρ be an m-mode state with $\mathrm{Tr}(\rho H_0) < \infty$, where $H_0 = \sum_{j=1}^m a_j^\dagger a_j$. Set $\rho_t = \mathcal{N}_t(\rho)$ and $I_\rho(t) = S(\rho_t) - S(\rho)$. Then $t \mapsto S(\rho_t)$ is continuously differentiable for $t > 0$, and*

$$\frac{d}{dt} I_\rho(t) = \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t)\, du}{t}. \tag{1.10}$$

Equivalently, for $0 < t_0 < t_1 < \infty$,

$$S(\rho_{t_1}) - S(\rho_{t_0}) = \int_{t_0}^{t_1} \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t)\, du}{t}\, dt. \tag{1.11}$$

The proof is in Section 4. This is an estimation representation of the known quantum de Bruijn identity [7, 8, 6]: Gaussian integration by parts identifies each ordered score with the optimal estimator of the displacement noise. Its classical counterpart, for a real X with finite second moment and an independent $G \sim \mathsf{N}(0,1)$, is

$$\frac{d}{dt} h(X + \sqrt{t}G) = \frac{1}{2} J(X + \sqrt{t}G) = \frac{1 - \mathrm{mmse}(G \mid X + \sqrt{t}G)}{2t}, \tag{1.12}$$

where h is differential entropy, J is Fisher information, and $\mathrm{mmse}(G \mid Y) = \mathbb{E}|G - \mathbb{E}[G \mid Y]|^2$ is the classical MMSE [1, 9]. The factor $1/2$ corresponds to one real Gaussian coordinate; our convention uses one complex coordinate per mode.

Classically, rescaling the observation connects the signal I–MMSE relation with de Bruijn's identity [1]. Simultaneously rescaling both quadratures of a quantum mode changes their commutator, so that classical change of variables is not a canonical transformation on the fixed bosonic system. We therefore treat the signal and noise experiments separately: the first fixes the thermal noise while varying the transmitted signal, and the second fixes the input state while varying the added Gaussian displacement. Their common estimation structure will allow us to apply the same comparison principles to both.

The noise identity leads to an estimation proof of the known quantum entropy power inequality (qEPI). Let ρ_A, ρ_B be independent m-mode finite-energy states, and let ρ_C be the retained output of a beam splitter with transmissivity $0 < \eta < 1$. Following the unconditional beam-splitter case of De Palma and Trevisan's interpolation [8, Thm. 6], we add independent Gaussian noises with variances $\lambda t/\eta$ and $(1-\lambda)t/(1-\eta)$ to the inputs. The output then has added variance t, for any entropy weight $0 < \lambda < 1$ chosen independently of η. Optimizing over this weight proves the exponential qEPI, and the estimation argument identifies its logarithmic deficit:

$$\begin{aligned} &S(\rho_C) - m \log\Bigl(\eta e^{S(\rho_A)/m} + (1-\eta) e^{S(\rho_B)/m}\Bigr) \\ &\qquad = \int_0^\infty \frac{dt}{t} \int_0^1 \mathcal{G}_{\lambda_*,u}(t)\, du, \qquad \lambda_* = \frac{\eta e^{S(\rho_A)/m}}{\eta e^{S(\rho_A)/m} + (1-\eta) e^{S(\rho_B)/m}}. \end{aligned} \tag{1.13}$$

Here $\mathcal{G}_{\lambda,u}(t)$ is the increase in ordered MMSE when one beam-splitter output is discarded, with input noise levels $\lambda t/\eta$ and $(1-\lambda)t/(1-\eta)$. Proposition 15 expresses this nonnegative gap as a squared difference between the optimal input estimator and the pulled-back output estimator. The derivation of (1.13) is in Subsection 5.2. For displacements of a common thermal noise state, the signal identity also gives a direct interpolation in SNR. Section 5 develops both applications.

1.2 Proof techniques

The proof rests on expressing the derivative of the operator logarithm as an average over the same operator orderings that define the estimation errors. This makes it possible to use the familiar orthogonality principle after entropy differentiation. Specifically, the ordered loss uses

$$\mathcal{K}_{\rho,u}(B) = (1-u)\rho B + uB\rho,$$

whereas entropy differentiation uses the derivative $\mathcal{T}_\rho$ of the operator logarithm. The proof connects them in three steps.

First, the thermal identity $a_j\tau_\nu = \frac{N}{N+1}\tau_\nu a_j$ (derived in Section 3) identifies the optimal ordered estimator, denoted $\widehat{Z}_{j,u}$, with an ordered score. With $\Gamma_j = \mathbb{E}[Z_j\rho_{s,Z}]$ and $\mathcal{S}_{j,u}(\rho) = \mathcal{K}_{\rho,u}^{-1}([a_j,\rho])$, where $[B,C] = BC - CB$ is the commutator, it gives

$$\widehat{Z}_{j,u} = \mathcal{K}_{\rho_s,u}^{-1}(\Gamma_j) = \frac{a_j + (N+1-u)\mathcal{S}_{j,u}(\rho_s)}{\sqrt{s}}. \tag{1.14}$$

Second, the logarithmic divided difference yields

$$\mathcal{T}_\rho = \int_0^1 \mathcal{K}_{\rho,u}^{-1}\,du, \qquad [a_j,\log\rho] = \int_0^1 \mathcal{S}_{j,u}(\rho)\,du. \tag{1.15}$$

This ordered representation is a reparameterization of [6, Sec. 3.1, Eqs. (26)–(31)]; Subsection 2.5 specifies the quadratic-form and pairing interpretation used here.

Third, differentiating the displacement ensemble and using (1.15) gives

$$I'(s) = -\frac{1}{\sqrt{s}}\int_0^1 \mathrm{Re}\sum_j \mathrm{Tr}\big(\Gamma_j^\dagger \mathcal{S}_{j,u}(\rho_s)\big)\,du. \tag{1.16}$$

Substitution of (1.14), followed by the conditional-projection identity (Lemma 4), turns each integrand into $\mathsf{M}_u(Z \mid A_s)/(N+1-u)$. The proof in Section 3 displays this algebra step by step. Appendix B uses bounded logarithmic regularization and trace-class displacement derivatives to justify the pairings under the finite-second-moment assumption. For Theorem 2, Gaussian integration by parts replaces the thermal commutation step.

1.3 Relation to previous work

The classical starting point is [1]; estimation proofs of the classical EPI were developed in [2, 3, 4]. Dytso, Bustin, Poor, and Shamai [10] further expressed the classical entropy-concavity deficit as an integral of squared differences between conditional-mean estimators and used this expression to characterize equality. Their result is the classical antecedent of the estimation-deficit interpretation in Section 5. Han and Song extended I–MMSE methods to Gaussian models with feedback or memory, where parameter dependence introduces explicit correction terms [9]. Equation (1.9) has a related structure: retaining ordinary operator MMSE introduces a correction because entropy differentiation uses KMB rather than SLD geometry. The source of this correction is different from feedback or memory; the two score maps agree on tangents that commute with the state.

Quantum Bayesian quadratic estimation goes back to Personick [11]. Tsang's generalized conditional expectations provide a common framework for different operator weightings, projection identities, and estimation regret [12, 13]. The ordered losses and their data-processing properties used here follow the same framework. The general rigorous treatments in [12, 13] are formulated in finite dimensions. We therefore give the cq construction and the completion and continuity arguments

needed for the bosonic setting, rather than infer the infinite-dimensional limits from matrix formulas. The scalar Gaussian-prior identity relating Bayesian error and SLD information is also established: Macieszczak, Fraas, and Demkowicz-Dobrzański give an exact version in [14, Eq. (6)]. That scalar mechanism is an antecedent of the Gaussian-noise calculation; the extension to our complex vector target and ordering parameter follows from the same integration-by-parts argument. Tsang's information–estimation identities for quantum filtering concern classical measurement records [15]; the information considered here is the unmeasured cq Holevo quantity.

Quantum entropy production and beam-splitter convolution have a separate development. König and Smith proved the entropy-concavity form of the qEPI and a balanced exponential qEPI [7]; De Palma, Mari, and Giovannetti established the exponential form at arbitrary transmissivity [16]. De Palma and Trevisan developed conditional entropy-production methods and the conditional qEPI [8]; their full linear form and its optimization provide the interpolation used here. Huber, König, and Vershynina established Fisher-information and entropy-power inequalities for quantum–classical convolutions [17]. Beigi and Mehrabi used score and lifting methods in the quantum central limit theorem [5] and proved entropy monotonicity under quantum convolution [6]. Their ordered representation of KMB information and their score-projection identity are direct antecedents of the calculations here. The Gaussian joint-measurement benchmark is also known: its one-mode Bayesian optimum is given in [18, Sec. 7.5]. Appendix C translates that result to our normalization and extends it to independent modes.

Our contributions are as follows.

(i) The arbitrary-prior thermal I–MMSE identity (1.4) (Theorem 1), its estimation interpretation, and its proof under a finite-second-moment assumption. This is the principal contribution. Its model-specific step is the thermal posterior–score relation (1.14), which connects the arbitrary signal prior to entropy differentiation. The ordered geometry and conditional-projection tools used with it are established ingredients.

(ii) Its consequences for a fixed input distribution: concavity in the SNR parameter and the exact initial slope (Corollary 10), and the KMB–SLD correction to ordinary operator MMSE.

(iii) The Gaussian-noise identity (Theorem 2), which expresses established heat entropy production in terms of the same ordered-estimation family.

(iv) An estimation derivation of the known full linear and exponential qEPI (Theorem 14 and Corollary 5.3), together with the linear and logarithmic deficits written as integrated squared projection errors (Proposition 15).

Organization. Section 2 presents the model, the bosonic channels and measurements used later, ordered estimation, the logarithmic derivative, and the known entropy facts for the heat semigroup; results taken from the literature are stated there with references. Section 3 proves Theorem 1, derives its consequences and ordinary-MMSE correction. Section 4 derives Theorem 2 from de Bruijn's identity. Section 5 presents the estimation proof of the full linear qEPI, derives its exponential form and estimation deficit, and gives the direct thermal-mixture interpolation. Section 6 discusses open questions. Appendices A and B give the weighted-space and entropy-differentiation arguments. Appendix C gives the normalization and multimode reduction for the cited measurement optimum.

2 Preliminaries

This section fixes the notation and collects the tools used in the rest of the paper. Subsection 2.1 specifies the bosonic model, Subsection 2.2 the Gaussian channels and measurements, Subsections 2.3–2.4 the ordered estimation errors and their comparison properties, Subsection 2.5 the logarithmic derivative, and Subsection 2.6 the entropy facts for the heat semigroup. The projection and data-processing properties are cq specializations of quantum Bayesian estimation and generalized conditional expectation [11, 12, 13]. The logarithmic derivative is standard in quantum information geometry [19]; its ordered decomposition appears in the bosonic analysis of Beigi and Mehrabi [6, Sec. 3.1]. The de Bruijn identity and the entropy asymptotics of the heat semigroup are due to König and Smith [7] and De Palma and Trevisan [8]. Results taken from the literature are stated with references and without proof; proofs are given only for the extensions to unbounded observables and finite-energy states, mostly in the appendices.

2.1 Bosonic displacement experiments

We work on $\mathcal{H}_m = L^2(\mathbb{R}^m)$ and use natural logarithms. Write $R = (Q_1, P_1, \dots, Q_m, P_m)^T$ for the vector of quadrature operators and $\Omega = \bigoplus_{j=1}^m \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ for the symplectic form. The Schwartz space $\mathscr{S}(\mathbb{R}^m)$ consists of smooth functions whose derivatives decay faster than every inverse polynomial. On this space, $(Q_j\psi)(x) = x_j\psi(x)$ and $(P_j\psi)(x) = -i\partial_j\psi(x)$ satisfy $[R_j, R_k]\psi = i\Omega_{jk}\psi$, with the commutator $[B, C] = BC - CB$ introduced in Section 1; we also write $\{B, C\} = BC + CB$ for the anticommutator. It is invariant under these operators and is a common core: taking the closure of their restrictions to $\mathscr{S}(\mathbb{R}^m)$ recovers the self-adjoint position and momentum operators [20, Sec. 9.8 and Def. A.15]. Define

$$a_j = \frac{Q_j + iP_j}{\sqrt{2}}, \qquad H_0 = \sum_{j=1}^m a_j^\dagger a_j, \qquad \sum_{k=1}^{2m} R_k^2 = 2H_0 + mI. \tag{2.1}$$

Thus a_j and $a_j^\dagger$ are the annihilation and creation operators of mode j. On $\mathscr{S}(\mathbb{R}^m)$ their canonical commutation relations are

$$[a_j, a_k^\dagger] = \delta_{jk} I, \qquad [a_j, a_k] = [a_j^\dagger, a_k^\dagger] = 0.$$

We take the self-adjoint closure of the operator H_0 in (2.1), initially defined on $\mathscr{S}(\mathbb{R}^m)$; it is the total photon-number operator. Products and commutators of these unbounded operators are initially evaluated on $\mathscr{S}(\mathbb{R}^m)$; any subsequent extension or quadratic-form interpretation is specified where used.

For $\boldsymbol{n} = (n_1, \dots, n_m) \in \mathbb{N}_0^m$, write $|\boldsymbol{n}\rangle$ for the normalized photon-number basis vector, with vacuum $|\boldsymbol{0}\rangle = |0\rangle^{\otimes m}$ and $|\boldsymbol{n}| = \sum_j n_j$. In the Schrödinger representation these are the products of normalized Hermite functions; their completeness follows from the one-mode result in [20, Thm. 11.4] by tensor products. They satisfy

$$a_j|\boldsymbol{n}\rangle = \sqrt{n_j}\,|\boldsymbol{n} - \boldsymbol{e}_j\rangle, \qquad a_j^\dagger|\boldsymbol{n}\rangle = \sqrt{n_j + 1}\,|\boldsymbol{n} + \boldsymbol{e}_j\rangle,$$
$$H_0|\boldsymbol{n}\rangle = |\boldsymbol{n}|\,|\boldsymbol{n}\rangle.$$

where $\boldsymbol{e}_j$ is the jth coordinate vector and the first expression is zero when $n_j = 0$. We use $\mathcal{D}_{\text{fin}} = \operatorname{span}\{|\boldsymbol{n}\rangle : \boldsymbol{n} \in \mathbb{N}_0^m\}$ for their finite linear span. Its elements are called finite photon-number vectors (or finite Fock vectors). This dense subspace lies in $\mathscr{S}(\mathbb{R}^m)$ and is invariant under a_j, $a_j^\dagger$, and H_0.

A state is a positive trace-class operator of trace one. Its support is $\operatorname{supp}\rho = (\ker\rho)^\perp$, and it is faithful when $\ker\rho = \{0\}$. For an operator T, write $|T| = (T^\dagger T)^{1/2}$, $\|T\|_1 = \operatorname{Tr}|T|$ for the trace norm, $\|T\|_2 = (\operatorname{Tr} T^\dagger T)^{1/2}$ for the Hilbert–Schmidt norm, and $\|T\|_{\mathrm{op}}$ for the operator norm. An unsubscripted norm of an operator also means its operator norm; for vectors it means the Euclidean norm (or the specified Hilbert-space norm). The energy is the nonnegative spectral sum

$$E_\rho = \operatorname{Tr}(\rho H_0) = \sum_{\boldsymbol{n}} |\boldsymbol{n}|\langle\boldsymbol{n}|\rho|\boldsymbol{n}\rangle,$$

possibly infinite. Equation (2.1) gives $\sum_k \operatorname{Tr}(\rho R_k^2) = 2E_\rho + m$ with these nonnegative trace conventions. Thus finite energy is equivalent to finite quadrature second moments. Since $\operatorname{Tr} e^{-\beta H_0} = (1-e^{-\beta})^{-m} < \infty$ for every $\beta > 0$, the Gibbs maximum-entropy principle [21, Sec. IV, Gibbs Hypothesis] applies and yields $S(\rho) \le mg(E_\rho/m) < \infty$ when $0 < E_\rho < \infty$, with g defined in (2.8). If $E_\rho = 0$, the energy sum forces ρ to be the vacuum state, and the same bound holds with both sides zero. States with finite energy are called states with finite second-order moments in [5, 6]; we use the two phrases interchangeably. We use the covariance convention

$$d_j = \operatorname{Tr}(\rho R_j), \qquad V_{jk} = \tfrac{1}{2}\operatorname{Tr}\rho\{R_j - d_j I, R_k - d_k I\}, \tag{2.2}$$

For finite energy, the product in this formula is understood as a quadratic-form trace: with $T_j = (R_j - d_j I)\rho^{1/2}$,

$$V_{jk} = \operatorname{Re}\operatorname{Tr}(T_j^\dagger T_k), \qquad \|T_j\|_2^2 = \operatorname{Tr}\rho(R_j - d_j I)^2 < \infty.$$

When the state needs to be indicated, we write $d_\rho = (d_j)_j$ and $V_\rho = (V_{jk})_{jk}$.

Background on bosonic states and Gaussian channels can be found in [22]; that review normalizes the quadratures so that the vacuum covariance matrix is I_{2m}, and its covariance matrices are therefore twice ours.

For $z \in \mathbb{C}^m$ the Weyl displacement is

$$D_z = \exp\left(\sum_j z_j a_j^\dagger - \overline{z}_j a_j\right), \qquad D_z^\dagger a_j D_z = a_j + z_j I. \tag{2.3}$$

These unitaries satisfy the Weyl multiplication rule [20, Eq. (14.10) and Example 14.4], which in our normalization is

$$D_z D_w = \exp\left(\frac{1}{2}\sum_j (z_j\overline{w}_j - \overline{z}_j w_j)\right) D_{z+w}, \qquad D_z^\dagger = D_{-z}.$$

One easily checks that $D_z D_w \rho D_w^\dagger D_z^\dagger = D_{z+w}\rho D_{z+w}^\dagger$ for every state ρ.

If $x \in \mathbb{R}^{2m}$ and $z_j(x) = (x_{2j-1} + i x_{2j})/\sqrt{2}$, then $D_{z(x)} = \exp(-i x^T \Omega R)$ and $D_{z(x)}^\dagger R D_{z(x)} = R + x$. In particular $\|x\|^2 = 2\|z(x)\|^2$.

The characteristic function of a state ρ is

$$\chi_\rho(w) = \operatorname{Tr}(\rho D_w), \qquad w \in \mathbb{C}^m; \tag{2.4}$$

it determines ρ uniquely [22, Sec. II.A.1]. The Weyl multiplication rule gives $D_z^\dagger D_w D_z = e^{2i\operatorname{Im}\sum_j \overline{z}_j w_j} D_w$, and hence, for every state ρ,

$$\chi_{D_z \rho D_z^\dagger}(w) = e^{2i\operatorname{Im}\sum_j \overline{z}_j w_j}\chi_\rho(w). \tag{2.5}$$

For quantum Gaussian states and their characteristic functions please check [22, Sec. II.A.1]. The coherent states are the displaced vacua $|\alpha\rangle = D_\alpha|\mathbf{0}\rangle$, $\alpha \in \mathbb{C}^m$, with the photon-number expansion [22, Sec. II.B.2, Eq. (29)]

$$|\alpha\rangle = e^{-\|\alpha\|^2/2} \sum_{\boldsymbol{n}} \frac{\prod_j \alpha_j^{n_j}}{\sqrt{\prod_j n_j!}} |\boldsymbol{n}\rangle. \tag{2.6}$$

In the signal model, the classical displacement specifies the information-bearing input and the state before displacement describes the thermal noise. The family of conditional output states, together with the input distribution, specifies the estimation problem. We call such a family, with its prior, an *experiment*; A_s denotes the thermal signal experiment at SNR parameter s, and $I(Z; A_s)$ its Holevo information. The value z of the classical variable is called the *label* of the conditional state ($\rho_{s,z}$ here, ω_z in general); in [6] the label is carried by a classical register. Our thermal parameter is the mean photon number $N > 0$ per mode; we retain $\nu = N + 1/2$ as a shorthand for the quadrature variance. The centered thermal noise state is

$$\tau_\nu = (N+1)^{-m} \left(\frac{N}{N+1}\right)^{H_0}, \qquad \nu = N + \tfrac{1}{2}, \tag{2.7}$$

with covariance νI_{2m}, energy mN, and entropy $mg(N)$, where

$$g(x) = (x+1)\log(x+1) - x\log x, \qquad g(0) = 0. \tag{2.8}$$

Its characteristic function is $\chi_{\tau_\nu}(w) = e^{-\nu\|w\|^2}$ [22, Sec. II.B.1], and (2.6) with $\xi = N/(N+1)$ gives its coherent-state matrix elements

$$\langle\alpha|\tau_\nu|\alpha\rangle = (N+1)^{-m} e^{-\|\alpha\|^2} \sum_{\boldsymbol{n}} \frac{\xi^{|\boldsymbol{n}|} \prod_j |\alpha_j|^{2n_j}}{\prod_j n_j!} = (N+1)^{-m} \exp\left(-\frac{\|\alpha\|^2}{N+1}\right). \tag{2.9}$$

The value $N = 0$, equivalently $\nu = 1/2$, denotes the vacuum $\tau_{1/2} = |\mathbf{0}\rangle\langle\mathbf{0}|$ and is used only where explicitly allowed. For the thermal signal experiment (1.2), set

$$\Gamma_j(s) = \mathbb{E}[Z_j \rho_{s,Z}], \qquad V_Z = \mathbb{E}\|Z\|^2 < \infty. \tag{2.10}$$

All expectations of operators are Bochner integrals in trace norm, that is, trace-norm limits of integrals of simple operator-valued functions. Here they exist because the displacement orbit is trace-norm continuous and

$$\mathbb{E}\|\rho_{s,Z}\|_1 = 1, \qquad \mathbb{E}\|Z_j \rho_{s,Z}\|_1 = \mathbb{E}|Z_j| \le (\mathbb{E}|Z_j|^2)^{1/2} < \infty.$$

Using $D^\dagger_{\sqrt{s}z} a_j D_{\sqrt{s}z} = a_j + \sqrt{s} z_j I$ and $\mathrm{Tr}(\tau_\nu a_j) = 0$ gives $\mathrm{Tr}(\rho_s H_0) = mN + sV_Z$. The mutual information between Z and A_s is

$$I(s) = I(Z; A_s) = S(\rho_s) - mg(N). \tag{2.11}$$

The states $\rho_{s,z}$ have entropy $mg(N)$, and ρ_s has finite entropy by its finite energy. Tonelli's theorem for the nonnegative cross-entropy term gives

$$\int \mathrm{Tr}[\rho_{s,z}(-\log\rho_s)]\, P_Z(dz) = \mathrm{Tr}[\rho_s(-\log\rho_s)] = S(\rho_s),$$

which proves the last equality in (2.11) for discrete, continuous, and mixed priors.

We use $s \ge 0$ as the SNR parameter in the encoding $D_{\sqrt{s}z} \tau_\nu D^\dagger_{\sqrt{s}z}$, with the prior and thermal noise held fixed.

Table 1 summarizes the notation used repeatedly below. The weight map and scores are defined in Subsections 2.3 and 2.5; the Jordan midpoint is used in Subsection 3.3.

Notation	Meaning
$s;\ N, \nu$	SNR parameter; thermal mean photon number and covariance parameter, $\nu = N + 1/2$.
$\omega_z, \omega, \Gamma_j$	Conditional state, average state, and first moment $\Gamma_j = \mathbb{E}[Z_j \omega_Z]$ of the specified experiment.
$\mathsf{M}_u, \widehat{Z}_{j,u}$	Minimum ordered risk and optimal estimator of coordinate Z_j.
$\mathcal{K}_{\omega,u}, \mathcal{J}_\omega$	Ordered weight map and its Jordan midpoint $\mathcal{J}_\omega = \mathcal{K}_{\omega,1/2}$.
$\mathcal{T}_\omega$	KMB logarithmic derivative.
$\mathcal{S}_{j,u}, J_u, J_{\mathrm{KMB}}$	Ordered score, its squared norm summed over modes, and its ordering average.
$\mathcal{N}_t,\ \mathcal{B}_\eta,\ \mathcal{A}_{\eta,N}$	Heat channel adding noise variance t, beam-splitter channel of transmissivity η, and thermal attenuator of transmissivity η with environment τ_ν (Subsection 2.2).
$G,\ I_\rho(t),\ \mathsf{M}_{\rho,u}(t)$	Standard complex Gaussian label of the noise experiment, its Holevo information, and its ordered error.
$\lambda,\ d(\lambda\|\eta),\ \Phi_\lambda,\ \mathcal{G}_{\lambda,u}$	Entropy weight, binary relative entropy, entropy difference along the unequal-time interpolation, and ordered-risk gap (Section 5).

Table 1: Notation for the displacement and estimation formulas.

2.2 Gaussian channels and measurements

We define the three channels and the two measurements used later and record their elementary properties.

Beam splitter. Let A and B be two copies of the m-mode system, with annihilation operators $a_{A,j} = a_j \otimes I$ and $a_{B,j} = I \otimes a_j$ on $\mathcal{H}_m \otimes \mathcal{H}_m$. For $0 < \eta < 1$, the beam-splitter unitary of transmissivity η is

$$U_\eta = \exp\Big[\vartheta \sum_{j=1}^{m} (a_{A,j}^\dagger a_{B,j} - a_{A,j} a_{B,j}^\dagger)\Big], \qquad \cos\vartheta = \sqrt{\eta}, \quad \sin\vartheta = \sqrt{1-\eta},$$

see [22] and [7, Sec. III]. Write $a_{C,j} = a_{A,j}$ and $a_{E,j} = a_{B,j}$ for the same operators regarded as the output modes, and $\Lambda_{AB} = \sum_j (a_{A,j}^\dagger a_{B,j} - a_{A,j} a_{B,j}^\dagger)$, so that $U_\eta = e^{\vartheta \Lambda_{AB}}$. The operator Λ_{AB} commutes with $H_A + H_B$ and leaves each of its finite-dimensional eigenspaces invariant, so $e^{\vartheta \Lambda_{AB}}$ preserves $\mathcal{D}_{\mathrm{fin}} \otimes \mathcal{D}_{\mathrm{fin}}$. There the canonical commutation relations give $[a_{A,j}, \Lambda_{AB}] = a_{B,j}$ and $[a_{B,j}, \Lambda_{AB}] = -a_{A,j}$; hence the pair $(e^{-\vartheta\Lambda_{AB}} a_{A,j} e^{\vartheta\Lambda_{AB}}, e^{-\vartheta\Lambda_{AB}} a_{B,j} e^{\vartheta\Lambda_{AB}})$ solves $\frac{d}{d\vartheta}(x, y) = (y, -x)$ with initial value $(a_{A,j}, a_{B,j})$, which is the rotation

$$\begin{aligned} U_\eta^\dagger a_{C,j} U_\eta &= \sqrt{\eta}\, a_{A,j} + \sqrt{1-\eta}\, a_{B,j}, \\ U_\eta^\dagger a_{E,j} U_\eta &= -\sqrt{1-\eta}\, a_{A,j} + \sqrt{\eta}\, a_{B,j}. \end{aligned} \tag{2.12}$$

Let $H_L = \sum_j a_{L,j}^\dagger a_{L,j}$ for $L \in \{A, B, C, E\}$; operators on different systems are tensored with identities. With $T_{AB} = \sum_j (a_{A,j}^\dagger a_{B,j} + a_{B,j}^\dagger a_{A,j})$, substitution of (2.12) on the finite photon-number

span gives

$$\begin{aligned} U_\eta^\dagger(H_C + H_E)U_\eta &= \eta H_A + (1-\eta)H_B + \sqrt{\eta(1-\eta)}\,T_{AB} \\ &\quad + (1-\eta)H_A + \eta H_B - \sqrt{\eta(1-\eta)}\,T_{AB} \\ &= H_A + H_B. \end{aligned}$$

Approximation by finite photon-number vectors extends this equality of nonnegative quadratic forms to $\mathrm{Dom}((H_A+H_B)^{1/2})$; applying the inverse beam splitter gives the reverse domain inclusion. Hence the closed forms, and their associated self-adjoint operators, agree. This is the photon-number conservation used below.

For a trace-class operator ω on a tensor product, the partial trace over a factor is defined by duality; for example, the partial trace over the second factor E is the trace-class operator $\mathrm{Tr}_E\,\omega$ with $\mathrm{Tr}[(\mathrm{Tr}_E\,\omega)B] = \mathrm{Tr}[\omega(B\otimes I)]$ for all bounded B, and $\mathrm{Tr}_A, \mathrm{Tr}_B, \mathrm{Tr}_C$ are defined likewise. Define the beam-splitter channel

$$\mathcal{B}_\eta(\omega) = \mathrm{Tr}_E(U_\eta \omega U_\eta^\dagger), \qquad \rho_C = \mathcal{B}_\eta(\rho_A \otimes \rho_B). \tag{2.13}$$

In the notation of [6], $\rho_C = \rho_A \boxplus_\eta \rho_B$ is the quantum convolution of the two inputs. At the endpoints, define $\mathcal{B}_0(\omega) = \mathrm{Tr}_A\,\omega$ and $\mathcal{B}_1(\omega) = \mathrm{Tr}_B\,\omega$, identifying the retained system with C. The displacement covariance of this channel is

$$\mathcal{B}_\eta(D_\alpha \rho_A D_\alpha^\dagger \otimes D_\beta \rho_B D_\beta^\dagger) = D_{\sqrt{\eta}\alpha + \sqrt{1-\eta}\beta}\,\rho_C\, D^\dagger_{\sqrt{\eta}\alpha+\sqrt{1-\eta}\beta}. \tag{2.14}$$

If $\rho_{CE} = U_\eta(\rho_A\otimes\rho_B)U_\eta^\dagger$ and $\rho_E = \mathrm{Tr}_C\,\rho_{CE}$, the photon-number conservation established above gives

$$E_{\rho_C} + E_{\rho_E} = \mathrm{Tr}[\rho_{CE}(H_C+H_E)] = \mathrm{Tr}[(\rho_A\otimes\rho_B)(H_A+H_B)] = E_{\rho_A} + E_{\rho_B}, \tag{2.15}$$

where $E_\rho = \mathrm{Tr}(\rho H_0)$ is the energy of a state on one copy of the system. Since $E_{\rho_E} \geq 0$, the output ρ_C has finite energy whenever ρ_A and ρ_B do.

The product of two thermal states with the same mean photon number is invariant under beam splitter

$$\mathcal{B}_\eta(\tau_\nu \otimes \tau_\nu) = \tau_\nu. \tag{2.16}$$

Thermal attenuator. For $0 < \eta < 1$ and $N > 0$, the thermal attenuator of transmissivity η with environment τ_ν is

$$\mathcal{A}_{\eta,N}(\rho) = \mathcal{B}_\eta(\rho \otimes \tau_\nu); \tag{2.17}$$

it is the one-mode lossy channel with a thermal environment of [22, Sec. V.B], applied to each mode. By (2.14) with $\beta = 0$ and (2.16),

$$\mathcal{A}_{\eta,N}(\tau_\nu) = \tau_\nu, \qquad \mathcal{A}_{\eta,N}(D_z \tau_\nu D_z^\dagger) = D_{\sqrt{\eta}z}\,\tau_\nu\, D^\dagger_{\sqrt{\eta}z}, \qquad z \in \mathbb{C}^m. \tag{2.18}$$

Heat channel. For $t \geq 0$, let $G \sim \mathcal{CN}(0, I_m)$ have density $\pi^{-m}\exp(-\|g\|^2)$ and define

$$\mathcal{N}_t(\rho) = \mathbb{E} D_{\sqrt{t}G}\,\rho\, D^\dagger_{\sqrt{t}G} = \int_{\mathbb{C}^m} \pi^{-m} e^{-\|g\|^2} D_{\sqrt{t}g}\,\rho\, D^\dagger_{\sqrt{t}g}\, d^{2m}g. \tag{2.19}$$

This is the additive classical-noise channel of Gaussian quantum information [22]; we call it the heat channel.

The displacement rule (2.3) and $\mathbb{E}G_j = 0$, $\mathbb{E}|G_j|^2 = 1$ give

$$d_{\rho_t} = d_\rho, \qquad V_{\rho_t} = V_\rho + tI_{2m}, \qquad \mathrm{Tr}(\rho_t H_0) = \mathrm{Tr}(\rho H_0) + mt.$$

Writing $\rho_t = \mathcal{N}_t(\rho)$, the rule (2.5) gives

$$\chi_{\rho_t}(z) = \mathbb{E}\, e^{2i\sqrt{t}\,\mathrm{Im}\sum_j \overline{G}_j z_j}\chi_\rho(z) = e^{-t\|z\|^2}\chi_\rho(z), \tag{2.20}$$

because $\mathrm{Im}\sum_j \overline{G}_j z_j$ is a real Gaussian variable with variance $\|z\|^2/2$. By Lemma 8 below, the trajectory is differentiable in trace norm for $t > 0$, with

$$\dot{\rho}_t = -\sum_j [a_j^\dagger, [a_j, \rho_t]]. \tag{2.21}$$

The family $\{\mathcal{N}_t\}_{t\geq 0}$ is the quantum heat semigroup of [7, 8, 6]. In the notation of [6], $\mathcal{N}_t(\rho)$ is the quantum–classical convolution $\rho \star_t G$ of ρ with the Gaussian label G. The beam splitter is compatible with the heat semigroup. For $s, r \geq 0$ and independent $G_A, G_B \sim \mathcal{CN}(0, I_m)$,

$$\mathcal{B}_\eta(\mathcal{N}_s(\rho_A) \otimes \mathcal{N}_r(\rho_B)) = \mathcal{N}_{\eta s+(1-\eta)r}(\mathcal{B}_\eta(\rho_A \otimes \rho_B)), \tag{2.22}$$

which is [8, Lemma 2] and [7, Lemma III.2].

For later reference we summarize the beam-splitter facts.

Lemma 3 (Beam-splitter covariance, thermal invariance, and heat compatibility [22, 7, 8])**.** *Let $0 < \eta < 1$ and let ρ_A, ρ_B be m-mode states. Then $\mathcal{B}_\eta$ satisfies the displacement covariance* (2.14)*, the energy identity* (2.15)*, the thermal invariance* (2.16)*, and the heat compatibility* (2.22)*; and the thermal attenuator* (2.17) *satisfies* (2.18)*.*

Measurements. A measurement with outcomes in a standard Borel space $\mathcal{Y}$ is a positive operator-valued measure (POVM) Π: a map from Borel sets to positive bounded operators, countably additive in the weak operator topology, with $\Pi(\mathcal{Y}) = I$. The outcome distribution for the state ω is $P(dy) = \mathrm{Tr}(\omega\Pi(dy))$. Two measurements are used below. Heterodyne measurement is the coherent-state POVM

$$\Pi(d^{2m}\alpha) = \pi^{-m}|\alpha\rangle\langle\alpha|\, d^{2m}\alpha, \qquad Y = \sqrt{2}(\mathrm{Re}\,\alpha_1, \mathrm{Im}\,\alpha_1, \ldots, \mathrm{Re}\,\alpha_m, \mathrm{Im}\,\alpha_m), \tag{2.23}$$

where Y is the real outcome in the normalization of the quadratures. The polar-coordinate integral

$$\frac{1}{\pi}\int_{\mathbb{C}} e^{-|\alpha|^2}\frac{\alpha^n\overline{\alpha}^{\,k}}{\sqrt{n!k!}}\, d^2\alpha = \delta_{nk}$$

and (2.6) give $\int \Pi(d^{2m}\alpha) = I$ entrywise in the photon-number basis, hence in the weak operator sense. Homodyne measurement of a quadrature R_k is its projection-valued spectral measure; its outcome is a real number.

2.3 Ordered quadratic estimation

Consider a classical variable encoded in a quantum system. Let $Z \in \mathbb{C}^m$ have finite second moment, let $z \mapsto \omega_z$ be a family of density operators measurable in trace norm, and write

$$\omega = \mathbb{E}\omega_Z, \qquad \Gamma_j = \mathbb{E}[Z_j\omega_Z]. \tag{2.24}$$

We seek operator estimates of the coordinates of Z from the quantum system A. A complex coordinate has an operator estimate B_j that need not be self-adjoint. Consequently, its two squared errors, $(Z_jI - B_j)(Z_jI - B_j)^\dagger$ and $(Z_jI - B_j)^\dagger(Z_jI - B_j)$, can differ. The parameter u below assigns a weight to each order. This family will allow us to match the operator logarithm appearing in the entropy derivative. The discussion after Lemma 4 spells out the classical dictionary: when the conditional states commute, the construction is ordinary conditional expectation, whatever the value of u.

Write $\mathcal{B}(\mathcal{H})$ for the bounded operators on the encoding Hilbert space. We define the infimum over $\mathcal{B}(\mathcal{H})^m$ and solve the minimization in its weighted completion.

For $0 < u < 1$, define

$$\mathcal{K}_{\omega,u}(B) = (1-u)\omega B + uB\omega, \qquad \langle B, C\rangle_{\omega,u} = \operatorname{Tr}[B^\dagger \mathcal{K}_{\omega,u}(C)]. \tag{2.25}$$

In particular,

$$\|B\|^2_{\omega,u} = (1-u)\operatorname{Tr}(\omega BB^\dagger) + u\operatorname{Tr}(\omega B^\dagger B). \tag{2.26}$$

This is a seminorm when ω is not faithful. Two bounded operators are identified if their difference has zero seminorm; the resulting quotient is then completed to the Hilbert space $\mathsf{L}^2(\omega, u)$. Its inner product is conjugate-linear in the first argument, as in (2.25). Its elements are mean-square limits of bounded estimators. For the first-moment operators in (2.24), inverse weight maps take values in this completion, as constructed in Appendix A.

In particular, $\mathcal{K}^{-1}_{\omega,u}(C)$ denotes the completed solution of the normal equation on the range of the extended weight map; it does not assert the existence of a bounded inverse on all trace-class operators.

For a bounded vector estimator $B = (B_1, \dots, B_m)$, consider the positive quadratic loss

$$\mathcal{R}_u(B) = \mathbb{E}\sum_{j=1}^m \operatorname{Tr}\omega_Z\big[(1-u)(Z_jI - B_j)(\overline{Z}_jI - B_j^\dagger) + u(\overline{Z}_jI - B_j^\dagger)(Z_jI - B_j)\big], \tag{2.27}$$

and its minimum

$$\mathsf{M}_u(Z \mid A) = \inf_{B\in\mathcal{B}(\mathcal{H})^m} \mathcal{R}_u(B). \tag{2.28}$$

Both orderings in (2.27) are nonnegative. Moreover, their dependence on the conditional state is linear. This makes conditional projection and data processing available for every u. The risk extends continuously to $\mathsf{L}^2(\omega, u)^m$ and has the same infimum there. We use *completed estimator* to mean an element of this space. An unbounded observable belongs to it when bounded approximations converge in both terms of (2.26).

Lemma 4 (Conditional projection)**.** *For the finite-second-moment ensemble in* (2.24)*, $\mathcal{K}_{\omega,u}$ extends continuously from $\mathsf{L}^2(\omega, u)$ to trace class. The risk has a unique minimizer in $\mathsf{L}^2(\omega, u)^m$, characterized by*

$$\mathcal{K}_{\omega,u}(\widehat{Z}_{j,u}) = \Gamma_j, \qquad \widehat{Z}_{j,u} = \mathcal{K}^{-1}_{\omega,u}(\Gamma_j). \tag{2.29}$$

For every completed estimator B,

$$\mathcal{R}_u(B) = \mathsf{M}_u(Z \mid A) + \sum_j \|B_j - \widehat{Z}_{j,u}\|^2_{\omega,u}, \tag{2.30}$$

and

$$\mathsf{M}_u(Z \mid A) = \mathbb{E}\|Z\|^2 - \sum_j \operatorname{Tr}[\Gamma_j^\dagger \mathcal{K}^{-1}_{\omega,u}(\Gamma_j)]. \tag{2.31}$$

Each trace pairing is the squared norm $\|\widehat{Z}_{j,u}\|^2_{\omega,u}$. In particular, $0 \le \mathsf{M}_u(Z \mid A) \le \mathbb{E}\|Z - \mathbb{E}Z\|^2$.

Proof. Appendix A constructs the completed solution of (2.29) and proves that, for bounded B_j,

$$\mathrm{Tr}(\Gamma_j^\dagger B_j) = \langle \widehat{Z}_{j,u}, B_j \rangle_{\omega,u}. \tag{2.32}$$

This equation defines the pairing for completed estimators. For bounded B,

$$\mathcal{R}_u(B) \overset{\text{(a)}}{=} \mathbb{E}\|Z\|^2 - 2\,\mathrm{Re}\sum_j \mathrm{Tr}(\Gamma_j^\dagger B_j) + \sum_j \|B_j\|_{\omega,u}^2 \tag{2.33}$$

$$\overset{\text{(b)}}{=} \mathbb{E}\|Z\|^2 - \sum_j \|\widehat{Z}_{j,u}\|_{\omega,u}^2 + \sum_j \|B_j - \widehat{Z}_{j,u}\|_{\omega,u}^2, \tag{2.34}$$

where (a) multiplies out the two products in (2.27) and uses $\mathrm{Tr}\,\omega_Z = 1$, $\mathbb{E}[\overline{Z}_j \omega_Z] = \Gamma_j^\dagger$, and $\omega = \mathbb{E}\omega_Z$, and (b) is (2.32) together with $\|B_j - \widehat{Z}_{j,u}\|_{\omega,u}^2 = \|B_j\|_{\omega,u}^2 - 2\,\mathrm{Re}\langle \widehat{Z}_{j,u}, B_j\rangle_{\omega,u} + \|\widehat{Z}_{j,u}\|_{\omega,u}^2$. The last sum vanishes exactly at $B = \widehat{Z}_{\cdot,u}$. Density of bounded estimators and continuity of the displayed quadratic form give the same infimum in (2.28). This proves (2.30)–(2.31). Positivity of the loss and the constant estimator $B_j = (\mathbb{E}Z_j)I$ give the stated bounds. □

This is the generalized conditional-expectation construction of [12, Sec. III.B] and [13, Eqs. (2.15)–(2.17)]. Those references develop the general formulas in finite dimensions; Appendix A supplies the completion and trace-class extension used here. In an eigenbasis $\omega = \sum_k \mu_k |k\rangle\langle k|$, the normal equation reads

$$(\widehat{Z}_{j,u})_{k\ell} = \frac{(\Gamma_j)_{k\ell}}{(1-u)\mu_k + u\mu_\ell} \quad \text{when the denominator is positive.} \tag{2.35}$$

Zero denominators correspond to null directions and may be assigned zero entries. The trace pairing in (2.31) is therefore the convergent nonnegative sum

$$\|\widehat{Z}_{j,u}\|_{\omega,u}^2 = \sum_{(1-u)\mu_k + u\mu_\ell > 0} \frac{|(\Gamma_j)_{k\ell}|^2}{(1-u)\mu_k + u\mu_\ell}.$$

When the conditional states commute, this construction reduces to ordinary conditional expectation. Indeed, in a common eigenbasis, write $p(k \mid z) = \langle k|\omega_z|k\rangle$, which is the likelihood of the outcome k; then ω and Γ_j are diagonal with entries $p(k) = \mathbb{E}[p(k \mid Z)]$ and $\mathbb{E}[Z_j\, p(k \mid Z)]$. Formula (2.35) then gives $(\widehat{Z}_{j,u})_{kk} = \mathbb{E}[Z_j \mid k] = \mathbb{E}[Z_j\, p(k \mid Z)]/p(k)$ whenever $p(k) > 0$, independently of u. Values at outcomes with $p(k) = 0$ are immaterial to the risk and may be set to zero. The minimum risk is the classical mean-square error for observing k. Moreover, for an estimator that is diagonal in this basis, $B_j = \sum_k b_j(k)|k\rangle\langle k|$ with b_j bounded, identity (2.30) becomes, coordinate by coordinate, the familiar decomposition

$$\mathbb{E}|Z_j - b_j(Y)|^2 = \mathbb{E}|Z_j - \mathbb{E}[Z_j \mid Y]|^2 + \mathbb{E}|\mathbb{E}[Z_j \mid Y] - b_j(Y)|^2,$$

where Y is the outcome k, drawn with probability $p(k \mid Z)$. Its cross term vanishes by the tower property of conditional expectation; in the quantum case the normal equation $\mathcal{K}_{\omega,u}(\widehat{Z}_{j,u}) = \Gamma_j$ plays that role. In the noncommuting case the ordering matters. The completion and spectral interpretation in Lemma 4 make the same projection construction applicable to the unbounded observables needed for bosonic displacement models.

The midpoint recovers ordinary operator squared error. Write $Z_j = (X_{2j-1} + iX_{2j})/\sqrt{2}$ and $B_j = (C_{2j-1} + iC_{2j})/\sqrt{2}$, with self-adjoint C_k. At $u = 1/2$, the loss is half the sum of the ordinary coordinatewise squared errors. Thus

$$\mathsf{M}_{1/2}(Z \mid A) = \tfrac{1}{2}\,\mathrm{mmse}_J(X \mid A), \tag{2.36}$$

where $\mathrm{mmse}_J(X \mid A)$ is the coordinatewise Jordan operator error defined in (3.24), and the coordinate estimators solve $\{\omega, \widehat{X}_k\}/2 = \mathbb{E}[X_k \omega_X]$. For a scalar real target, this minimum equals the infimum of the risk over measurements: the spectral measurement of a bounded candidate attains its operator risk, and bounded approximation gives equality of infima. For a vector, one measurement need not attain the separate coordinate minima, so its optimal risk can be strictly larger. The target Z_j is complex, so its estimator need not be self-adjoint. More generally, adjoint conjugates the target and reverses the ordering:

$$\widehat{\overline{Z_j}}_{\,1-u} = (\widehat{Z}_{j,u})^\dagger, \qquad \|B^\dagger\|_{\omega,1-u} = \|B\|_{\omega,u}. \tag{2.37}$$

The first equality follows by taking adjoints in the normal equation; both equalities extend by continuity to the completions. Thus a real target has a self-adjoint completed estimator at $u = 1/2$, whereas the conditional-estimator map generally fails to preserve adjoints at a fixed $u \neq 1/2$. Each ordering defines a positive quadratic loss and its own conditional projection.

2.4 Data processing and independent experiments

Two properties of classical conditional mean estimation remain valid for every ordered loss. Processing an observation cannot reduce the minimum error. For independent experiments, the minimum error in a linear combination of the unknown variables is the corresponding sum of their separate errors. These are the only estimation comparisons needed for the entropy power inequality in Section 5. Data processing is known for the Jordan-product Bayesian error in finite dimensions [13, Cor. 1]; the short argument below covers every ordering and the unbounded estimators of the bosonic setting. The independence property is the operator analogue of an elementary classical fact.

Lemma 5 (Data processing; cf. [13, Cor. 1])**.** *For the ensemble in* (2.24)*, if a quantum channel* Φ *(a completely positive trace-preserving map on trace-class operators) maps the encoding system* A *to* C*, then*

$$\mathsf{M}_u(Z \mid C) \geq \mathsf{M}_u(Z \mid A), \qquad 0 < u < 1. \tag{2.38}$$

Proof. The adjoint channel is unital and completely positive, with $\mathrm{Tr}[\Phi(T)B] = \mathrm{Tr}[T\Phi^\dagger(B)]$. For every bounded output estimator B, Kadison–Schwarz gives

$$\Phi^\dagger(B_j)\Phi^\dagger(B_j)^\dagger \leq \Phi^\dagger(B_j B_j^\dagger), \qquad \Phi^\dagger(B_j)^\dagger\Phi^\dagger(B_j) \leq \Phi^\dagger(B_j^\dagger B_j).$$

Thus, with $\omega_C = \Phi(\omega_A)$, trace duality yields

$$\|\Phi^\dagger(B_j)\|^2_{\omega_A,u} \leq (1-u)\,\mathrm{Tr}(\omega_C B_j B_j^\dagger) + u\,\mathrm{Tr}(\omega_C B_j^\dagger B_j) = \|B_j\|^2_{\omega_C,u}.$$

The linear terms agree because

$$\Gamma_j^C = \Phi(\Gamma_j^A), \qquad \mathrm{Tr}[(\Gamma_j^C)^\dagger B_j] = \mathrm{Tr}[(\Gamma_j^A)^\dagger \Phi^\dagger(B_j)]. \tag{2.39}$$

Substitution in (2.33) gives

$$\mathsf{M}_u(Z \mid A) \leq \mathcal{R}_u^A(\Phi^\dagger(B)) \leq \mathcal{R}_u^C(B).$$

Taking the infimum over bounded B proves (2.38). The norm bound also extends $\Phi^\dagger$ as a contraction between the completed estimation spaces. □

Lemma 6 (Independent experiments)**.** *Suppose $Z_1, Z_2 \in \mathbb{C}^m$ are independent, have finite second moments, and their joint encoding is $\omega_{z_1}^A \otimes \omega_{z_2}^B$. For $c, d \in \mathbb{C}$,*

$$\mathsf{M}_u(cZ_1 + dZ_2 \mid AB) = |c|^2 \mathsf{M}_u(Z_1 \mid A) + |d|^2 \mathsf{M}_u(Z_2 \mid B). \tag{2.40}$$

Proof. Set $Y = cZ_1 + dZ_2$. Independence gives $\omega_{AB} = \omega_A \otimes \omega_B$ and $\Gamma_j^{AB} = c\Gamma_j^A \otimes \omega_B + d\omega_A \otimes \Gamma_j^B$. The tensor rule with one identity factor is

$$\mathcal{K}_{\omega_A \otimes \omega_B, u}(T \otimes I) = \mathcal{K}_{\omega_A, u}(T) \otimes \omega_B; \tag{2.41}$$

see also [13, Eq. (2.5)]. The norm identities in Lemma 20 and continuity of $\mathcal{K}$ extend this rule to completed estimators. Hence

$$\begin{aligned} &\mathcal{K}_{\omega_{AB}, u}(c\widehat{Z}_{1,j,u} \otimes I + dI \otimes \widehat{Z}_{2,j,u}) \\ &\qquad = c\Gamma_j^A \otimes \omega_B + d\omega_A \otimes \Gamma_j^B = \Gamma_j^{AB}, \end{aligned}$$

which identifies the argument as $\widehat{Y}_{j,u}$. Writing $E_{i,j} = Z_{i,j} I - \widehat{Z}_{i,j,u}$, the normal equations tested against I give $\mathbb{E}\,\mathrm{Tr}(\omega_{Z_1}^A E_{1,j}) = \mathbb{E}\,\mathrm{Tr}(\omega_{Z_2}^B E_{2,j}) = 0$. In particular,

$$\mathbb{E}\,\mathrm{Tr}[(\omega_{Z_1}^A \otimes \omega_{Z_2}^B)(E_{1,j} \otimes E_{2,j}^\dagger)] = \mathbb{E}\,\mathrm{Tr}(\omega_{Z_1}^A E_{1,j})\, \overline{\mathbb{E}\,\mathrm{Tr}(\omega_{Z_2}^B E_{2,j})} = 0.$$

The conjugate cross term also vanishes; the second operator ordering has the same two cross terms. These field pairings extend by Cauchy–Schwarz to completed estimators. Expanding the ordered loss of $Y_j I - \widehat{Y}_{j,u} = cE_{1,j} \otimes I + dI \otimes E_{2,j}$ therefore gives

$$\mathsf{M}_u(Y \mid AB) = |c|^2 \mathsf{M}_u(Z_1 \mid A) + |d|^2 \mathsf{M}_u(Z_2 \mid B).$$

□

2.5 Logarithmic derivatives and KMB geometry

Entropy differentiation introduces the operator logarithm. To relate that derivative to estimation, we need to express its derivative using the inverse weight maps from (2.29). The key is a scalar identity for the logarithmic divided difference. Applied entry by entry in an eigenbasis, it writes the KMB map as the average of all ordered inverse maps. Ordinary operator squared error uses only their midpoint, which explains why it does not generally give the entropy derivative by itself.

For a faithful state with spectral decomposition $\omega = \sum_k \mu_k |k\rangle\langle k|$, and with $C_{k\ell} = \langle k|C|\ell\rangle$, let $\mathcal{T}_\omega$ denote the divided-difference form

$$(\mathcal{T}_\omega C)_{k\ell} = \frac{\log \mu_k - \log \mu_\ell}{\mu_k - \mu_\ell} C_{k\ell}, \tag{2.42}$$

with coefficient $1/\mu_k$ whenever $\mu_k = \mu_\ell$. In finite dimensions, $\mathcal{T}_\omega$ is the derivative of the matrix logarithm at ω: for self-adjoint C, $\frac{d}{d\varepsilon} \log(\omega + \varepsilon C)|_{\varepsilon=0} = \mathcal{T}_\omega(C)$. This is why it governs entropy derivatives: they involve $\log \omega$, and differentiating $\log \omega$ along a state path brings in $\mathcal{T}_\omega$ (Lemma 7). The elementary identity, for $x, y > 0$,

$$\frac{\log x - \log y}{x - y} = \int_0^1 \frac{du}{(1-u)x + uy} \tag{2.43}$$

implies

$$\mathcal{T}_\omega = \int_0^1 \mathcal{K}_{\omega,u}^{-1}\, du \tag{2.44}$$

as an identity of nonnegative quadratic forms: for a trace-class C, with either side allowed to be $+\infty$,

$$q_\omega(C) := \sum_{k,\ell} |C_{k\ell}|^2 \frac{\log \mu_k - \log \mu_\ell}{\mu_k - \mu_\ell} = \int_0^1 \sum_{k,\ell} \frac{|C_{k\ell}|^2}{(1-u)\mu_k + u\mu_\ell}\, du.$$

The coefficients at equal eigenvalues use the convention in (2.42). Thus (2.44) asserts equality of these spectral sums, rather than convergence in the norm of bounded operators. The decomposition is the one in [6, Sec. 3.1, Eqs. (26)–(31)], with the changes of variables $u = v/(1+v)$ and $u = 1/(1+v)$ on the two halves of $[0,1]$. Half of their measure, $dv/(1+v)^2$, becomes du on each half, with reversed orientation on the second. Thus the ordering integral itself is an established KMB representation; the thermal weight in Theorem 1 is the additional feature of our signal identity.

For a faithful finite-energy bosonic state ω, define the ordered score and information

$$\mathcal{S}_{j,u}(\omega) = \mathcal{K}^{-1}_{\omega,u}([a_j, \omega]), \qquad J_u(\omega) = \sum_j \|\mathcal{S}_{j,u}(\omega)\|^2_{\omega,u}. \tag{2.45}$$

Put $e_j = \operatorname{Tr}(\omega a_j^\dagger a_j) < \infty$. The commutator $C_j = [a_j, \omega]$ is trace class, because Hilbert–Schmidt factorization gives $\|a_j\omega\|_1 \le \|a_j\omega^{1/2}\|_2\|\omega^{1/2}\|_2 = \sqrt{e_j}$ and $\|\omega a_j\|_1 = \|a_j^\dagger\omega\|_1 \le \sqrt{e_j+1}$. To check that the inverse in (2.45) belongs to the estimation completion, note that for bounded B, Cauchy–Schwarz gives

$$\begin{aligned} |\operatorname{Tr}(B^\dagger C_j)| &\le \sqrt{e_j}\sqrt{\operatorname{Tr}(\omega B^\dagger B)} + \sqrt{e_j+1}\sqrt{\operatorname{Tr}(\omega B B^\dagger)} \\ &\le \left(\frac{e_j}{u} + \frac{e_j+1}{1-u}\right)^{1/2} \|B\|_{\omega,u}. \end{aligned}$$

The functional therefore has a Riesz representative in $\mathsf{L}^2(\omega, u)$, which is $\mathcal{S}_{j,u}$ by its weak normal equation. Equivalently,

$$(\mathcal{S}_{j,u})_{k\ell} = \frac{(C_j)_{k\ell}}{(1-u)\mu_k + u\mu_\ell}, \qquad \|\mathcal{S}_{j,u}\|^2_{\omega,u} = \sum_{k,\ell} \frac{|(C_j)_{k\ell}|^2}{(1-u)\mu_k + u\mu_\ell} < \infty.$$

This interior-u bound does not imply integrability at $u = 0, 1$; J_{KMB} below may still be infinite. The commutator is the combination $[a_j, \omega] = ([Q_j, \omega] + i[P_j, \omega])/\sqrt{2}$ of the quadrature commutators; the real displacement derivatives themselves are $-i[P_j, \omega]$ and $i[Q_j, \omega]$. Applying the inverse weight map plays the role of dividing a classical density derivative by its density. The squared norm J_u measures the size of this score in the estimation geometry with parameter u.

Then

$$J_{\mathrm{KMB}}(\omega) := \sum_j \operatorname{Tr}[[a_j, \omega]^\dagger \mathcal{T}_\omega([a_j, \omega])] = \int_0^1 J_u(\omega)\, du. \tag{2.46}$$

Tonelli's theorem justifies the integral even before finiteness is known. This normalization uses one complex annihilation direction per mode; it is half the sum over the $2m$ real displacement directions.

Lemma 7 (Commutator chain rule)**.** *Let ω be a faithful finite-energy bosonic state and set $C_j = [a_j, \omega]$. For $\varepsilon > 0$,*

$$[a_j, \log(\omega + \varepsilon I)] = \mathcal{T}_{\omega+\varepsilon I}(C_j). \tag{2.47}$$

For $f = \log(\omega + \varepsilon I)$, the left side is initially the sesquilinear form

$$\langle \psi, [a_j, f]\varphi\rangle := \langle a_j^\dagger \psi, f\varphi\rangle - \langle f\psi, a_j\varphi\rangle$$

on the finite span of eigenvectors of ω. The trace-class right side specifies its extension. This weak definition does not assume that $a_j\varphi$ lies in the domain of an unregularized logarithm. Without regularization, the corresponding identity is

$$[a_j, \log\omega] = \mathcal{T}_\omega(C_j) = \int_0^1 \mathcal{S}_{j,u}(\omega)\, du, \tag{2.48}$$

in the following precise pairing sense, using $q_\omega(C) = \mathrm{Tr}[C^\dagger \mathcal{T}_\omega(C)]$ with the spectral definition above. If a trace-class test operator X and C_j have finite q_ω, then pairing each term of (2.48) *against X gives an absolutely convergent scalar identity, and the regularized pairing converges to it as $\varepsilon \downarrow 0$.*

The identity (2.48) is the chain rule for the operator logarithm [19, Sec. 2.5, Example 1], in the bosonic form of [6, Sec. 3.1]. Its pairing form for the unbounded a_j is proved in Appendix B.2.

In our entropy derivatives, the test operator X is the first-moment operator. Lemma 22(c) justifies the associated cyclic trace identity, and Appendix B proves the finite quadratic form bounds needed for the limiting pairing.

2.6 Entropy along the heat semigroup

The following two results about $\rho_t = \mathcal{N}_t(\rho)$ are known. We state them here for later use.

Lemma 8 (Heat smoothing and integrated de Bruijn identity; cf. [7, Thm. V.1], [8, Thms. 1–2 and Prop. 1], [6, Eq. (22)])**.** *For every finite-energy m-mode state ρ, the heat trajectory $\rho_t = \mathcal{N}_t(\rho)$ is faithful and smooth in trace norm for $t > 0$, with trace-norm derivative* (2.21)*. Its entropy satisfies*

$$S(\rho_{t_1}) - S(\rho_{t_0}) = \int_{t_0}^{t_1} J_{\mathrm{KMB}}(\rho_t)\, dt, \qquad 0 < t_0 < t_1 < \infty. \tag{2.49}$$

Identity (2.49) is the integrated form of the quantum de Bruijn identity $\frac{d}{dt}S(\rho_t) = J_{\mathrm{KMB}}(\rho_t)$ of [7, Thm. V.1]; in the annihilator normalization used here it is [6, Sec. 2.2, Eq. (22)], with the Fisher information written as the spectral sum (2.46). The derivations in [7, 6] differentiate $\log\rho_t$ formally; see the discussion in [17, Sec. 3.2]. The rigorous finite-energy treatment of [8, Thms. 1–2 and Prop. 1] defines the Fisher information as the entropy derivative [8, Def. 7]. Since Theorem 2 evaluates the spectral form (2.46) of J_{KMB}, we prove the lemma as stated, by entropy regularization, in Appendix B.3.

Lemma 9 (Entropy at small and large noise; cf. [21, Lemma 15], [8, Thm. 5])**.** *For every finite-energy m-mode state ρ,*

$$\lim_{t\downarrow 0} S(\mathcal{N}_t(\rho)) = S(\rho), \tag{2.50}$$

and

$$S(\mathcal{N}_t(\rho)) = m\log t + m + o(1), \qquad t \longrightarrow \infty. \tag{2.51}$$

For (2.50), strong continuity of the Weyl unitaries and dominated convergence in (2.19) give

$$\|\mathcal{N}_t(\rho) - \rho\|_1 \le \mathbb{E}\|D_{\sqrt{t}G}\rho D_{\sqrt{t}G}^\dagger - \rho\|_1 \longrightarrow 0 \qquad (t \downarrow 0),$$

while $\mathrm{Tr}(\mathcal{N}_t(\rho)H_0) = E_\rho + mt \le E_\rho + m$ for $0 \le t \le 1$; the energy-constrained entropy-continuity theorem [21, Lemma 15] then gives (2.50). Equation (2.51) is [8, Theorem 5, Eq. (98)] with a trivial memory system; the heat parameter there agrees with ours, since in both cases the added quadrature covariance is tI_{2m}, equivalently $\chi_{\mathcal{N}_t(\rho)}(\zeta) = e^{-t\|\zeta\|^2}\chi_\rho(\zeta)$ as in (2.20), so no rescaling of the constant term is required. It also follows from the entropy bounds of [7, Thm. III.3 and Cor. III.4].

3 The thermal signal I–MMSE identity

We first prove the relation between information and estimation for the transmitted signal. We then derive its zero-SNR slope and concavity, compare it with ordinary quadratic error. Throughout this section, the thermal noise level $N > 0$ is fixed, $\nu = N + 1/2$ is the noise variance in each quadrature, and s is the SNR parameter defined in the introduction. For a signal $Z \in \mathbb{C}^m$ with finite second moment, recall that

$$\rho_{s,z} = D_{\sqrt{s}z}\tau_\nu D_{\sqrt{s}z}^\dagger, \qquad \rho_s = \mathbb{E}\rho_{s,Z}, \qquad I(s) = S(\rho_s) - S(\tau_\nu). \tag{3.1}$$

Although each conditional output is Gaussian, the average state generally is not. All entropies are finite because the output energy is $mN + s\mathbb{E}\|Z\|^2$.

In the classical I–MMSE relation [1], one posterior mean determines the derivative. Here the derivative averages the errors of the ordered estimators introduced in Section 2.3. The thermal noise determines the weight of each error; the signal distribution and SNR determine the error itself. The proof has three steps. The thermal commutation relation expresses each estimator through a displacement score. Entropy differentiation averages these scores over the ordering parameter. The normal equation then converts the resulting expression into a minimum quadratic risk.

3.1 Proof of Theorem 1

Proof. We use the completed estimation spaces of Section 2.3. Lemma 23 justifies the trace-norm derivatives and spectral pairings below under the stated second-moment assumption.

Step 1: compute the ordered posterior estimator. Fix $s > 0$ and $0 < u < 1$, and put $n_u = N+1-u$ and $\Gamma_j = \mathbb{E}[Z_j\rho_{s,Z}]$.

First derive the thermal commutation relation from (2.7). Put $\xi = N/(N+1)$ and $p_{\boldsymbol{n}} = (1-\xi)^m\xi^{|\boldsymbol{n}|}$ in the photon-number basis of Section 2.1; $\boldsymbol{e}_j$ is the jth standard basis vector. For $n_j \geq 1$,

$$a_j\tau_\nu|\boldsymbol{n}\rangle = p_{\boldsymbol{n}}\sqrt{n_j}|\boldsymbol{n}-\boldsymbol{e}_j\rangle = \xi p_{\boldsymbol{n}-\boldsymbol{e}_j}\sqrt{n_j}|\boldsymbol{n}-\boldsymbol{e}_j\rangle = \xi\tau_\nu a_j|\boldsymbol{n}\rangle;$$

both sides vanish if $n_j = 0$. Thus $(N+1)a_j\tau_\nu - N\tau_\nu a_j = 0$ on $\mathcal{D}_{\mathrm{fin}}$. The products have trace-class extensions, since $\|a_j\tau_\nu\|_1 \leq \sqrt{N}$ and $\|\tau_\nu a_j\|_1 \leq \sqrt{N+1}$ by Hilbert–Schmidt Cauchy–Schwarz, so the equality also holds between those extensions. For $D = D_{\sqrt{s}z}$, this identity and $D^\dagger a_j D = a_j + \sqrt{s}z_j I$ give

$$\begin{aligned}
&(N+1)a_j\rho_{s,z} - N\rho_{s,z}a_j \\
&\quad = D\big[(N+1)(a_j + \sqrt{s}z_j I)\tau_\nu - N\tau_\nu(a_j + \sqrt{s}z_j I)\big]D^\dagger \\
&\quad = \sqrt{s}z_j\rho_{s,z}.
\end{aligned} \tag{3.2}$$

Taking expectations yields

$$\begin{aligned}
\sqrt{s}\,\Gamma_j &= (N+1)a_j\rho_s - N\rho_s a_j \\
&= \big[(1-u)\rho_s a_j + u a_j\rho_s\big] + (N+1-u)(a_j\rho_s - \rho_s a_j) \\
&= \mathcal{K}_{\rho_s,u}(a_j) + n_u[a_j,\rho_s].
\end{aligned} \tag{3.3}$$

Finite output energy gives $\|a_j\|_{\rho_s,u}^2 = N+1-u+s\mathbb{E}|Z_j|^2 < \infty$. Applying the inverse in the weighted completion, and using (2.29), we obtain

$$\widehat{Z}_{j,u} = \mathcal{K}_{\rho_s,u}^{-1}(\Gamma_j) = \frac{a_j + n_u\mathcal{S}_{j,u}(\rho_s)}{\sqrt{s}}, \tag{3.4}$$

where $\mathcal{S}_{j,u}(\rho_s) = \mathcal{K}^{-1}_{\rho_s,u}([a_j,\rho_s])$.

Step 2: write the entropy increment as a score pairing. Let $L_z = \sum_j (z_j a_j^\dagger - \overline{z}_j a_j)$. Since $D_{\sqrt{s}z} = e^{\sqrt{s}L_z}$,

$$\frac{d}{ds}\rho_{s,z} = \frac{L_z\rho_{s,z} - \rho_{s,z}L_z}{2\sqrt{s}} = \frac{1}{2\sqrt{s}}[L_z, \rho_{s,z}]. \tag{3.5}$$

Taking expectations, it then follows from Lemma 23(a) that

$$\rho_s' = \frac{1}{2\sqrt{s}} \sum_j \left([a_j^\dagger, \Gamma_j] - [a_j, \Gamma_j^\dagger]\right). \tag{3.6}$$

For $\varepsilon > 0$, put $L_\varepsilon = \log(\rho_s + \varepsilon I)$ and $A_{j,\varepsilon} = \mathcal{T}_{\rho_s+\varepsilon I}([a_j,\rho_s])$. The regularized orbit identity (B.11) with $B = \Gamma_j^\dagger$, the relation $[a_j^\dagger,\Gamma_j] = -[a_j,\Gamma_j^\dagger]^\dagger$, and the self-adjointness of L_ε give

$$\begin{aligned} \operatorname{Tr}([a_j,\Gamma_j^\dagger]L_\varepsilon) &= -\operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon}), \\ \operatorname{Tr}([a_j^\dagger,\Gamma_j]L_\varepsilon) &= -\overline{\operatorname{Tr}([a_j,\Gamma_j^\dagger]L_\varepsilon)} = \overline{\operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon})}. \end{aligned} \tag{3.7}$$

Let F_ε be the trace-class entropy regularizer defined in (B.3); by Lemma 21(a), it satisfies $F_\varepsilon(\omega) \uparrow S(\omega) + 1$ as $\varepsilon \downarrow 0$. Consequently,

$$\begin{aligned} \frac{d}{ds}F_\varepsilon(\rho_s) &\overset{\text{(a)}}{=} -\operatorname{Tr}(\rho_s' L_\varepsilon) \overset{\text{(b)}}{=} -\frac{1}{2\sqrt{s}} \sum_j \left(\overline{\operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon})} + \operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon})\right) \\ &= -\frac{1}{\sqrt{s}} \operatorname{Re} \sum_j \operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon}), \end{aligned} \tag{3.8}$$

where (a) is the regularized chain rule (B.4), applied to the locally C^1 path of Lemma 23(a), and (b) substitutes (3.6) and (3.7). By Lemma 23(b), we may integrate this identity and then let $\varepsilon \downarrow 0$. The scalar pairings below are the absolutely convergent eigenbasis sums specified in Lemma 7, with the Cauchy–Schwarz bound (B.18). Lemma 7 and (2.44) give

$$[a_j, \log\rho_s] = \mathcal{T}_{\rho_s}([a_j,\rho_s]) = \int_0^1 \mathcal{S}_{j,u}(\rho_s)\, du. \tag{3.9}$$

Define the entropy-production pairing $h(s)$ by

$$\begin{aligned} h(s) &:= -\frac{1}{\sqrt{s}} \operatorname{Re} \sum_j \operatorname{Tr}(\Gamma_j^\dagger [a_j, \log\rho_s]) \\ &= -\frac{1}{\sqrt{s}} \int_0^1 \operatorname{Re} \sum_j \operatorname{Tr}(\Gamma_j^\dagger \mathcal{S}_{j,u}(\rho_s))\, du. \end{aligned} \tag{3.10}$$

Then (B.23) reads

$$I(s_1) - I(s_0) = \int_{s_0}^{s_1} h(s)\, ds, \qquad 0 \le s_0 < s_1 < \infty.$$

At this point h is defined by the pairing, without assuming that I is differentiable.

Step 3: identify the pairing with the minimum risk. Solving (3.4) for the score, $\mathcal{S}_{j,u}(\rho_s) = (\sqrt{s}\,\widehat{Z}_{j,u} - a_j)/n_u$, and substituting into (3.10), we obtain

$$h(s) = \int_0^1 \frac{1}{n_u}\left[\frac{1}{\sqrt{s}} \operatorname{Re} \sum_j \operatorname{Tr}(\Gamma_j^\dagger a_j) - \sum_j \operatorname{Tr}(\Gamma_j^\dagger \widehat{Z}_{j,u})\right] du; \tag{3.11}$$

here $\mathrm{Tr}(\Gamma_j^\dagger \widehat{Z}_{j,u})$ is real, by (2.32) with $B_j = \widehat{Z}_{j,u}$. The two terms in this bracket satisfy

$$\mathrm{Tr}(\Gamma_j^\dagger a_j) \overset{\text{(a)}}{=} \mathbb{E}[\overline{Z}_j \,\mathrm{Tr}(\rho_{s,Z} a_j)] \overset{\text{(b)}}{=} \mathbb{E}[\overline{Z}_j \,\mathrm{Tr}(\tau_\nu(a_j + \sqrt{s} Z_j I))] \overset{\text{(c)}}{=} \sqrt{s}\,\mathbb{E}|Z_j|^2, \tag{3.12}$$

where (a) is the continuity of the annihilator pairing in Lemma 23(c), (b) is $D^\dagger a_j D = a_j + \sqrt{s} z_j I$ and the cyclicity of the trace, and (c) is $\mathrm{Tr}(\tau_\nu a_j) = 0$; and

$$\mathrm{Tr}(\Gamma_j^\dagger \widehat{Z}_{j,u}) = \langle \widehat{Z}_{j,u}, \widehat{Z}_{j,u} \rangle_{\rho_s,u} = \|\widehat{Z}_{j,u}\|^2_{\rho_s,u}, \tag{3.13}$$

by (2.32). Thus

$$\frac{1}{\sqrt{s}} \,\mathrm{Re} \sum_j \mathrm{Tr}(\Gamma_j^\dagger a_j) - \sum_j \mathrm{Tr}(\Gamma_j^\dagger \widehat{Z}_{j,u}) \overset{\text{(d)}}{=} \mathbb{E}\|Z\|^2 - \sum_j \|\widehat{Z}_{j,u}\|^2_{\rho_s,u} \overset{\text{(e)}}{=} \mathsf{M}_u(Z \mid A_s), \tag{3.14}$$

where (d) is (3.12)–(3.13) and (e) is the minimum-risk formula (2.31). Since $I(0) = 0$, substitution gives

$$I(s) = \int_0^s h(v)\,dv = \int_0^s \int_0^1 \frac{\mathsf{M}_u(Z \mid A_v)}{N + 1 - u}\,du\,dv.$$

The inner integral is continuous in $v \geq 0$ by Lemma 23(c). The fundamental theorem of calculus therefore proves (1.4) at every $s > 0$ and the integrated identity (1.5). Its value at $v = 0$ gives the right derivative stated in Corollary 10. Appendix B.5 gives the real-coordinate version of this calculation. □

The positive thermal mean photon number N is used in this proof. At the vacuum endpoint the weight becomes $(1 - u)^{-1}$, and the domination argument above no longer applies. No general-prior vacuum extension of (1.4) is asserted here.

3.2 Consequences for a fixed input distribution

Corollary 10 (Zero-SNR slope and concavity)**.** *For each fixed prior satisfying the theorem,*

$$I'(0+) = \log \frac{N+1}{N}\,\mathbb{E}\|Z - \mathbb{E}Z\|^2. \tag{3.15}$$

Furthermore, I is nondecreasing and concave as a function of the SNR parameter $s \geq 0$.

Proof. Put $V = \mathbb{E}\|Z - \mathbb{E}Z\|^2$. At $s = 0$ the state is independent of Z, so the normal equation gives $\widehat{Z}_{j,u} = (\mathbb{E}Z_j)I$ and $\mathsf{M}_u(Z \mid A_0) = V$. Then

$$I'(0+) \overset{\text{(a)}}{=} \lim_{s \downarrow 0} \frac{1}{s} \int_0^s \int_0^1 \frac{\mathsf{M}_u(Z \mid A_v)}{N + 1 - u}\,du\,dv \overset{\text{(b)}}{=} V \int_0^1 \frac{du}{N + 1 - u} = V \log \frac{N+1}{N},$$

where (a) is (1.5) and (b) is the continuity at zero SNR of the inner integral, Corollary 19, Eq. (A.5), together with the mean value theorem for the outer integral.

For $s_2 > s_1 > 0$, put $\eta = s_1/s_2 \in (0, 1)$. The thermal attenuator $\mathcal{A}_{\eta,N}$ of (2.17) satisfies $\mathcal{A}_{\eta,N}(\rho_{s_2,z}) = \rho_{\eta s_2,z} = \rho_{s_1,z}$ for every z by (2.18); it is a quantum channel from the experiment A_{s_2} to the experiment A_{s_1} with the same label Z. Lemma 5 therefore gives, for $0 < u < 1$,

$$0 \leq \mathsf{M}_u(Z \mid A_{s_2}) \leq \mathsf{M}_u(Z \mid A_{s_1}),$$

and this inequality also holds for $s_1 = 0$, because $\mathsf{M}_u(Z \mid A_{s_2}) \le V = \mathsf{M}_u(Z \mid A_0)$ by the variance bound in Lemma 4. Hence, for $s_2 > s_1 > 0$,

$$0 \le I'(s_2) \overset{\text{(a)}}{=} \int_0^1 \frac{\mathsf{M}_u(Z \mid A_{s_2})}{N+1-u}\,du \le \int_0^1 \frac{\mathsf{M}_u(Z \mid A_{s_1})}{N+1-u}\,du \overset{\text{(a)}}{=} I'(s_1),$$

where (a) is (1.4) and the inequality uses $N+1-u > 0$. Thus I is nondecreasing and concave on $(0,\infty)$; continuity of I at zero, Lemma 23(c), extends both conclusions to $[0,\infty)$. □

One can normalize the weight by setting $\beta_N = \log((N+1)/N)$ and $w_N(u) = 1/[\beta_N(N+1-u)]$. Thus $I'(s)$ is β_N times a probability-weighted average of ordered errors. Equivalently,

$$I'(s) = \inf_{\{B^{(u)}\}} \int_0^1 \frac{\mathcal{R}_u(B^{(u)})}{N+1-u}\,du, \tag{3.16}$$

where the estimator field is measurable and may depend on u.

To verify (3.16), let $\{B^{(r)}\}_{r\ge1}$ be a countable dense family of finite-rank estimators with rational coefficients.

One common family for all u consists of matrices with finitely many nonzero entries in the photon-number basis, each entry having rational real and imaginary parts. Indeed, for the projections P_L onto $\{|\boldsymbol{n}\rangle : |\boldsymbol{n}| \le L\}$ and any bounded B, $P_L B P_L \to B$ and $P_L B^\dagger P_L \to B^\dagger$ strongly, with uniformly bounded norms. Pairing their squared differences with the trace-class state ρ_s gives $\|P_L B P_L - B\|_{\rho_s,u} \to 0$. Rational approximation of each finite matrix then gives the stated density in the estimator completion.

For $\varepsilon > 0$, choose the first $r = r_\varepsilon(u)$ such that $\mathcal{R}_u(B^{(r)}) \le \mathsf{M}_u(Z \mid A_s) + \varepsilon$. The risks are measurable in u, so this choice is measurable and

$$I'(s) \le \int_0^1 \frac{\mathcal{R}_u(B^{(r_\varepsilon(u))})}{N+1-u}\,du \le I'(s) + \varepsilon \log\frac{N+1}{N}.$$

Letting $\varepsilon \downarrow 0$ proves the equality. In particular, any measurable estimator field supplies an upper bound on $I'(s)$.

Proposition 11 (The Gaussian input bound from linear estimators)**.** *Let $\mathbb{E}Z = 0$ and $\sigma_j^2 = \mathbb{E}|Z_j|^2 < \infty$. For $s > 0$ in the first bound and $s \ge 0$ in the second,*

$$I'(s) \le \sum_{j=1}^m \sigma_j^2 \log\frac{N + s\sigma_j^2 + 1}{N + s\sigma_j^2}, \tag{3.17}$$

$$I(s) \le \sum_{j=1}^m \bigl[g(N + s\sigma_j^2) - g(N)\bigr]. \tag{3.18}$$

Independent circularly symmetric complex Gaussian coordinates with these variances attain both bounds. The information bound is the usual Gaussian maximum-entropy bound (the Gibbs principle recalled in Section 2.1), here obtained through estimation.

Proof. For each u, choose $B_j = c_j a_j$ and put $n_u = N+1-u$. Finite output energy places a_j in the estimator completion. The output moments and (3.12) give

$$\|a_j\|_{\rho_s,u}^2 = n_u + s\sigma_j^2, \qquad \operatorname{Tr}(\Gamma_j^\dagger a_j) = \sqrt{s}\,\sigma_j^2.$$

Expanding (2.27) and completing the square yields

$$\begin{aligned}\mathcal{R}_u(B) &= \sum_j [\sigma_j^2 - 2\sqrt{s}\,\sigma_j^2 \operatorname{Re} c_j + (n_u + s\sigma_j^2)|c_j|^2] \\ &= \sum_j \left[(n_u + s\sigma_j^2) \left| c_j - \frac{\sqrt{s}\,\sigma_j^2}{n_u + s\sigma_j^2} \right|^2 + \frac{\sigma_j^2 n_u}{n_u + s\sigma_j^2} \right].\end{aligned}$$

Choose $c_j = \sqrt{s}\,\sigma_j^2/(n_u + s\sigma_j^2)$, the same for every u. Then

$$I'(s) \overset{\text{(a)}}{\le} \int_0^1 \frac{\mathcal{R}_u(B)}{N+1-u}\,du \overset{\text{(b)}}{=} \int_0^1 \sum_j \frac{\sigma_j^2}{N+1-u+s\sigma_j^2}\,du = \sum_j \sigma_j^2 \log \frac{N+s\sigma_j^2+1}{N+s\sigma_j^2},$$

where (a) is (3.16) and (b) is the completed square with the chosen c_j. Since $g'(x) = \log((x+1)/x)$ and $I(0) = 0$, integration of this bound gives

$$I(s) = \int_0^s I'(v)\,dv \le \sum_j \int_0^s \sigma_j^2 g'(N + v\sigma_j^2)\,dv = \sum_j [g(N+s\sigma_j^2) - g(N)].$$

For independent circularly symmetric Gaussian coordinates $Z_j \sim \mathcal{CN}(0, \sigma_j^2)$, the rule (2.5) and $\chi_{\tau_\nu}(w) = e^{-\nu\|w\|^2}$ give

$$\chi_{\rho_s}(w) = \mathbb{E}\chi_{\rho_{s,Z}}(w) = e^{-\nu\|w\|^2}\mathbb{E}e^{2i\sqrt{s}\operatorname{Im}\sum_j \overline{Z}_j w_j} = \prod_j e^{-(\nu+s\sigma_j^2)|w_j|^2},$$

because $\operatorname{Im}(\overline{Z}_j w_j)$ is a real Gaussian variable with variance $\sigma_j^2|w_j|^2/2$. By (2.4) and (2.7), this is the characteristic function of the product of thermal states with mean photon numbers $N + s\sigma_j^2$, so $I(s) = \sum_j [g(N+s\sigma_j^2) - g(N)]$ by (2.11); its derivative is the right side of (3.17). □

This is the same use of linear estimation that underlies classical I–MMSE information bounds [1].

A deterministic displacement does not change the Holevo information, so the bound extends to general priors with $\sigma_j^2 = \mathbb{E}|Z_j - \mathbb{E}Z_j|^2$. By concavity of g, under the energy constraint $\mathbb{E}\|Z\|^2 \le \mathcal{E}$ the maximum Holevo information is $C_{\mathcal{E}}(s) = m[g(N + s\mathcal{E}/m) - g(N)]$; this is the energy-constrained classical capacity of the additive classical-noise channel with mean photon number N per mode, attained by coherent-state encodings [23]. Corollary 10 shows that every fixed zero-mean prior using the full energy $\mathbb{E}\|Z\|^2 = \mathcal{E}$ satisfies $C_{\mathcal{E}}(s) - I(s) = o(s)$ as $s \downarrow 0$; such inputs are therefore first-order optimal at low SNR.

3.3 Comparison with ordinary operator MMSE

Theorem 1 expresses the entropy derivative through positive ordered risks. A natural question is how this derivative compares with the minimum of the ordinary squared error for the real signal coordinates. That minimum is governed by the symmetric logarithmic derivative (SLD), whereas entropy differentiation is governed by the KMB derivative. We now isolate the difference as an explicit correction term. This gives a second form of the identity and explains why replacing the ordered average by ordinary operator MMSE would lose information.

Ordinary quadratic estimation uses the Jordan map

$$\mathcal{J}_\omega(B) = \tfrac{1}{2}(\omega B + B\omega) = \mathcal{K}_{\omega,1/2}(B). \tag{3.19}$$

For a faithful state ω and a self-adjoint trace-class state perturbation Θ with $\operatorname{Tr}\Theta = 0$ (a state tangent), define

$$\begin{aligned} F_{\mathrm{KMB},\omega}(\Theta) &= \operatorname{Tr}[\Theta \mathcal{T}_\omega(\Theta)], \\ F_{\mathrm{SLD},\omega}(\Theta) &= \operatorname{Tr}[\Theta \mathcal{J}_\omega^{-1}(\Theta)]. \end{aligned} \tag{3.20}$$

These expressions denote nonnegative spectral sums, possibly infinite. For example the first is $\sum_{k,\ell} |\Theta_{k\ell}|^2 (\log \mu_k - \log \mu_\ell)/(\mu_k - \mu_\ell)$. Jensen's inequality applied to the reciprocal function gives

$$\int_0^1 \frac{du}{(1-u)x + uy} \geq \frac{2}{x+y}, \qquad F_{\mathrm{KMB},\omega}(\Theta) \geq F_{\mathrm{SLD},\omega}(\Theta). \tag{3.21}$$

The second inequality follows by multiplying the first by each $|\Theta_{k\ell}|^2$ and summing. It compares the two quadratic forms on the *same tangent*. The metric structure is standard background in quantum information geometry [19]; its KMB form enters the quantum de Bruijn identity [7, 6].

Define the real displacement derivations and their scores by

$$\begin{aligned} \delta_k \omega &= -i[(\Omega R)_k, \omega], \\ L_k^{\mathrm{KMB}}(\omega) &= \mathcal{T}_\omega(\delta_k \omega), \qquad\qquad \mathcal{J}_\omega(L_k^{\mathrm{SLD}}(\omega)) = \delta_k \omega. \end{aligned} \tag{3.22}$$

Thus $\delta_{2j-1} = -i[P_j, \cdot]$ and $\delta_{2j} = i[Q_j, \cdot]$.

The expression $L_k^{\mathrm{KMB}} = -i[(\Omega R)_k, \log \omega]$ has the following meaning: for every trace-class test operator C with $q_\omega(C), q_\omega(\delta_k \omega) < \infty$, its pairing against C is the eigenbasis sum $\operatorname{Tr}[C^\dagger \mathcal{T}_\omega(\delta_k \omega)]$ of Lemma 7, or equivalently the limit of the regularized logarithmic pairings. Here $q_\omega(C) = \operatorname{Tr}[C^\dagger \mathcal{T}_\omega(C)]$. The SLD inverse denotes the unique solution of the Jordan normal equation in the completion with norm $\| \cdot \|_{\omega,1/2}$, as in Lemma 4. No bounded inverse on all trace-class operators is asserted.

For the thermal experiment, write $Z = z(X)$, $\rho_{s,x} = \rho_{s,z(x)}$, and $\Gamma_k^X(s) = \mathbb{E}[X_k \rho_{s,X}]$. The real-coordinate form of the entropy calculation gives

$$I'(s) = \frac{1}{2\sqrt{s}} \sum_{k=1}^{2m} \operatorname{Tr}[\Gamma_k^X L_k^{\mathrm{KMB}}(\rho_s)]. \tag{3.23}$$

Appendix B.5 gives the change of coordinates and the corresponding conditional identity for more general initial states.

For a real target, denote the coordinatewise Jordan operator error by

$$\operatorname{mmse}_J(X \mid A) = \inf_{C_k = C_k^\dagger} \mathbb{E} \sum_{k=1}^{2m} \operatorname{Tr} \omega_X (X_k I - C_k)^2. \tag{3.24}$$

The infimum is taken in the real weighted Hilbert-space completion. The Jordan normal equation is the standard Bayesian least-squares equation [14, Sec. 3, Eq. (5)]; see also [12, Sec. III.B]. Its optimal coordinates satisfy $\mathcal{J}_\omega(\widehat{X}_k) = \mathbb{E}[X_k \omega_X]$, and Lemma 4 gives

$$\operatorname{mmse}_J(X \mid A) = \mathbb{E}\|X\|^2 - \sum_k \operatorname{Tr}(\Gamma_k^X \widehat{X}_k). \tag{3.25}$$

This is coordinatewise operator estimation: it permits a separate observable for each coordinate. Appendix C records, for a Gaussian prior, the error attainable by a single measurement (positive operator-valued measure, POVM), which must produce all coordinates in one classical outcome.

Proposition 12 (The KMB–SLD correction for thermal noise)**.** *Let $N > 0$, set $\nu = N + 1/2$, let $\mathbb{E}\|X\|^2 < \infty$, and encode X by $D_{\sqrt{s}z(X)}\tau_\nu D^\dagger_{\sqrt{s}z(X)}$. For every $s > 0$, the optimal Jordan coordinates are*

$$\widehat{X}_k = \frac{R_k - \nu L_k^{\mathrm{SLD}}(\rho_s)}{\sqrt{s}}, \tag{3.26}$$

and

$$I'(s) = \frac{1}{2\nu}\,\mathrm{mmse}_J(X \mid A_s) + \mathcal{Q}(s), \tag{3.27}$$

$$\mathcal{Q}(s) = \frac{1}{2\sqrt{s}}\sum_k \Big\{\mathrm{Tr}[\Gamma_k^X L_k^{\mathrm{KMB}}(\rho_s)] - \mathrm{Tr}[\Gamma_k^X L_k^{\mathrm{SLD}}(\rho_s)]\Big\}. \tag{3.28}$$

Both scalar pairings in each summand of (3.28) *are finite, with each score interpreted in its corresponding weighted space.*

Proof. The thermal relation derived at the start of the proof of Theorem 1 and its adjoint give $[a_j, \tau_\nu] = -\tau_\nu a_j/(N+1)$ and $[a_j^\dagger, \tau_\nu] = a_j^\dagger \tau_\nu/(N+1)$. Hence

$$-i[P_j, \tau_\nu] = \frac{a_j^\dagger\tau_\nu + \tau_\nu a_j}{\sqrt{2}(N+1)} = \frac{\{Q_j, \tau_\nu\}}{2\nu},$$
$$i[Q_j, \tau_\nu] = \frac{i(a_j^\dagger\tau_\nu - \tau_\nu a_j)}{\sqrt{2}(N+1)} = \frac{\{P_j, \tau_\nu\}}{2\nu},$$

where $2\nu = 2N + 1$. Conjugating by $D = D_{\sqrt{s}z(x)}$ and using $DR_kD^\dagger = R_k - \sqrt{s}x_kI$ yields

$$\delta_k\rho_{s,x} = D(\delta_k\tau_\nu)D^\dagger = \frac{1}{2\nu}\{R_k - \sqrt{s}x_kI, \rho_{s,x}\}. \tag{3.29}$$

Taking expectations gives

$$\nu\delta_k\rho_s = \frac{1}{2}\{R_k, \rho_s\} - \sqrt{s}\,\Gamma_k^X = \mathcal{J}_{\rho_s}(R_k) - \sqrt{s}\,\Gamma_k^X. \tag{3.30}$$

Finite energy places R_k in the Jordan completion. Applying $\mathcal{J}_{\rho_s}^{-1}$ there gives

$$\nu L_k^{\mathrm{SLD}}(\rho_s) = R_k - \sqrt{s}\,\widehat{X}_k,$$

which proves (3.26). Moreover,

$$\mathrm{Tr}(\Gamma_k^X R_k) = \mathbb{E}\big[X_k\,\mathrm{Tr}(\tau_\nu(R_k + \sqrt{s}X_kI))\big] = \sqrt{s}\,\mathbb{E}X_k^2. \tag{3.31}$$

Using (3.26), (3.31), and (3.25), we obtain

$$\begin{aligned}\sum_k \mathrm{Tr}(\Gamma_k^X L_k^{\mathrm{SLD}}(\rho_s)) &= \frac{1}{\nu}\sum_k\Big[\mathrm{Tr}(\Gamma_k^X R_k) - \sqrt{s}\;\mathrm{Tr}(\Gamma_k^X\widehat{X}_k)\Big]\\ &= \frac{\sqrt{s}}{\nu}\left[\mathbb{E}\|X\|^2 - \sum_k \mathrm{Tr}(\Gamma_k^X\widehat{X}_k)\right]\\ &= \frac{\sqrt{s}}{\nu}\,\mathrm{mmse}_J(X \mid A_s).\end{aligned} \tag{3.32}$$

Therefore (3.23) implies

$$\begin{aligned} I'(s) &= \frac{1}{2\sqrt{s}} \sum_k \operatorname{Tr}(\Gamma_k^X L_k^{\mathrm{SLD}}(\rho_s)) \\ &\quad + \frac{1}{2\sqrt{s}} \sum_k \{\operatorname{Tr}[\Gamma_k^X L_k^{\mathrm{KMB}}(\rho_s)] - \operatorname{Tr}[\Gamma_k^X L_k^{\mathrm{SLD}}(\rho_s)]\} \\ &= \frac{1}{2\nu} \operatorname{mmse}_J(X \mid A_s) + \mathcal{Q}(s). \end{aligned}$$

All pairings are defined in the weighted spaces above. In particular, the KMB pairing bound (B.18), the real–complex conversion (B.28), and (B.25) give

$$\sum_k |\operatorname{Tr}(\Gamma_k^X L_k^{\mathrm{KMB}}(\rho_s))| \le 2\sqrt{\mathbb{E}\|Z\|^2 \, J_{\mathrm{KMB}}(\rho_s)} < \infty.$$

For the SLD pairings, (3.31), $\operatorname{Tr}(\Gamma_k^X \widehat{X}_k) = \|\widehat{X}_k\|^2_{\rho_s,1/2} \le \mathbb{E}X_k^2$, and (3.26) give finiteness. Lemma 23 supplies the entropy-differentiation justification under the stated moment assumption. □

The correction in (3.28) is a mixed pairing between a first moment and a difference of scores. Inequality (3.21) compares two quadratic forms on a common tangent and does not by itself determine the sign of this mixed pairing.

4 An estimation identity for Gaussian displacement noise

We now exchange the roles of the two assumptions in the thermal model. The initial quantum state may be arbitrary, while the displacement is Gaussian. The quantity to be estimated is the added Gaussian noise. This model will let us compare the entropies of arbitrary input states at a beam splitter in Section 5. Let ρ be an m-mode state with $\operatorname{Tr}(\rho H_0) < \infty$, and let $G \sim \mathcal{CN}(0, I_m)$ have density $\pi^{-m} \exp(-\|g\|^2)$. Define

$$\rho_{t,g} = D_{\sqrt{t}g} \rho D_{\sqrt{t}g}^\dagger, \qquad \rho_t = \mathcal{N}_t(\rho) = \mathbb{E}\rho_{t,G}. \tag{4.1}$$

The channel $\mathcal{N}_t$ is the heat channel of Subsection 2.2; see (2.19)–(2.21). With A_t now denoting the noise experiment $g \mapsto \rho_{t,g}$ with prior G, that is, the quantum system of the classical–quantum state

$$\rho_{GA_t} = \int_{\mathbb{C}^m} \pi^{-m} e^{-\|g\|^2} \, |g\rangle\langle g| \otimes \rho_{t,g} \, d^{2m}g,$$

whose marginal on A_t is the average state $\rho_t = \mathbb{E}\rho_{t,G}$. Then we have

$$I_\rho(t) = I(G; A_t) = S(\rho_t) - S(\rho), \qquad \mathsf{M}_{\rho,u}(t) = \mathsf{M}_u(G \mid A_t). \tag{4.2}$$

Here t measures the added noise variance, rather than the signal SNR of Section 3. The conditional states all have entropy $S(\rho)$, so their Holevo information is precisely the entropy increase caused by the heat channel. The identification of this Gaussian-label information with a heat-flow entropy increment is already contained in [8, Definition 6 and Theorem 1], with trivial quantum memory. The calculation below expresses its derivative in ordered estimation risks. Gaussian integration by parts takes the place of the thermal commutation relation in the signal I–MMSE proof.

4.1 The Gaussian posterior mean and displacement score

Proposition 13 (Gaussian posterior-mean/score identity)**.** *At every $t > 0$ and $0 < u < 1$, the optimal ordered estimator $\widehat{G}_{j,u}$ of G_j in the noise experiment and the minimum risk satisfy*

$$\widehat{G}_{j,u} = -\sqrt{t}\,\mathcal{S}_{j,u}(\rho_t), \qquad \mathsf{M}_{\rho,u}(t) = m - tJ_u(\rho_t). \tag{4.3}$$

In particular, $0 \le J_u(\rho_t) \le m/t$.

The total prior quadratic moment of G is m. Thus (4.3) says that the reduction in its estimation error is t times the ordered Fisher information of the average output. The same relation holds for every ordering parameter, which allows us to average it when differentiating entropy. For a scalar Gaussian prior, the analogous SLD relation follows by Gaussian integration by parts in [14, Sec. 3, Eq. (6)]. The following calculation extends that mechanism to the complex vector target and the full ordered family.

Proof. Put $p(g) = \pi^{-m} e^{-\|g\|^2}$ and $\partial_{\overline{h}_j} = \frac{1}{2}(\partial_{\mathrm{Re}\, h_j} + i\partial_{\mathrm{Im}\, h_j})$. Weyl multiplication and the change of variables $g \mapsto g + h/\sqrt{t}$ give

$$\begin{aligned} D_h \rho_t D_h^\dagger &= \int_{\mathbb{C}^m} p(g) D_{h+\sqrt{t}g} \rho D_{h+\sqrt{t}g}^\dagger \, d^{2m}g \\ &= \int_{\mathbb{C}^m} p(g - h/\sqrt{t}) \rho_{t,g} \, d^{2m}g. \end{aligned} \tag{4.4}$$

Since p and its first derivatives are integrable, we may differentiate the last integral in trace norm. Moreover,

$$\partial_{\overline{h}_j} p(g - h/\sqrt{t})\Big|_{h=0} = \frac{g_j}{\sqrt{t}} p(g), \qquad \partial_{\overline{h}_j} (D_h \rho_t D_h^\dagger)\Big|_{h=0} = -[a_j, \rho_t].$$

The commutator is trace class by finite energy and Hilbert–Schmidt Cauchy–Schwarz. Differentiation of (4.4) therefore yields

$$\Gamma_j(t) := \mathbb{E}[G_j \rho_{t,G}] = \int g_j p(g) \rho_{t,g} \, d^{2m}g = -\sqrt{t}\,[a_j, \rho_t]. \tag{4.5}$$

Then

$$\begin{aligned} \widehat{G}_{j,u} &\overset{\text{(a)}}{=} \mathcal{K}_{\rho_t,u}^{-1}(\Gamma_j(t)) \overset{\text{(b)}}{=} -\sqrt{t}\,\mathcal{K}_{\rho_t,u}^{-1}([a_j, \rho_t]) \overset{\text{(c)}}{=} -\sqrt{t}\,\mathcal{S}_{j,u}(\rho_t), \\ \mathsf{M}_{\rho,u}(t) &\overset{\text{(d)}}{=} \mathbb{E}\|G\|^2 - \sum_j \|\widehat{G}_{j,u}\|_{\rho_t,u}^2 = m - t\sum_j \|\mathcal{S}_{j,u}(\rho_t)\|_{\rho_t,u}^2 \overset{\text{(e)}}{=} m - tJ_u(\rho_t), \end{aligned}$$

where (a) is the normal equation (2.29) in the weighted completion, (b) is (4.5), (c) and (e) are the definitions (2.45), and (d) is the minimum-risk formula (2.31) with $\mathbb{E}\|G\|^2 = m$. Finally, $0 \le \mathsf{M}_{\rho,u}(t) \le m$ (Lemma 4) implies $0 \le J_u(\rho_t) \le m/t$. □

The classical counterpart is $J(Y_t) = [1 - \mathrm{mmse}(G \mid Y_t)]/t$ for $Y_t = X + \sqrt{t}\,G$ with a real standard Gaussian G independent of X; the real de Bruijn derivative is $J(Y_t)/2$; see [1, Sec. II] and [9, Sec. II-B, Remark 2]. Bayesian estimation of Gaussian random displacements is also studied in [26]. Here (4.3) holds for the full ordered family, allowing its KMB average to enter the entropy derivative.

4.2 Proof of Theorem 2

Proof. For $0 < t_0 < t_1 < \infty$,

$$S(\rho_{t_1}) - S(\rho_{t_0}) \overset{\text{(a)}}{=} \int_{t_0}^{t_1} J_{\mathrm{KMB}}(\rho_t)\, dt \overset{\text{(b)}}{=} \int_{t_0}^{t_1} \int_0^1 J_u(\rho_t)\, du\, dt \overset{\text{(c)}}{=} \int_{t_0}^{t_1} \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t)\, du}{t}\, dt,$$

where (a) is the integrated de Bruijn identity (2.49) of Lemma 8, (b) is (2.46), and (c) is (4.3). In particular,

$$J_{\mathrm{KMB}}(\rho_t) = \int_0^1 J_u(\rho_t)\, du = \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t)\, du}{t} \le \frac{m}{t}. \tag{4.6}$$

To obtain the derivative at every $t > 0$, fix $0 < u < 1$ and let $t_n \to t$. Strong continuity of the Weyl unitaries gives

$$\Delta_n(g) := \|D_{\sqrt{t_n}g}\rho D^\dagger_{\sqrt{t_n}g} - D_{\sqrt{t}g}\rho D^\dagger_{\sqrt{t}g}\|_1 \longrightarrow 0, \qquad \Delta_n(g) \le 2.$$

For $G^{(R)} = G\mathbf{1}_{\{\|G\| \le R\}}$ and $\Gamma_j^{(R)}(v) = \mathbb{E}[G_j^{(R)}\rho_{v,G}]$, dominated convergence therefore gives

$$\|\rho_{t_n} - \rho_t\|_1 \le \mathbb{E}\Delta_n(G) \to 0, \qquad \|\Gamma_j^{(R)}(t_n) - \Gamma_j^{(R)}(t)\|_1 \le \mathbb{E}\big[|G_j^{(R)}|\Delta_n(G)\big] \to 0,$$

while the scalar second moments of $G^{(R)}$ are independent of n. Lemma 18, including its L^2 target extension, now implies $\mathsf{M}_{\rho,u}(t_n) \to \mathsf{M}_{\rho,u}(t)$. Since $0 \le \mathsf{M}_{\rho,u}(t_n) \le m$, dominated convergence also gives

$$\lim_{n\to\infty} \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t_n)\, du}{t_n} = \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t)\, du}{t}.$$

The fundamental theorem of calculus now yields, at every $t > 0$,

$$\frac{d}{dt}S(\rho_t) = J_{\mathrm{KMB}}(\rho_t) = \frac{m - \int_0^1 \mathsf{M}_{\rho,u}(t)\, du}{t}. \tag{4.7}$$

Together with $I_\rho(t) = S(\rho_t) - S(\rho)$, the integral and derivative formulas prove (1.11) and (1.10). □

5 Applications to quantum entropy inequalities

Let ρ_A, ρ_B be independent m-mode states of finite energy, let $0 < \eta < 1$, and let $\rho_C = \mathcal{B}_\eta(\rho_A \otimes \rho_B)$ be the retained output of the beam splitter of Subsection 2.2; ρ_C has finite energy by (2.15). With the binary relative entropy

$$d(\lambda\|\eta) = \lambda \log \frac{\lambda}{\eta} + (1-\lambda)\log\frac{1-\lambda}{1-\eta}, \qquad 0\log 0 = 0, \tag{5.1}$$

the quantum entropy power inequality (qEPI) has the linear form

$$S(\rho_C) \ge \lambda S(\rho_A) + (1-\lambda)S(\rho_B) - m\, d(\lambda\|\eta), \qquad 0 \le \lambda \le 1, \tag{5.2}$$

and the exponential form

$$e^{S(\rho_C)/m} \ge \eta e^{S(\rho_A)/m} + (1-\eta)e^{S(\rho_B)/m}. \tag{5.3}$$

Both are known. The family (5.2) is the unconditional beam-splitter case of [8, Theorem 6, Eq. (106)], and its member $\lambda = \eta$ is the entropy-concavity inequality of [7]; the exponential form (5.3) was

proved in [7] for $\eta = 1/2$, in [16] for every η, and in [8, Theorem 6, Eq. (107)] with a memory system. The family (5.2) and the single inequality (5.3) are equivalent. Indeed, write $\bar{S}_L = S(\rho_L)/m$ and $Z_\eta = \eta e^{\bar{S}_A} + (1-\eta)e^{\bar{S}_B}$. By (5.1),

$$\lambda \bar{S}_A + (1-\lambda)\bar{S}_B - d(\lambda\|\eta) = \log Z_\eta - d(\lambda\|\lambda_*), \qquad \lambda_* = \frac{\eta e^{\bar{S}_A}}{Z_\eta}, \tag{5.4}$$

and $d(\lambda\|\lambda_*) \geq 0$ with equality exactly at $\lambda = \lambda_*$. Hence the maximum over $0 \leq \lambda \leq 1$ of the right side of (5.2) is $m \log Z_\eta$, attained at λ_*: the member $\lambda = \lambda_*$ of (5.2) is (5.3) after exponentiation, and conversely (5.3) implies every member of (5.2). This is the optimization in [8, Eqs. (107) and (115)]; it requires no further proof. At $\eta \in \{0, 1\}$, $\rho_C \in \{\rho_B, \rho_A\}$ and (5.3) holds with equality.

This section proves the linear family (5.2) from the Gaussian-noise I–MMSE identity (Theorem 2) and the estimation comparisons of Lemmas 5 and 6, along the unequal-time interpolation of [8, proof of Theorem 6] (Theorem 14). The proof keeps an exact remainder. Corollary 15 writes the deficit of (5.2) as an integral of squared differences between optimal ordered estimators; the deficit of (5.3) is its member $\lambda = \lambda_*$. Corollary 16 is the corresponding Fisher-information inequality for each ordering parameter. Subsection 5.3 gives a direct SNR interpolation for common thermal mixtures.

Throughout, for $L \in \{A, B, C\}$ and $r \geq 0$, L_r denotes the Gaussian-noise experiment $g \mapsto \rho_{L,r,g} := D_{\sqrt{r}g}\rho_L D_{\sqrt{r}g}^\dagger$ of Section 4, with label $G_L \sim \mathcal{CN}(0, I_m)$, average state $\rho_L(r) = \mathcal{N}_r(\rho_L)$, minimum ordered risk $\mathsf{M}_{L,u}(r) = \mathsf{M}_u(G_L \mid L_r)$, and optimal ordered estimators $\widehat{G}_{L,j,u}(r)$ of the coordinates $G_{L,j}$. Since the three states have finite energy, Theorem 2 applies to each of them.

5.1 The full linear qEPI

Theorem 14 (Full linear qEPI [8, Theorem 6, Eq. (106)])**.** *Let ρ_A, ρ_B be m-mode states of finite energy and let $\rho_C = \mathcal{B}_\eta(\rho_A \otimes \rho_B)$ with $0 < \eta < 1$. Then (5.2) holds for every $0 \leq \lambda \leq 1$. In particular, $\lambda = \eta$ gives the entropy-concavity form*

$$S(\rho_C) \geq \eta S(\rho_A) + (1-\eta)S(\rho_B). \tag{5.5}$$

Proof. Fix $0 < \lambda < 1$. Define the noise-variance multipliers

$$\kappa_A = \frac{\lambda}{\eta}, \qquad \kappa_B = \frac{1-\lambda}{1-\eta}, \qquad \eta\kappa_A + (1-\eta)\kappa_B = 1, \tag{5.6}$$

and, for independent standard complex Gaussian vectors G_A, G_B, set

$$G_C = \sqrt{\lambda}\, G_A + \sqrt{1-\lambda}\, G_B. \tag{5.7}$$

Then G_C is standard complex Gaussian.

Step 1: compare the risks for the common target G_C. By (5.6) and the displacement covariance (2.14),

$$\begin{aligned} \sqrt{\eta\kappa_A t}\, G_A + \sqrt{(1-\eta)\kappa_B t}\, G_B &= \sqrt{t}(\sqrt{\lambda}G_A + \sqrt{1-\lambda}G_B) = \sqrt{t}\, G_C, \\ \mathcal{B}_\eta(\rho_{A,\kappa_A t, G_A} \otimes \rho_{B,\kappa_B t, G_B}) &= \rho_{C,t,G_C}. \end{aligned} \tag{5.8}$$

Averaging the last equality over (G_A, G_B) gives

$$\mathcal{B}_\eta(\rho_A(\kappa_A t) \otimes \rho_B(\kappa_B t)) = \rho_C(t), \tag{5.9}$$

which is (2.22) with $s = \kappa_A t$ and $r = \kappa_B t$, since $\eta s + (1-\eta) r = t$ by (5.6). Consequently,

$$\begin{aligned} \mathsf{M}_{C,u}(t) &\overset{\text{(a)}}{\geq} \mathsf{M}_u(G_C \mid A_{\kappa_A t} B_{\kappa_B t}) \\ &\overset{\text{(b)}}{=} \lambda \mathsf{M}_{A,u}(\kappa_A t) + (1-\lambda)\mathsf{M}_{B,u}(\kappa_B t), \end{aligned} \tag{5.10}$$

where (a) is Lemma 5 for the channel $\mathcal{B}_\eta$, which maps the product experiment $(g_A, g_B) \mapsto \rho_{A,\kappa_A t, g_A} \otimes \rho_{B,\kappa_B t, g_B}$ to C_t by (5.8), with the common target G_C; and (b) is Lemma 6 with coefficients $\sqrt{\lambda}$ and $\sqrt{1-\lambda}$. Define

$$\begin{aligned} \mathcal{G}_{\lambda,u}(t) &:= \mathsf{M}_{C,u}(t) - \lambda \mathsf{M}_{A,u}(\kappa_A t) - (1-\lambda)\mathsf{M}_{B,u}(\kappa_B t) \\ &= \mathsf{M}_u(G_C \mid C_t) - \mathsf{M}_u(G_C \mid A_{\kappa_A t} B_{\kappa_B t}) \geq 0. \end{aligned} \tag{5.11}$$

Step 2: differentiate the entropy difference. Define

$$\Phi_\lambda(t) = S(\rho_C(t)) - \lambda S(\rho_A(\kappa_A t)) - (1-\lambda) S(\rho_B(\kappa_B t)). \tag{5.12}$$

The chain rule and (1.10) give

$$\begin{aligned} \frac{d}{dt} S(\rho_A(\kappa_A t)) &= \frac{\kappa_A}{\kappa_A t}\left(m - \int_0^1 \mathsf{M}_{A,u}(\kappa_A t)\, du\right) = \frac{m}{t} - \frac{1}{t}\int_0^1 \mathsf{M}_{A,u}(\kappa_A t)\, du, \\ \frac{d}{dt} S(\rho_B(\kappa_B t)) &= \frac{\kappa_B}{\kappa_B t}\left(m - \int_0^1 \mathsf{M}_{B,u}(\kappa_B t)\, du\right) = \frac{m}{t} - \frac{1}{t}\int_0^1 \mathsf{M}_{B,u}(\kappa_B t)\, du, \\ \frac{d}{dt} S(\rho_C(t)) &= \frac{m}{t} - \frac{1}{t}\int_0^1 \mathsf{M}_{C,u}(t)\, du. \end{aligned}$$

Substitution in (5.12) and (5.11) yield

$$\begin{aligned} \Phi'_\lambda(t) &= \underbrace{\frac{m}{t} - \lambda\frac{m}{t} - (1-\lambda)\frac{m}{t}}_{=0} - \frac{1}{t}\int_0^1 \left[\mathsf{M}_{C,u}(t) - \lambda \mathsf{M}_{A,u}(\kappa_A t) - (1-\lambda)\mathsf{M}_{B,u}(\kappa_B t)\right] du \\ &= -\frac{1}{t}\int_0^1 \mathcal{G}_{\lambda,u}(t)\, du \leq 0, \qquad t > 0. \end{aligned} \tag{5.13}$$

Since $t \mapsto S(\rho_L(t))$ is continuously differentiable on $(0,\infty)$ by Theorem 2, integration gives

$$\Phi_\lambda(t_0) - \Phi_\lambda(t_1) = \int_{t_0}^{t_1} \frac{dt}{t}\int_0^1 \mathcal{G}_{\lambda,u}(t)\, du, \qquad 0 < t_0 < t_1 < \infty. \tag{5.14}$$

Step 3: the two endpoints. Lemma 9 applies to $\rho_A(\kappa_A t)$, $\rho_B(\kappa_B t)$, and $\rho_C(t)$. Substituting (2.51) into (5.12),

$$\Phi_\lambda(t) = \underbrace{(1 - \lambda - (1-\lambda))(m \log t + m)}_{=0} - m\lambda \log \kappa_A - m(1-\lambda)\log \kappa_B + o(1), \qquad t \to \infty,$$

so (5.6) and (5.1) give

$$\lim_{t\to\infty} \Phi_\lambda(t) = -m\big[\lambda \log \kappa_A + (1-\lambda)\log \kappa_B\big] = -m\, d(\lambda \| \eta). \tag{5.15}$$

At the other endpoint, (2.50) gives

$$\lim_{t \downarrow 0} \Phi_\lambda(t) = S(\rho_C) - \lambda S(\rho_A) - (1-\lambda) S(\rho_B). \tag{5.16}$$

Combining (5.16), (5.13), and (5.15),

$$S(\rho_C) - \lambda S(\rho_A) - (1-\lambda)S(\rho_B) = \lim_{t\downarrow 0} \Phi_\lambda(t) \overset{\text{(c)}}{\geq} \lim_{t\to\infty} \Phi_\lambda(t) = -m\, d(\lambda\|\eta),$$

where (c) holds because Φ_λ is nonincreasing by (5.13). This proves (5.2) for $0 < \lambda < 1$; continuity of both sides in λ proves the cases $\lambda = 0, 1$. □

When $\lambda = \eta$, both multipliers in (5.6) equal one and the proof reduces to the equal-time interpolation of [7] for (5.5). Unequal times supply the free entropy weight; this is the interpolation of [8, proof of Theorem 6], expressed here through Gaussian-label estimation.

5.2 Two corollaries: the deficit and the ordered Stam inequality

The proof gives an exact remainder as well as an entropy inequality. For $0 < \eta, \lambda < 1$, define the corrected deficit

$$\Delta_{\lambda,\eta} = S(\rho_C) - \lambda S(\rho_A) - (1-\lambda)S(\rho_B) + m\, d(\lambda\|\eta) \geq 0, \tag{5.17}$$

and the deficit of the exponential form,

$$\Delta_{\exp} = S(\rho_C) - m \log(\eta e^{S(\rho_A)/m} + (1-\eta)e^{S(\rho_B)/m}) \geq 0. \tag{5.18}$$

By (5.4) at $\lambda = \lambda_*$,

$$\Delta_{\exp} = \Delta_{\lambda_*,\eta}, \qquad \lambda_* = \frac{\eta e^{S(\rho_A)/m}}{\eta e^{S(\rho_A)/m} + (1-\eta)e^{S(\rho_B)/m}}. \tag{5.19}$$

In the next statement, κ_A, κ_B are the multipliers (5.6) and $\mathcal{G}_{\lambda,u}$ is the risk gap (5.11); further, with U_η the beam-splitter unitary and E its discarded output,

$$\omega_t = \rho_A(\kappa_A t) \otimes \rho_B(\kappa_B t), \qquad \iota_\eta(B) = U_\eta^\dagger (B \otimes I_E) U_\eta. \tag{5.20}$$

For bounded B, ι_η is the channel adjoint: $\mathrm{Tr}[\omega_t \iota_\eta(B)] = \mathrm{Tr}[\mathcal{B}_\eta(\omega_t)B] = \mathrm{Tr}[\rho_C(t)B]$ by (5.9). Its isometric extension from $\mathsf{L}^2(\rho_C(t), u)$ to $\mathsf{L}^2(\omega_t, u)$ is supplied by Lemma 20; it is this extension that acts on the possibly unbounded estimators below. Tensor embeddings are interpreted by the same lemma.

Corollary 15 (Integrated risk and projection identities)**.** *Under the assumptions of Theorem 14, for every $0 < \lambda < 1$,*

$$\Delta_{\lambda,\eta} = \int_0^\infty \frac{dt}{t} \int_0^1 \mathcal{G}_{\lambda,u}(t)\, du. \tag{5.21}$$

Moreover, for every $t > 0$ and $0 < u < 1$, in the ordered Hilbert space $\mathsf{L}^2(\omega_t, u)$,

$$\widehat{G}_{AB,j,u}(t) := \sqrt{\lambda}\, \widehat{G}_{A,j,u}(\kappa_A t) \otimes I + \sqrt{1-\lambda}\, I \otimes \widehat{G}_{B,j,u}(\kappa_B t) \tag{5.22}$$

is the optimal ordered estimator of $G_{C,j}$ based on both inputs, and

$$\mathcal{G}_{\lambda,u}(t) = \sum_{j=1}^m \|\iota_\eta(\widehat{G}_{C,j,u}(t)) - \widehat{G}_{AB,j,u}(t)\|^2_{\omega_t,u}. \tag{5.23}$$

Consequently,

$$\Delta_{\lambda,\eta} = \int_0^\infty \frac{dt}{t} \int_0^1 \sum_{j=1}^m \|\iota_\eta(\widehat{G}_{C,j,u}(t)) - \widehat{G}_{AB,j,u}(t)\|^2_{\omega_t,u}\, du, \tag{5.24}$$

and, by (5.19)*, the deficit of the exponential qEPI is the member* $\lambda = \lambda_*$ *of* (5.21)–(5.24)*:*

$$\Delta_{\exp} = \int_0^\infty \frac{dt}{t} \int_0^1 \mathcal{G}_{\lambda_*,u}(t)\, du, \qquad \kappa_A = \frac{\lambda_*}{\eta}, \quad \kappa_B = \frac{1-\lambda_*}{1-\eta}. \tag{5.25}$$

Proof. By (5.11), the integrand in (5.14) is nonnegative; Lemma 18 makes it measurable. Thus

$$\int_0^\infty \frac{dt}{t} \int_0^1 \mathcal{G}_{\lambda,u}(t)\, du \overset{\text{(a)}}{=} \lim_{n\to\infty} \int_{1/n}^n \frac{dt}{t} \int_0^1 \mathcal{G}_{\lambda,u}(t)\, du \overset{\text{(b)}}{=} \lim_{n\to\infty} \big[\Phi_\lambda(1/n) - \Phi_\lambda(n)\big] \overset{\text{(c)}}{=} \Delta_{\lambda,\eta},$$

where (a) is monotone convergence, (b) is (5.14), and (c) is (5.16), (5.15), and the definition (5.17). This proves (5.21); its value is finite because the three entropies are finite.

Fix $t > 0$ and $0 < u < 1$. Lemma 6 identifies the optimal estimator of $G_{C,j} = \sqrt{\lambda} G_{A,j} + \sqrt{1-\lambda} G_{B,j}$ in the product experiment as (5.22). Put

$$\Gamma_{AB,j} = \mathbb{E}[G_{C,j}\omega_{t,G_A,G_B}], \qquad \Gamma_{C,j} = \mathbb{E}[G_{C,j}\rho_{C,t,G_C}], \qquad \omega_{t,g_A,g_B} = \rho_{A,\kappa_A t,g_A} \otimes \rho_{B,\kappa_B t,g_B}.$$

By (5.8), $\mathcal{B}_\eta(\omega_{t,G_A,G_B}) = \rho_{C,t,G_C}$, and hence $\mathcal{B}_\eta(\Gamma_{AB,j}) = \Gamma_{C,j}$. For bounded B, preservation of products by ι_η and (5.9) give

$$\|\iota_\eta(B)\|^2_{\omega_t,u} = (1-u)\operatorname{Tr}\omega_t \iota_\eta(BB^\dagger) + u \operatorname{Tr}\omega_t \iota_\eta(B^\dagger B) = \|B\|^2_{\rho_C(t),u}. \tag{5.26}$$

Consequently the extended range $\mathscr{V}_{t,u} = \iota_\eta(\mathsf{L}^2(\rho_C(t), u))$ is closed in $\mathsf{L}^2(\omega_t, u)$. For bounded B,

$$\begin{aligned} \langle \iota_\eta(B), \widehat{G}_{AB,j,u}(t)\rangle_{\omega_t,u} &\overset{\text{(d)}}{=} \operatorname{Tr}[\iota_\eta(B)^\dagger \Gamma_{AB,j}] \overset{\text{(e)}}{=} \operatorname{Tr}[B^\dagger \Gamma_{C,j}] \\ &\overset{\text{(d)}}{=} \langle B, \widehat{G}_{C,j,u}(t)\rangle_{\rho_C(t),u} \overset{\text{(f)}}{=} \langle \iota_\eta(B), \iota_\eta(\widehat{G}_{C,j,u}(t))\rangle_{\omega_t,u}, \end{aligned} \tag{5.27}$$

where (d) is the normal equation (2.32) in the respective experiment, (e) is trace duality for ι_η together with $\mathcal{B}_\eta(\Gamma_{AB,j}) = \Gamma_{C,j}$, and (f) is the polarization of (5.26). The bounded-operator density and continuous pairings in Lemmas 4 and 20 extend (5.27) to every $B \in \mathsf{L}^2(\rho_C(t), u)$. Thus, if $P_{t,u}$ is the orthogonal projection onto $\mathscr{V}_{t,u}$,

$$P_{t,u}\widehat{G}_{AB,j,u}(t) = \iota_\eta(\widehat{G}_{C,j,u}(t)). \tag{5.28}$$

Both experiments have prior moment $\mathbb{E}\|G_C\|^2 = m$. Therefore

$$\begin{aligned} \mathcal{G}_{\lambda,u}(t) &\overset{\text{(g)}}{=} \left[m - \sum_j \|\widehat{G}_{C,j,u}(t)\|^2_{\rho_C(t),u}\right] - \left[m - \sum_j \|\widehat{G}_{AB,j,u}(t)\|^2_{\omega_t,u}\right] \\ &\overset{\text{(h)}}{=} \sum_j \Big(\|\widehat{G}_{AB,j,u}(t)\|^2_{\omega_t,u} - \|P_{t,u}\widehat{G}_{AB,j,u}(t)\|^2_{\omega_t,u}\Big) \overset{\text{(i)}}{=} \sum_j \|(I - P_{t,u})\widehat{G}_{AB,j,u}(t)\|^2_{\omega_t,u}, \end{aligned}$$

where (g) is the minimum-risk formula (2.31) for the two experiments and the definition (5.11), (h) is (5.26) and (5.28), and (i) is the Pythagorean theorem for the orthogonal projection $P_{t,u}$. This is (5.23). Substitution into (5.21) proves (5.24), and (5.19) gives (5.25). □

The projection step (5.28)–(5.23) is the generalized conditional-expectation principle of [13, Theorem 2], with the completed spaces supplying the infinite-dimensional extension. The classical counterpart of (5.24) is [10, Theorem 1, Eq. (19b)]; its estimators lie in a classical L^2 space, whereas (5.24) averages ordered operator-space residuals over u. Because the integrand in (5.24) is nonnegative, $\Delta_{\lambda,\eta} = 0$ if and only if $\iota_\eta(\widehat{G}_{C,j,u}(t)) = \widehat{G}_{AB,j,u}(t)$ in $\mathsf{L}^2(\omega_t, u)$ for every j and for

almost every (t,u) with respect to $dt\,du/t$ on $(0,\infty)\times(0,1)$. This criterion concerns estimators. For $\lambda=\eta$ and for the exponential inequality, the state-level equality classification is given in the companion work [27, Theorems 1–2]: the independent finite-energy inputs are Gaussian with the same covariance matrix, with arbitrary displacements. Deriving that classification directly from the estimation residual, or obtaining a quantitative stability bound from it, is a separate question.

The second corollary rewrites the gap through the ordered scores (2.45); its score form is related to [6, Corollary 8 and Sec. 4.2, Eq. (41)]. Define

$$V_{j,u}(t)=\frac{\lambda}{\sqrt{\eta}}\mathcal{S}_{j,u}(\rho_A(\kappa_A t))\otimes I+\frac{1-\lambda}{\sqrt{1-\eta}}I\otimes\mathcal{S}_{j,u}(\rho_B(\kappa_B t)). \tag{5.29}$$

Corollary 16 (Ordered Stam inequality along the heat flow)**.** *Under the assumptions of Theorem 14, for every $0<\lambda<1$, $t>0$, and $0<u<1$,*

$$\frac{\lambda^2}{\eta}J_u(\rho_A(\kappa_A t))+\frac{(1-\lambda)^2}{1-\eta}J_u(\rho_B(\kappa_B t))-J_u(\rho_C(t))=\frac{\mathcal{G}_{\lambda,u}(t)}{t}=\sum_{j=1}^{m}\|(I-P_{t,u})V_{j,u}(t)\|^2_{\omega_t,u}\geq 0. \tag{5.30}$$

Proof. Substituting (4.3) for the three risks in (5.11) and using (5.6),

$$\begin{aligned}\mathcal{G}_{\lambda,u}(t)&=[m-tJ_u(\rho_C(t))]-\lambda[m-\kappa_A tJ_u(\rho_A(\kappa_A t))]-(1-\lambda)[m-\kappa_B tJ_u(\rho_B(\kappa_B t))]\\&=\underbrace{m-\lambda m-(1-\lambda)m}_{=0}+t\left[\frac{\lambda^2}{\eta}J_u(\rho_A(\kappa_A t))+\frac{(1-\lambda)^2}{1-\eta}J_u(\rho_B(\kappa_B t))-J_u(\rho_C(t))\right],\end{aligned}$$

which proves the first equality in (5.30) after division by $t>0$. For the second, Proposition 13 and (5.6) give

$$\widehat{G}_{AB,j,u}(t)=-\sqrt{\lambda\kappa_A t}\,\mathcal{S}_{j,u}(\rho_A(\kappa_A t))\otimes I-\sqrt{(1-\lambda)\kappa_B t}\,I\otimes\mathcal{S}_{j,u}(\rho_B(\kappa_B t))=-\sqrt{t}\,V_{j,u}(t),$$

so (5.23) with (5.28) reads $\mathcal{G}_{\lambda,u}(t)=t\sum_j\|(I-P_{t,u})V_{j,u}(t)\|^2_{\omega_t,u}$. □

At $\lambda=\eta$, (5.30) is the score projection behind entropy concavity in the symmetric-convolution formulation of [6, Sec. 4.2], here in the Gaussian-label formulation and for arbitrary η,λ; integration over u gives the unconditional KMB Stam inequality of [8, Theorem 3].

5.3 A direct SNR interpolation for common thermal mixtures

The signal I–MMSE identity gives a complementary interpolation for displacement mixtures with the same thermal noise state. Its entropy inequality is a special case of Theorem 14; the direct interpolation and its deficit formula explain what the signal identity contributes. Related entropy inequalities for quantum–classical convolution are studied in [17].

Example 17 (Common thermal mixtures). Fix $N>0$, $\nu=N+1/2$, and $0<\eta<1$. Let $Z_A,Z_B\in\mathbb{C}^m$ be independent random vectors with finite second moments, and use the same thermal noise state τ_ν in the two displacement experiments:

$$\rho_L(s)=\mathbb{E}[D_{\sqrt{s}Z_L}\tau_\nu D^\dagger_{\sqrt{s}Z_L}],\qquad L\in\{A,B\},\quad s\geq 0. \tag{5.31}$$

In the notation of [6], $\rho_L(s)=\tau_\nu\star_s Z_L$ is the quantum–classical convolution of the thermal state with the label Z_L; here the law of Z_L replaces a probability density. In this example, $\rho_L(s)$ and

the experiments A_s, B_s, C_s refer to the thermal mixtures (5.31), not to the noise experiments of Section 4 used in Subsection 5.1.

For

$$W = \sqrt{\eta} Z_A + \sqrt{1-\eta} Z_B, \tag{5.32}$$

the displacement covariance (2.14) and the thermal invariance (2.16) give

$$\begin{aligned} &\mathcal{B}_\eta(D_{\sqrt{s}Z_A} \tau_\nu D^\dagger_{\sqrt{s}Z_A} \otimes D_{\sqrt{s}Z_B} \tau_\nu D^\dagger_{\sqrt{s}Z_B}) \\ &\quad = D_{\sqrt{s}W} \mathcal{B}_\eta(\tau_\nu \otimes \tau_\nu) D^\dagger_{\sqrt{s}W} = D_{\sqrt{s}W} \tau_\nu D^\dagger_{\sqrt{s}W}. \end{aligned}$$

Averaging yields

$$\rho_C(s) = \mathcal{B}_\eta(\rho_A(s) \otimes \rho_B(s)) = \mathbb{E}[D_{\sqrt{s}W} \tau_\nu D^\dagger_{\sqrt{s}W}]. \tag{5.33}$$

Write A_s, B_s, C_s for the three experiments and set $Z_C = W$. The Cauchy–Schwarz inequality $\|W\|^2 \le (\eta + (1-\eta))(\|Z_A\|^2 + \|Z_B\|^2)$ and the energy formula after (2.10) give

$$\mathbb{E}\|W\|^2 \le \mathbb{E}\|Z_A\|^2 + \mathbb{E}\|Z_B\|^2 < \infty, \qquad \mathrm{Tr}(\rho_L(s) H_0) = mN + s\mathbb{E}\|Z_L\|^2 \quad (L = A, B, C).$$

Hence Theorem 1 applies to all three experiments. For their common target W,

$$\begin{aligned} \mathsf{M}_u(W \mid C_s) &\overset{\text{(a)}}{\ge} \mathsf{M}_u(W \mid A_s B_s) \\ &\overset{\text{(b)}}{=} \eta \mathsf{M}_u(Z_A \mid A_s) + (1-\eta) \mathsf{M}_u(Z_B \mid B_s), \end{aligned} \tag{5.34}$$

where (a) is Lemma 5 for $\mathcal{B}_\eta$, and (b) is Lemma 6. That lemma permits nonzero input means. Define

$$\begin{aligned} \Delta_u(s) &:= \mathsf{M}_u(W \mid C_s) - \eta \mathsf{M}_u(Z_A \mid A_s) - (1-\eta) \mathsf{M}_u(Z_B \mid B_s) \\ &= \mathsf{M}_u(W \mid C_s) - \mathsf{M}_u(W \mid A_s B_s) \ge 0 \end{aligned} \tag{5.35}$$

and

$$d_{\mathrm{th}}(s) = S(\rho_C(s)) - \eta S(\rho_A(s)) - (1-\eta) S(\rho_B(s)). \tag{5.36}$$

Since $I(Z_L; L_s) = S(\rho_L(s)) - S(\tau_\nu)$ for $L = A, B, C$, Theorem 1 gives

$$\begin{aligned} \frac{d}{ds} S(\rho_A(s)) &= \int_0^1 \frac{\mathsf{M}_u(Z_A \mid A_s)}{N+1-u}\, du, \\ \frac{d}{ds} S(\rho_B(s)) &= \int_0^1 \frac{\mathsf{M}_u(Z_B \mid B_s)}{N+1-u}\, du, \\ \frac{d}{ds} S(\rho_C(s)) &= \int_0^1 \frac{\mathsf{M}_u(W \mid C_s)}{N+1-u}\, du. \end{aligned}$$

Substituting these three expressions in (5.36) gives

$$\begin{aligned} d'_{\mathrm{th}}(s) &= \int_0^1 \frac{\mathsf{M}_u(W \mid C_s) - \eta \mathsf{M}_u(Z_A \mid A_s) - (1-\eta) \mathsf{M}_u(Z_B \mid B_s)}{N+1-u}\, du \\ &= \int_0^1 \frac{\Delta_u(s)}{N+1-u}\, du \ge 0, \qquad s > 0, \end{aligned} \tag{5.37}$$

where the last step uses (5.35) and $N + 1 - u \ge N > 0$. At zero SNR,

$$d_{\mathrm{th}}(0) = \underbrace{[1 - \eta - (1-\eta)]}_{=0} S(\tau_\nu) = 0.$$

Subtracting the three integrated identities in Theorem 1 therefore proves

$$d_{\mathrm{th}}(s) = d_{\mathrm{th}}(0) + \int_0^s \int_0^1 \frac{\Delta_u(r)}{N+1-u}\, du\, dr = \int_0^s \int_0^1 \frac{\Delta_u(r)}{N+1-u}\, du\, dr. \tag{5.38}$$

For completeness, let $V_L = \mathbb{E}\|Z_L - \mathbb{E}Z_L\|^2$ for $L = A, B$. The constant estimator $\mathbb{E}W$ and independence give

$$\begin{aligned} 0 \le \Delta_u(s) \le \mathsf{M}_u(W \mid C_s) &\le \mathbb{E}\|W - \mathbb{E}W\|^2 \\ &= \eta V_A + (1-\eta) V_B. \end{aligned} \tag{5.39}$$

Thus the integrand in (5.38) is bounded by $[\eta V_A + (1-\eta)V_B]/N$ on every finite SNR interval. Equations (5.38) and (5.35) yield

$$S(\rho_C(s)) \ge \eta S(\rho_A(s)) + (1-\eta) S(\rho_B(s)). \tag{5.40}$$

Finally,

$$\begin{aligned} &I(W; C_s) - \eta I(Z_A; A_s) - (1-\eta) I(Z_B; B_s) \\ &\quad = d_{\mathrm{th}}(s) + \underbrace{[-1 + \eta + (1-\eta)]}_{=0} S(\tau_\nu) = d_{\mathrm{th}}(s) \ge 0, \end{aligned}$$

which proves

$$I(W; C_s) \ge \eta I(Z_A; A_s) + (1-\eta) I(Z_B; B_s). \tag{5.41}$$

6 Final remarks

The thermal I–MMSE identity relates the change in Holevo information with SNR to optimal quadratic estimation. Its two structural features have separate origins: the thermal commutation relation determines the weight, while entropy differentiation requires an average over operator orderings. The KMB–SLD correction provides an equivalent expression in terms of Jordan operator error. For a Gaussian prior, comparing the ordered risks in (1.7) with the joint-measurement optimum (C.5) shows that a joint measurement error cannot generally replace the ordered risks.

The two entropy interpolations illustrate complementary uses of the identities. For mixtures with a common thermal noise state, the signal identity starts at zero SNR and accumulates a nonnegative estimation loss. For arbitrary finite-energy inputs, the Gaussian-noise identity compares unequal noise levels and uses the entropy asymptotic $S(\mathcal{N}_t(\rho)) = m \log t + m + o(1)$ from (2.51). Combining this identity with the interpolation of De Palma and Trevisan gives the full linear qEPI; optimizing its free entropy weight gives the exponential form. Retaining the estimation remainder expresses both the weighted linear deficit and the logarithmic exponential deficit as integrals of squared projection errors.

These identities also give an estimation criterion for equality: a vanishing deficit forces agreement between the optimal estimator based on both inputs and the estimator accessible at the retained output for almost every noise level and ordering. For the entropy-concavity form, with entropy weight $\lambda = \eta$, and for the exponential qEPI, the state-level equality class is known: independent finite-energy inputs attain equality precisely when they are Gaussian with the same covariance matrix, allowing arbitrary displacements [27, Thms. 1–2]. A further question is how to recover this classification directly from the estimator identity. Quantitative stability would require an additional estimate relating the integrated squared estimator difference to a specified distance from that equality class, under suitable energy or regularity assumptions.

Several questions also remain for the signal identity. At zero thermal photon number, the ordering weight becomes singular, so a result for arbitrary priors requires additional endpoint control. For nonisotropic Gaussian noise states, the formulation must account for mode-dependent thermal parameters and the corresponding changes of quadratures $R \mapsto TR$, where a real symplectic matrix T satisfies $T\Omega T^T = \Omega$. To interpret the full ordered average in (1.4) through a measurement experiment, one would need a specified measurement and estimator whose expected loss equals that average for general ordering parameters. These extensions may lead to applications in entropy stability and the optimization of bosonic communication channels.

A Estimation spaces and continuity

This appendix supplies the Hilbert-space justification for the least-squares calculations in the main text. In infinite dimensions, the optimal estimator need not be bounded. We therefore complete the bounded estimators in the norm defined by their quadratic loss and construct the optimum as an orthogonal projection. We also prove the continuity statements used in entropy differentiation and at zero SNR.

For finitely many targets, take the Cartesian product of their Hilbert spaces and sum the squared norms and risks over the target coordinates.

Construction for Lemma 4. Fix one target $f \in L^2(P_Z)$; the vector result follows by taking finite Cartesian products. Let $\mathscr{E}$ be the completion, after removing null vectors, of uniformly bounded measurable operator fields under

$$\langle F, G\rangle_{\mathscr{E}} = \mathbb{E}[(1-u)\operatorname{Tr}(F(Z)^\dagger \omega_Z G(Z)) + u \operatorname{Tr}(\omega_Z F(Z)^\dagger G(Z))].$$

Here measurability means measurability of all matrix entries in a fixed countable orthonormal basis. Constant fields satisfy $\|z \mapsto B\|_{\mathscr{E}} = \|B\|_{\omega,u}$, so their closure $\mathscr{C}$ is isometric to $\mathsf{L}^2(\omega, u)$. With $f_R = f 1_{\{|f|\le R\}}$,

$$\|fI - f_R I\|_{\mathscr{E}}^2 = \mathbb{E}[|f|^2 1_{\{|f|>R\}}] \longrightarrow 0.$$

Thus $fI \in \mathscr{E}$. Its orthogonal projection $\widehat{f}_u$ onto $\mathscr{C}$ satisfies, for every bounded constant field B,

$$\begin{aligned} \langle B, \widehat{f}_u\rangle_{\omega,u} &= \langle B, fI\rangle_{\mathscr{E}} = \mathbb{E}[f \operatorname{Tr}(B^\dagger \omega_Z)] = \operatorname{Tr}(B^\dagger \Gamma_f), \\ |\operatorname{Tr}(B^\dagger \Gamma_f)| &\le \|B\|_{\omega,u}\|f\|_{L^2(P_Z)}, \qquad \Gamma_f = \mathbb{E}[f(Z)\omega_Z]. \end{aligned}$$

The first identity is (2.32) after complex conjugation. To give the normal equation its trace-class meaning, apply Hilbert–Schmidt factorization and weighted Cauchy–Schwarz:

$$\begin{aligned} \|\mathcal{K}_{\omega,u}(B)\|_1 &\le (1-u)\|\omega B\|_1 + u\|B\omega\|_1 \\ &\le (1-u)\sqrt{\operatorname{Tr}(\omega B B^\dagger)} + u\sqrt{\operatorname{Tr}(\omega B^\dagger B)} \le \|B\|_{\omega,u}. \end{aligned}$$

Hence $\mathcal{K}_{\omega,u}$ extends continuously to trace class on the completion. For bounded $B_n \to \widehat{f}_u$ in weighted norm and every bounded B,

$$\begin{aligned} \operatorname{Tr}[B^\dagger \mathcal{K}_{\omega,u}(\widehat{f}_u)] &= \lim_n \operatorname{Tr}[B^\dagger \mathcal{K}_{\omega,u}(B_n)] \\ &= \lim_n \langle B, B_n\rangle_{\omega,u} = \operatorname{Tr}(B^\dagger \Gamma_f). \end{aligned}$$

Bounded operators separate trace-class operators, giving $\mathcal{K}_{\omega,u}(\widehat{f}_u) = \Gamma_f$. The map is injective on the completion: if $\mathcal{K}_{\omega,u}(T) = 0$, then $\langle B, T\rangle_{\omega,u} = 0$ for every bounded B, and density implies $T = 0$. Testing the normal equation against matrix units proves (2.35); zero denominators force zero entries of Γ_f and represent precisely the removed null directions. Finally, $\mathcal{R}_u(B) = \|fI - B\|^2_{\mathscr{E}}$ by (2.27). The square completion in the main-text proof therefore applies to every completed estimator, proving the claimed projection and minimum-risk formulas.

Lemma 18 (Bounded targets and continuous risks)**.** *If $|f| \le C_f$, the estimator has a bounded representative satisfying*

$$\|\widehat{f}_u\| \le \frac{C_f}{2\sqrt{u(1-u)}}. \tag{A.1}$$

For uniformly bounded targets, the minimum risk is continuous in the average state and first moment in trace norm, in the scalar second moment, and in the interior parameter u. For a fixed L^2 target, the same conclusion holds whenever bounded target truncations give the preceding trace-norm convergence.

Proof. No faithfulness assumption is needed. For vectors x, y, positivity and Cauchy–Schwarz give

$$\begin{aligned}
|\langle x, \Gamma_f y\rangle| &\le C_f \mathbb{E}|\langle x, \omega_Z y\rangle| \\
&\le C_f \mathbb{E}\sqrt{\langle x, \omega_Z x\rangle\langle y, \omega_Z y\rangle} \\
&\le C_f \sqrt{\langle x, \omega x\rangle\langle y, \omega y\rangle} = C_f \|\omega^{1/2} x\| \|\omega^{1/2} y\|.
\end{aligned}$$

Thus $\beta(\omega^{1/2}x, \omega^{1/2}y) = \langle x, \Gamma_f y\rangle$ is a well-defined bounded sesquilinear form. It extends to $\operatorname{supp}\omega = \overline{\operatorname{ran}\omega^{1/2}}$, where its representing operator C satisfies

$$\Gamma_f = \omega^{1/2} C \omega^{1/2}, \qquad \|C\| \le C_f.$$

On this support, define a weak integral

$$T = \int_0^\infty e^{-(1-u)v\omega} \Gamma_f e^{-uv\omega}\, dv$$

by integrating its matrix elements. The next estimate proves absolute convergence and boundedness.

If x lies in the support and $\alpha > 0$, spectral calculus and Tonelli's theorem give

$$\int_0^\infty \|\omega^{1/2} e^{-\alpha v\omega} x\|^2\, dv = \sum_{\mu_k > 0} |x_k|^2 \int_0^\infty \mu_k e^{-2\alpha v \mu_k}\, dv = \frac{\|x\|^2}{2\alpha}.$$

Consequently,

$$\begin{aligned}
|\langle x, Ty\rangle| &\le C_f \int_0^\infty \|\omega^{1/2} e^{-(1-u)v\omega} x\| \|\omega^{1/2} e^{-uv\omega} y\|\, dv \\
&\le C_f \left(\int_0^\infty \|\omega^{1/2} e^{-(1-u)v\omega} x\|^2\, dv\right)^{1/2} \left(\int_0^\infty \|\omega^{1/2} e^{-uv\omega} y\|^2\, dv\right)^{1/2} \\
&= \frac{C_f \|x\| \|y\|}{2\sqrt{u(1-u)}}.
\end{aligned}$$

Extending T by zero off the support makes it bounded on the full Hilbert space. For $\mu_k, \mu_\ell > 0$,

$$T_{k\ell} = (\Gamma_f)_{k\ell} \int_0^\infty e^{-v((1-u)\mu_k + u\mu_\ell)}\, dv = \frac{(\Gamma_f)_{k\ell}}{(1-u)\mu_k + u\mu_\ell}.$$

All entries of Γ_f off the support vanish by its factorization. Thus $\mathcal{K}_{\omega,u}(T) = \Gamma_f$, and uniqueness in Lemma 4 gives $T = \widehat{f}_u$ in the completion. This proves (A.1).

Consider experiments indexed by n, with targets f_n, average states ω_n, first moments Γ_{f_n}, the common bound C_f, and minimum risks $\mathsf{M}_u^{(n)}$. At fixed u, put $C_u = C_f/(2\sqrt{u(1-u)})$. By (A.1), both minima may be taken over $\|B\| \le C_u$. Formula (2.33) gives, on this ball,

$$\begin{aligned}|\mathcal{R}_u^{(n)}(B) - \mathcal{R}_u(B)| &\le |\mathbb{E}|f_n|^2 - \mathbb{E}|f|^2| + 2|\operatorname{Tr}[(\Gamma_{f_n} - \Gamma_f)^\dagger B]| \\ &\quad + (1-u)|\operatorname{Tr}[(\omega_n - \omega)BB^\dagger]| + u|\operatorname{Tr}[(\omega_n - \omega)B^\dagger B]| \\ &\le |\mathbb{E}|f_n|^2 - \mathbb{E}|f|^2| + 2C_u\|\Gamma_{f_n} - \Gamma_f\|_1 + C_u^2\|\omega_n - \omega\|_1.\end{aligned}$$

Taking the supremum over this ball and using $|\inf F - \inf G| \le \sup|F - G|$ proves

$$|\mathsf{M}_u^{(n)} - \mathsf{M}_u| \le |\mathbb{E}|f_n|^2 - \mathbb{E}|f|^2| + 2C_u\|\Gamma_{f_n} - \Gamma_f\|_1 + C_u^2\|\omega_n - \omega\|_1. \tag{A.2}$$

For $u, v \in [u_0, 1-u_0]$, where $0 < u_0 < 1/2$, the same ball can be chosen uniformly. Moreover,

$$|\mathcal{R}_u(B) - \mathcal{R}_v(B)| = |u - v|\,|\operatorname{Tr}[\omega(B^\dagger B - BB^\dagger)]| \le 2\|B\|^2|u - v|.$$

This estimate together with (A.2) proves joint continuity in the stated data and the interior ordering parameter.

For general square-integrable targets, use the projection space $\mathscr{C}$ from Lemma 4. The triangle inequality gives

$$\operatorname{dist}(fI, \mathscr{C}) \le \|fI - gI\|_{\mathscr{E}} + \operatorname{dist}(gI, \mathscr{C}).$$

Interchanging f and g proves

$$\left|\sqrt{\mathsf{M}_u(f \mid A)} - \sqrt{\mathsf{M}_u(g \mid A)}\right| \le \|f - g\|_{L^2(P_Z)}. \tag{A.3}$$

In particular, for $f_R = f1_{\{|f| \le R\}}$ under a fixed prior,

$$\begin{aligned}|\sqrt{\mathsf{M}_u^{(n)}(f)} - \sqrt{\mathsf{M}_u(f)}| &\le 2\|f - f_R\|_{L^2(P_Z)} + |\sqrt{\mathsf{M}_u^{(n)}(f_R)} - \sqrt{\mathsf{M}_u(f_R)}|, \\ \limsup_n |\sqrt{\mathsf{M}_u^{(n)}(f)} - \sqrt{\mathsf{M}_u(f)}| &\le 2\|f - f_R\|_{L^2(P_Z)} \longrightarrow 0 \quad (R \to \infty).\end{aligned}$$

The second line uses the bounded-target continuity just proved. The same estimates apply to varying targets when their truncation errors are uniformly small; convergence of second moments alone is not used in this step. □

In both displacement experiments of the paper, Weyl conjugation is continuous in trace norm. Dominated convergence supplies the average-state and bounded-target first-moment convergence required by Lemma 18. The risks are therefore jointly continuous in the positive channel parameter and $0 < u < 1$. The thermal experiment has the additional zero-SNR continuity proved below. All scalar risk integrands used below are measurable, as follows directly from their continuity.

Corollary 19 (Risk continuity at zero SNR)**.** *Let $Z \in \mathbb{C}^m$ satisfy $\mathbb{E}\|Z\|^2 < \infty$, let σ be a fixed state, and encode z as $\omega_{s,z} = D_{\sqrt{s}z}\sigma D_{\sqrt{s}z}^\dagger$ for $s \ge 0$. For every $0 < u < 1$,*

$$\lim_{s \downarrow 0} \mathsf{M}_u(Z \mid A_s) = \mathsf{M}_u(Z \mid A_0) = \mathbb{E}\|Z - \mathbb{E}Z\|^2. \tag{A.4}$$

If $\sigma = \tau_\nu$ with thermal mean photon number $N = \nu - \frac{1}{2} > 0$, then

$$\lim_{s \downarrow 0} \int_0^1 \frac{\mathsf{M}_u(Z \mid A_s)}{N + 1 - u}\, du = \log\frac{N+1}{N}\, \mathbb{E}\|Z - \mathbb{E}Z\|^2. \tag{A.5}$$

Proof. For each z, trace-norm continuity of Weyl conjugation gives $\|\omega_{s,z} - \sigma\|_1 \to 0$. This norm is at most 2, so for each bounded target f_R, dominated convergence yields

$$\omega_s \longrightarrow \sigma, \qquad \mathbb{E}[f_R(Z)\omega_{s,Z}] \longrightarrow (\mathbb{E}f_R)\sigma \quad \text{in trace norm.}$$

Lemma 18 and the uniform truncation bound (A.3) imply $\mathsf{M}_u(Z \mid A_s) \to \mathsf{M}_u(Z \mid A_0)$. At zero SNR,

$$\mathcal{K}_{\sigma,u}((\mathbb{E}Z_j)I) = (\mathbb{E}Z_j)\sigma = \mathbb{E}[Z_j\omega_{0,Z}],$$

so (2.31) gives $\mathsf{M}_u(Z \mid A_0) = \mathbb{E}\|Z\|^2 - \|\mathbb{E}Z\|^2$. For the weighted limit, use

$$0 \le \frac{\mathsf{M}_u(Z \mid A_s)}{N+1-u} \le \frac{\mathbb{E}\|Z - \mathbb{E}Z\|^2}{N}, \qquad \int_0^1 \frac{du}{N+1-u} = \log\frac{N+1}{N},$$

and apply dominated convergence in u. □

The optimal values are also measurable for merely measurable data. Fix an orthonormal basis, let P_n project onto its first n vectors, and let $\mathcal{D}$ be the countable set of finite matrices with rational real and imaginary entries. For bounded B, set $B_n = P_nBP_n$.

For every vector ψ,

$$\|(B_n - B)\psi\| \le \|B\|\|(P_n - I)\psi\| + \|(P_n - I)B\psi\| \longrightarrow 0.$$

This is strong operator convergence. Applying the same inequality to $B^\dagger$ proves $B_n^\dagger \to B^\dagger$ strongly as well, and $\|B_n\| \le \|B\|$.

If $\omega = \sum_k \mu_k |e_k\rangle\langle e_k|$,

$$\|B_n - B\|_{\omega,u}^2 = (1-u)\sum_k \mu_k \|(B_n - B)^\dagger e_k\|^2 + u\sum_k \mu_k \|(B_n - B)e_k\|^2 \longrightarrow 0.$$

Dominated convergence applies because each squared norm is at most $4\|B\|^2$ and $\sum_k \mu_k = 1$. Approximating each finite matrix B_n in operator norm by a member of $\mathcal{D}$, and using $\|T\|_{\omega,u} \le \|T\|$, proves that $\mathcal{D}$ is dense in every weighted completion, including when ω is not faithful. Risk continuity in that norm, from Lemma 4, gives

$$\mathsf{M}_u(f \mid A) = \inf_{B \in \mathcal{D}} \mathcal{R}_u(B).$$

For each fixed B, formula (2.33) expresses $\mathcal{R}_u(B)$ as a measurable function of the state, first-moment, second-moment, and ordering data. Their countable infimum is therefore measurable. For finitely many targets use the countable set $\mathcal{D}^m$.

Lemma 20 (Completed tensor embeddings and pullbacks)**.** *Let ι be a unital $*$-homomorphism (preserving the identity, products, and adjoints) satisfying $\mathrm{Tr}(\omega\iota(B)) = \mathrm{Tr}(\sigma B)$ for bounded B. Then the tensor embedding and ι extend to isometries with closed range between the corresponding completed estimator spaces:*

$$\|B \otimes I\|_{\omega_A \otimes \omega_B, u} = \|B\|_{\omega_A, u}, \qquad \|\iota(B)\|_{\omega,u} = \|B\|_{\sigma,u}. \tag{A.6}$$

Proof. For bounded B, $\mathrm{Tr}\,\omega_B = 1$ gives the first norm identity. For the second, multiplicativity and adjoint preservation give

$$\|\iota(B)\|_{\omega,u}^2 = (1-u)\,\mathrm{Tr}[\omega\iota(BB^\dagger)] + u\,\mathrm{Tr}[\omega\iota(B^\dagger B)] = (1-u)\,\mathrm{Tr}(\sigma BB^\dagger) + u\,\mathrm{Tr}(\sigma B^\dagger B).$$

Both identities extend by density; the image of a complete space under an isometry is closed. □

For the beam splitter, $\iota(B) = U_\eta^\dagger(B \otimes I)U_\eta$. Thus the pullbacks in Section 5 act on completed estimators by the isometric extension in Lemma 20.

Here "pullback" means expressing an output estimator as an operator on the input system through ι. In the ensemble Hilbert space $\mathscr{E}$, let $\mathscr{C}_{AB}$ be the closed space of all completed input estimators and let $\mathscr{C}_C = \iota(\mathsf{L}^2(\omega_C, u)) \subseteq \mathscr{C}_{AB}$. Here ω_C is the average retained-output state, and $\mathsf{L}^2(\omega_C, u)$ is its estimation completion. Write P_{AB} and P_C for their orthogonal projections. Since $P_C P_{AB} = P_C$, the Pythagorean theorem gives, for a target f,

$$\begin{aligned}\mathsf{M}_u(f \mid C) - \mathsf{M}_u(f \mid AB) &= \|fI - P_C(fI)\|_{\mathscr{E}}^2 - \|fI - P_{AB}(fI)\|_{\mathscr{E}}^2 \\ &= \|P_{AB}(fI) - P_C(fI)\|_{\mathscr{E}}^2.\end{aligned}$$

This is the squared-distance interpretation used in Section 5.

The tensor identity (2.41) extends by these isometries and trace-norm continuity of the weight map. This is the extension used in Lemma 6.

B Entropy differentiation for bosonic displacement ensembles

We justify the identities in Sections 3 and 4 under their stated moment assumptions. The issue is that finite energy does not make the quadratures or the logarithm of a state bounded. We first differentiate a bounded regularization of the logarithm and then pass to entropy. Subsection B.2 proves the pairing form of the commutator chain rule, Subsection B.3 proves the de Bruijn identity of Lemma 8 in the KMB form used in Section 4, and Subsection B.4 supplies the regularization estimates needed when an arbitrary signal distribution is scaled at fixed thermal noise. All logarithms are natural.

For a trace-class operator B, put $U_k(h) = D_{z(he_k)}$, where e_k is the kth coordinate vector of $\mathbb{R}^{2m}$. Saying that B has a trace-class displacement derivative $\delta_k B$ means

$$\lim_{h\to 0} \left\| \frac{U_k(h) B U_k(h)^\dagger - B}{h} - \delta_k B \right\|_1 = 0.$$

The notation δ_k in (3.22) has this meaning; when multiplication by the quadratures is justified, it equals the stated commutator. The unregularized logarithmic expressions are interpreted as the eigenbasis sums specified after Lemma 22.

B.1 Regularized entropy and displacement differentiation

The resolvent formula for the logarithmic derivative is standard [19, Sec. 2.5, Example 1]. We record a trace-class version in two lemmas. Lemma 21 concerns the regularized entropy, and Lemma 22 concerns the regularized logarithmic derivative and its behavior along Weyl orbits, including the displacement pairing used in the thermal proof.

Throughout this subsection, $\varepsilon > 0$ and σ is a state. Then $\sigma + \varepsilon I \geq \varepsilon I$ is invertible, and $\log(\sigma + \varepsilon I)$ is a bounded self-adjoint operator. We fix a spectral decomposition $\sigma = \sum_k \mu_k |k\rangle\langle k|$ whose orthonormal basis includes a basis of $\ker \sigma$ (with $\mu_k = 0$ there), and write $C_{k\ell} = \langle k|C|\ell\rangle$. The operator norm is denoted $\|\cdot\| = \|\cdot\|_\infty$ and the trace norm $\|\cdot\|_1$. For $p, q \geq 0$, define the scalar kernels

$$k_\varepsilon(p,q) = \int_0^1 \frac{du}{(1-u)p + uq + \varepsilon}, \qquad k_0(p,q) = \int_0^1 \frac{du}{(1-u)p + uq} \in (0,\infty]. \tag{B.1}$$

These are functions of two numbers only; they do not involve any operator. Both are symmetric in (p,q) (substitute $u \mapsto 1-u$). By (2.43), $k_0(p,q) = (\log p - \log q)/(p-q)$ for distinct $p, q > 0$, and

$k_0(p,p) = 1/p$, so k_0 is the coefficient in (2.42); moreover $k_0(p,q) = +\infty$ when $pq = 0$. Since the integrand in (B.1) increases as ε decreases, monotone convergence gives

$$k_\varepsilon(p,q) \le k_{\varepsilon'}(p,q) \le k_0(p,q) \quad (0 < \varepsilon' \le \varepsilon), \qquad k_\varepsilon(p,q) \uparrow k_0(p,q) \quad (\varepsilon \downarrow 0). \tag{B.2}$$

Lemma 21 (Regularized entropy)**.** *For $x \ge 0$, set*

$$g_\varepsilon(x) = \int_0^x \log \frac{1+\varepsilon}{r+\varepsilon}\, dr, \qquad F_\varepsilon(\sigma) = \operatorname{Tr} g_\varepsilon(\sigma). \tag{B.3}$$

(a) For every state σ, $0 \le F_\varepsilon(\sigma) \le \log \frac{1+\varepsilon}{\varepsilon} < \infty$, and $F_\varepsilon(\sigma) \uparrow S(\sigma) + 1$ as $\varepsilon \downarrow 0$, with a possibly infinite limit.

(b) For every path $r \mapsto \sigma_r$ of states that is C^1 in trace norm on an interval,

$$\frac{d}{dr} F_\varepsilon(\sigma_r) = -\operatorname{Tr}\big(\dot\sigma_r \log(\sigma_r + \varepsilon I)\big). \tag{B.4}$$

Proof. Step 1: scalar bounds. Fix $0 \le x \le 1$. For $0 \le r \le x$, the integrand in (B.3) satisfies $0 \le \log \frac{1+\varepsilon}{r+\varepsilon} \le \log \frac{1+\varepsilon}{\varepsilon}$, hence

$$0 \le g_\varepsilon(x) \le x \log \frac{1+\varepsilon}{\varepsilon}.$$

Moreover, $\frac{d}{d\varepsilon} \log \frac{1+\varepsilon}{r+\varepsilon} = \frac{1}{1+\varepsilon} - \frac{1}{r+\varepsilon} \le 0$ for $0 \le r \le 1$, so the integrand increases to $-\log r$ as $\varepsilon \downarrow 0$, and monotone convergence gives

$$g_\varepsilon(x) \uparrow \int_0^x (-\log r)\, dr = -x \log x + x \qquad (\varepsilon \downarrow 0),$$

with the value 0 at $x = 0$.

Step 2: proof of (a). The eigenvalues of σ lie in $[0,1]$ and $g_\varepsilon \ge 0$ there, so $g_\varepsilon(\sigma) = \sum_k g_\varepsilon(\mu_k)|k\rangle\langle k|$ is a positive operator with

$$F_\varepsilon(\sigma) = \sum_k g_\varepsilon(\mu_k) \le \log \frac{1+\varepsilon}{\varepsilon} \sum_k \mu_k = \log \frac{1+\varepsilon}{\varepsilon}.$$

Since $g_\varepsilon(0) = 0$, the kernel of σ contributes nothing. By Step 1, each term $g_\varepsilon(\mu_k)$ increases to $-\mu_k \log \mu_k + \mu_k$ as $\varepsilon \downarrow 0$. Monotone convergence for series gives $F_\varepsilon(\sigma) \uparrow \sum_k (-\mu_k \log \mu_k + \mu_k) = S(\sigma) + 1$.

Step 3: a series for g_ε. For $0 \le r \le 1$, $0 \le (1-r)/(1+\varepsilon) < 1$, so

$$\log \frac{1+\varepsilon}{r+\varepsilon} = -\log\Big(1 - \frac{1-r}{1+\varepsilon}\Big) = \sum_{k=1}^{\infty} \frac{(1-r)^k}{k(1+\varepsilon)^k}.$$

The terms are nonnegative, so termwise integration on $[0,x]$ is justified and gives

$$g_\varepsilon(x) = \sum_{k=1}^{\infty} \frac{1 - (1-x)^{k+1}}{k(k+1)(1+\varepsilon)^k}, \qquad 0 \le x \le 1. \tag{B.5}$$

Step 4: termwise differentiation. Let σ_r be as in (b), and put $p_k(x) = 1 - (1-x)^{k+1}$, so that $p_k(0) = 0$ and $p_k'(x) = (k+1)(1-x)^k$. For a polynomial p with $p(0) = 0$, the product rule in

trace norm and cyclicity of the trace give, monomial by monomial, $\frac{d}{dr}\operatorname{Tr} p(\sigma_r) = \operatorname{Tr}[p'(\sigma_r)\dot\sigma_r]$. Since $0 \le I - \sigma_r \le I$ and $\operatorname{Tr}\sigma_r = 1$,

$$0 \le \operatorname{Tr}\, p_k(\sigma_r) = \sum_{i=0}^{k} \operatorname{Tr}\big[(I-\sigma_r)^i \sigma_r\big] \le k+1,$$
$$\Big|\frac{d}{dr}\operatorname{Tr}\, p_k(\sigma_r)\Big| = (k+1)\big|\operatorname{Tr}[(I-\sigma_r)^k\dot\sigma_r]\big| \le (k+1)\|\dot\sigma_r\|_1.$$

Applying (B.5) to the eigenvalues of σ_r and summing (all terms are nonnegative),

$$F_\varepsilon(\sigma_r) = \sum_{k=1}^{\infty} \frac{\operatorname{Tr}\, p_k(\sigma_r)}{k(k+1)(1+\varepsilon)^k}.$$

On a compact subinterval, $\sup_r \|\dot\sigma_r\|_1 < \infty$, so the series of derivatives is dominated by $\sup_r \|\dot\sigma_r\|_1 \sum_{k\ge1}[k(1+\varepsilon)^k]^{-1} < \infty$ and converges uniformly. Termwise differentiation is therefore justified:

$$\frac{d}{dr}F_\varepsilon(\sigma_r) = \sum_{k=1}^{\infty} \frac{\operatorname{Tr}\big[(I-\sigma_r)^k\dot\sigma_r\big]}{k(1+\varepsilon)^k}.$$

Step 5: proof of (b). Since $\|(I-\sigma_r)/(1+\varepsilon)\| \le (1+\varepsilon)^{-1} < 1$, the operator series $\sum_{k\ge1}(I-\sigma_r)^k/[k(1+\varepsilon)^k]$ converges in operator norm, to

$$-\log\Big(I - \frac{I-\sigma_r}{1+\varepsilon}\Big) = -\log\frac{\sigma_r+\varepsilon I}{1+\varepsilon} = \log(1+\varepsilon)I - \log(\sigma_r+\varepsilon I).$$

Pairing with the trace-class operator $\dot\sigma_r$ and using $|\operatorname{Tr}(XY)| \le \|X\|_1\|Y\|$ to move the sum inside the trace,

$$\frac{d}{dr}F_\varepsilon(\sigma_r) = \operatorname{Tr}\Big[\dot\sigma_r\big(\log(1+\varepsilon)I - \log(\sigma_r+\varepsilon I)\big)\Big] = -\operatorname{Tr}\big(\dot\sigma_r \log(\sigma_r+\varepsilon I)\big),$$

because $\operatorname{Tr}\dot\sigma_r = \frac{d}{dr}\operatorname{Tr}\sigma_r = 0$. This is (B.4). □

Lemma 22 (Regularized logarithmic derivative along Weyl orbits)**.** *Let σ be a state and $\varepsilon > 0$.*

(a) Resolvent form. *Let $p \in \{1,\infty\}$ and let C satisfy $\|C\|_p < \infty$. The integral*

$$\mathcal{T}_{\sigma+\varepsilon I}(C) = \int_0^\infty (\sigma + (\varepsilon+v)I)^{-1} C (\sigma + (\varepsilon+v)I)^{-1}\, dv \tag{B.6}$$

converges absolutely in the p-norm, and for every state σ' and every C' with $\|C'\|_p < \infty$,

$$\|\mathcal{T}_{\sigma+\varepsilon I}(C)\|_p \le \varepsilon^{-1}\|C\|_p, \qquad \|\mathcal{T}_{\sigma'+\varepsilon I}(C') - \mathcal{T}_{\sigma+\varepsilon I}(C)\|_p \le \varepsilon^{-1}\|C'-C\|_p + \varepsilon^{-2}\|\sigma'-\sigma\|\,\|C\|_p. \tag{B.7}$$

The map $\mathcal{T}_{\sigma+\varepsilon I}$ is complex linear and preserves adjoints. In the fixed eigenbasis it acts entrywise,

$$\big(\mathcal{T}_{\sigma+\varepsilon I}(C)\big)_{k\ell} = k_\varepsilon(\mu_k,\mu_\ell)\, C_{k\ell}, \tag{B.8}$$

with the kernel k_ε of (B.1)*. In particular, $\operatorname{Tr}[B^\dagger \mathcal{T}_{\sigma+\varepsilon I}(C)] = \operatorname{Tr}[\mathcal{T}_{\sigma+\varepsilon I}(B)^\dagger C]$ for trace-class B and C.*

(b) Real directions. *Let B be trace class, and suppose that the Weyl orbits $\sigma_h = D_h \sigma D_h^\dagger$ and $B_h = D_h B D_h^\dagger$ are continuously differentiable in trace norm in $h \in \mathbb{C}^m \simeq \mathbb{R}^{2m}$. Then $h \mapsto \log(\sigma_h + \varepsilon I)$ is continuously differentiable in operator norm, and for every real direction $v \in \mathbb{R}^{2m}$,*

$$\begin{aligned} \partial_v \log(\sigma_h + \varepsilon I)\big|_{h=0} &= \mathcal{T}_{\sigma+\varepsilon I}(\partial_v \sigma_h|_{h=0}), \\ \operatorname{Tr}\big((\partial_v B_h|_{h=0}) \log(\sigma + \varepsilon I)\big) &= -\operatorname{Tr}\big(B\, \mathcal{T}_{\sigma+\varepsilon I}(\partial_v \sigma_h|_{h=0})\big). \end{aligned} \tag{B.9}$$

(c) Complex directions. *Put $\partial_{\overline{h}_j} = (\partial_{\operatorname{Re} h_j} + i\partial_{\operatorname{Im} h_j})/2$, and for a trace-class operator X with a C^1 Weyl orbit define $[a_j, X] := -\partial_{\overline{h}_j}(D_h X D_h^\dagger)|_{h=0}$ in trace norm; this definition requires no unbounded operator products. Under the assumptions of (b),*

$$-\,\partial_{\overline{h}_j} \log(D_h \sigma D_h^\dagger + \varepsilon I)\Big|_{h=0} = \mathcal{T}_{\sigma+\varepsilon I}([a_j, \sigma]) \tag{B.10}$$

in operator norm, and

$$\operatorname{Tr}([a_j, B] \log(\sigma + \varepsilon I)) = -\operatorname{Tr}\big(B \mathcal{T}_{\sigma+\varepsilon I}([a_j, \sigma])\big). \tag{B.11}$$

Proof. For a state ω and $c > 0$, write $R_c(\omega) = (\omega + cI)^{-1}$. Since $\omega + cI \geq cI$, and by multiplying out $R_c(\omega')[(\omega + cI) - (\omega' + cI)]R_c(\omega)$, we have, for states ω, ω',

$$\|R_c(\omega)\| \leq c^{-1}, \qquad R_c(\omega') - R_c(\omega) = -R_c(\omega')\,(\omega' - \omega)\,R_c(\omega). \tag{B.12}$$

We also use $\|XYZ\|_p \leq \|X\|\,\|Y\|_p\,\|Z\|$ for bounded X, Z and $p \in \{1, \infty\}$.

Step 1: convergence and the first bound in (B.7). The integrand of (B.6) is $R_{\varepsilon+v}(\sigma)\, C\, R_{\varepsilon+v}(\sigma)$. It is a continuous function of v in the p-norm, and by (B.12) its p-norm is at most $(\varepsilon + v)^{-2}\|C\|_p$. Hence the integral converges absolutely in the p-norm, and

$$\|\mathcal{T}_{\sigma+\varepsilon I}(C)\|_p \leq \|C\|_p \int_0^\infty (\varepsilon + v)^{-2}\, dv = \varepsilon^{-1}\|C\|_p.$$

Complex linearity in C is clear, and $\mathcal{T}_{\sigma+\varepsilon I}(C)^\dagger = \mathcal{T}_{\sigma+\varepsilon I}(C^\dagger)$ because every resolvent $R_c(\sigma)$ is self-adjoint.

Step 2: the Lipschitz bound in (B.7). Put $c = \varepsilon + v$ and expand

$$\begin{aligned} R_c(\sigma')C'R_c(\sigma') - R_c(\sigma)CR_c(\sigma) = {} & R_c(\sigma')\,(C' - C)\,R_c(\sigma') \\ & + [R_c(\sigma') - R_c(\sigma)]\,C\,R_c(\sigma') + R_c(\sigma)\,C\,[R_c(\sigma') - R_c(\sigma)]. \end{aligned}$$

By (B.12), $\|R_c(\sigma') - R_c(\sigma)\| \leq c^{-2}\|\sigma' - \sigma\|$, so the p-norm of the right side is at most $c^{-2}\|C' - C\|_p + 2c^{-3}\|\sigma' - \sigma\|\,\|C\|_p$. Integrating over $v \in (0, \infty)$, with $\int_0^\infty (\varepsilon + v)^{-2}\, dv = \varepsilon^{-1}$ and $\int_0^\infty 2(\varepsilon + v)^{-3}\, dv = \varepsilon^{-2}$, gives the second bound in (B.7).

Step 3: the entrywise form (B.8). Matrix entries are continuous in the p-norm, so they may be taken inside the integral. In the eigenbasis, $R_c(\sigma)|k\rangle = (\mu_k + c)^{-1}|k\rangle$, hence

$$\big(\mathcal{T}_{\sigma+\varepsilon I}(C)\big)_{k\ell} = C_{k\ell} \int_0^\infty \frac{dv}{(\mu_k + \varepsilon + v)(\mu_\ell + \varepsilon + v)}.$$

Put $x = \mu_k + \varepsilon$ and $y = \mu_\ell + \varepsilon$; both are positive. If $x \neq y$, partial fractions give

$$\begin{aligned} \int_0^\infty \frac{dv}{(x + v)(y + v)} &= \frac{1}{y - x} \int_0^\infty \Big(\frac{1}{x + v} - \frac{1}{y + v}\Big) dv = \frac{\log y - \log x}{y - x} \\ &= \int_0^1 \frac{du}{(1 - u)x + uy} = k_\varepsilon(\mu_k, \mu_\ell), \end{aligned}$$

where the third equality is (2.43) and the last is the definition (B.1), since $(1-u)x+uy = (1-u)\mu_k + u\mu_\ell + \varepsilon$. If $x = y$, both the v-integral and $k_\varepsilon(\mu_k,\mu_\ell)$ equal $1/x$. This proves (B.8). Finally, for trace-class B and C, the trace of a product of two Hilbert–Schmidt operators is the entrywise sum, so (B.8) and the symmetry and realness of k_ε give

$$\operatorname{Tr}[B^\dagger \mathcal{T}_{\sigma+\varepsilon I}(C)] = \sum_{k,\ell} \overline{B_{k\ell}}\, k_\varepsilon(\mu_k,\mu_\ell)\, C_{k\ell} = \operatorname{Tr}[\mathcal{T}_{\sigma+\varepsilon I}(B)^\dagger C].$$

This completes the proof of (a).

For (b) and (c), fix $v \in \mathbb{R}^{2m}$ and write $\sigma_\tau = \sigma_{\tau v}$ and $B_\tau = B_{\tau v}$ for $\tau \in \mathbb{R}$. Put

$$\Delta_\tau = \sigma_\tau - \sigma, \qquad \dot\sigma = \partial_v \sigma_h|_{h=0}, \qquad \dot B = \partial_v B_h|_{h=0}, \qquad L_\varepsilon = \log(\sigma+\varepsilon I).$$

By hypothesis, $\|\Delta_\tau - \tau\dot\sigma\|_1 = o(\tau)$ and $\|B_\tau - B - \tau\dot B\|_1 = o(\tau)$ as $\tau \to 0$; in particular, $\|\Delta_\tau\|_1 = O(\tau)$. Steps 4–7 use only the differentiability of $\tau \mapsto \sigma_\tau$ at $\tau = 0$ in trace norm; Step 8 uses the trace-norm continuity of $h \mapsto \sigma_h$ and $h \mapsto \partial_v\sigma_h$; Step 9 uses, in addition, the differentiability of $\tau \mapsto B_\tau$ at $\tau = 0$ in trace norm.

Step 4: integral form of the logarithm. For $x > 0$,

$$\int_0^\infty \Big[\frac{1}{1+t} - \frac{1}{x+t}\Big]\,dt = \lim_{T\to\infty} \log\frac{1+T}{x+T} - \log\frac{1}{x} = \log x.$$

For a state ω, the spectrum of $\omega + \varepsilon I$ lies in $[\varepsilon, 1+\varepsilon]$, where $\big|\frac{1}{1+t} - \frac{1}{x+t}\big| = \frac{|x-1|}{(1+t)(x+t)} \le \frac{1+\varepsilon}{(1+t)(\varepsilon+t)}$. This bound is integrable in t. For every vector ψ, Fubini's theorem for the spectral measure of $\omega+\varepsilon I$ therefore turns the scalar identity into $\langle\psi, \log(\omega+\varepsilon I)\psi\rangle = \int_0^\infty \langle\psi, [(1+t)^{-1} I - R_{\varepsilon+t}(\omega)]\psi\rangle\, dt$, and the same bound shows that the integrand is Bochner integrable in operator norm. Hence

$$\log(\omega+\varepsilon I) = \int_0^\infty \big[(1+t)^{-1} I - R_{\varepsilon+t}(\omega)\big]\,dt \tag{B.13}$$

in operator norm.

Step 5: exact increment. Apply (B.13) with $\omega = \sigma_\tau$ and with $\omega = \sigma$, and subtract; the terms $(1+t)^{-1}I$ cancel. The resolvent identity in (B.12), with $\omega' = \sigma_\tau$ and $\omega = \sigma$, reads $R_c(\sigma) - R_c(\sigma_\tau) = R_c(\sigma_\tau)\,\Delta_\tau\, R_c(\sigma)$. Hence

$$\log(\sigma_\tau+\varepsilon I) - \log(\sigma+\varepsilon I) = \int_0^\infty R_{\varepsilon+t}(\sigma_\tau)\,\Delta_\tau\, R_{\varepsilon+t}(\sigma)\,dt, \tag{B.14}$$

where the integrand has operator norm at most $(\varepsilon+t)^{-2}\|\Delta_\tau\|$.

Step 6: first-order term and remainder. Replacing $R_{\varepsilon+t}(\sigma_\tau)$ by $R_{\varepsilon+t}(\sigma)$ in (B.14) produces exactly $\mathcal{T}_{\sigma+\varepsilon I}(\Delta_\tau)$, by (B.6) with $v = t$. Applying the resolvent identity once more to the difference $R_{\varepsilon+t}(\sigma_\tau) - R_{\varepsilon+t}(\sigma)$,

$$\log(\sigma_\tau+\varepsilon I) - \log(\sigma+\varepsilon I) - \mathcal{T}_{\sigma+\varepsilon I}(\Delta_\tau) = -\int_0^\infty R_{\varepsilon+t}(\sigma_\tau)\,\Delta_\tau\, R_{\varepsilon+t}(\sigma)\,\Delta_\tau\, R_{\varepsilon+t}(\sigma)\,dt.$$

By (B.12) and $\|\Delta_\tau\| \le \|\Delta_\tau\|_1$, the right side has operator norm at most

$$\|\Delta_\tau\|_1^2 \int_0^\infty (\varepsilon+t)^{-3}\,dt = \frac{\|\Delta_\tau\|_1^2}{2\varepsilon^2}. \tag{B.15}$$

Step 7: the first identity in (B.9). By linearity of $\mathcal{T}_{\sigma+\varepsilon I}$,

$$\begin{aligned}&\log(\sigma_\tau+\varepsilon I) - \log(\sigma+\varepsilon I) - \tau\,\mathcal{T}_{\sigma+\varepsilon I}(\dot\sigma)\\ &\quad= \big[\log(\sigma_\tau+\varepsilon I) - \log(\sigma+\varepsilon I) - \mathcal{T}_{\sigma+\varepsilon I}(\Delta_\tau)\big] + \mathcal{T}_{\sigma+\varepsilon I}(\Delta_\tau - \tau\dot\sigma).\end{aligned}$$

The bracket is $O(\|\Delta_\tau\|_1^2) = O(\tau^2)$ in operator norm by (B.15), and the last term has operator norm at most $\varepsilon^{-1}\|\Delta_\tau - \tau\dot\sigma\|_1 = o(\tau)$ by the first bound in (B.7). Dividing by τ and letting $\tau \to 0$ gives, in operator norm,

$$\partial_v \log(\sigma_h + \varepsilon I)\big|_{h=0} = \lim_{\tau\to 0} \frac{\log(\sigma_\tau + \varepsilon I) - \log(\sigma + \varepsilon I)}{\tau} = \mathcal{T}_{\sigma+\varepsilon I}(\dot\sigma), \tag{B.16}$$

which is the first identity in (B.9).

Step 8: continuous differentiability in h. Each σ_h is a state, and $\tau \mapsto \sigma_{h+\tau v}$ is differentiable at $\tau = 0$ in trace norm with derivative $\partial_v\sigma_h$. Steps 4–7 at the base point σ_h therefore give $\partial_v \log(\sigma_h + \varepsilon I) = \mathcal{T}_{\sigma_h+\varepsilon I}(\partial_v\sigma_h)$ for every h. By the Lipschitz bound in (B.7) with $p = \infty$ and $\|\cdot\| \le \|\cdot\|_1$,

$$\|\mathcal{T}_{\sigma_{h'}+\varepsilon I}(\partial_v\sigma_{h'}) - \mathcal{T}_{\sigma_h+\varepsilon I}(\partial_v\sigma_h)\| \le \varepsilon^{-1}\|\partial_v\sigma_{h'} - \partial_v\sigma_h\|_1 + \varepsilon^{-2}\|\sigma_{h'} - \sigma_h\|_1\,\|\partial_v\sigma_h\|_1.$$

Since $h \mapsto \sigma_h$ and $h \mapsto \partial_v\sigma_h$ are continuous in trace norm, every partial derivative of $h \mapsto \log(\sigma_h+\varepsilon I)$ exists and is continuous in operator norm. Hence this map is continuously differentiable in operator norm.

Step 9: the second identity in (B.9). Since $D_{\tau v}$ is unitary, $\sigma_\tau + \varepsilon I = D_{\tau v}(\sigma + \varepsilon I)D_{\tau v}^\dagger$, and the functional calculus commutes with unitary conjugation: $\log(\sigma_\tau + \varepsilon I) = D_{\tau v}L_\varepsilon D_{\tau v}^\dagger$. With $B_\tau = D_{\tau v}BD_{\tau v}^\dagger$, cyclicity of the trace gives

$$f(\tau) := \operatorname{Tr}\big[B_\tau \log(\sigma_\tau + \varepsilon I)\big] = \operatorname{Tr}\big[D_{\tau v}BL_\varepsilon D_{\tau v}^\dagger\big] = \operatorname{Tr}(BL_\varepsilon) \qquad \text{for all } \tau,$$

so f is constant and $f'(0) = 0$. On the other hand, put $Y_\tau = \log(\sigma_\tau + \varepsilon I)$, so that $Y_0 = L_\varepsilon$, and write

$$\frac{f(\tau) - f(0)}{\tau} = \operatorname{Tr}\Big[\frac{B_\tau - B}{\tau}\,Y_\tau\Big] + \operatorname{Tr}\Big[B\,\frac{Y_\tau - Y_0}{\tau}\Big].$$

By $|\operatorname{Tr}(XY)| \le \|X\|_1\|Y\|$, the first term converges to $\operatorname{Tr}(\dot B L_\varepsilon)$, because $(B_\tau - B)/\tau \to \dot B$ in trace norm and $Y_\tau \to Y_0$ in operator norm by Step 7. The second term converges to $\operatorname{Tr}[B\,\mathcal{T}_{\sigma+\varepsilon I}(\dot\sigma)]$ by (B.16). Hence

$$0 = f'(0) = \operatorname{Tr}(\dot B L_\varepsilon) + \operatorname{Tr}(B\,\mathcal{T}_{\sigma+\varepsilon I}(\dot\sigma)),$$

which is the second identity in (B.9). This proves (b).

Step 10: proof of (c). Apply (B.9) to the two real coordinate directions of mode j and form the combination $\partial_{\overline{h_j}} = (\partial_{\operatorname{Re} h_j} + i\partial_{\operatorname{Im} h_j})/2$. Since the trace and $\mathcal{T}_{\sigma+\varepsilon I}$ are complex linear,

$$\partial_{\overline{h_j}} \log(\sigma_h + \varepsilon I)\big|_{h=0} = \mathcal{T}_{\sigma+\varepsilon I}(\partial_{\overline{h_j}}\sigma_h|_{h=0}), \qquad \operatorname{Tr}((\partial_{\overline{h_j}}B_h|_{h=0})L_\varepsilon) = -\operatorname{Tr}(B\,\mathcal{T}_{\sigma+\varepsilon I}(\partial_{\overline{h_j}}\sigma_h|_{h=0})).$$

By definition, $\partial_{\overline{h_j}}\sigma_h|_{h=0} = -[a_j, \sigma]$ and $\partial_{\overline{h_j}}B_h|_{h=0} = -[a_j, B]$. Substituting these gives (B.10) and (B.11). □

For a faithful state σ, the quadratic form q_σ of Subsection 2.5 reads, in the present notation,

$$q_\sigma(C) = \sum_{k,\ell} |C_{k\ell}|^2 k_0(\mu_k, \mu_\ell), \qquad C_{k\ell} = \langle k|C|\ell\rangle.$$

Whenever $q_\sigma(B)$ and $q_\sigma(C)$ are finite, the notation $\operatorname{Tr}(B^\dagger\mathcal{T}_\sigma(C))$ means the scalar sum

$$\sum_{k,\ell} \overline{B_{k\ell}}C_{k\ell}k_0(\mu_k, \mu_\ell), \qquad \sum_{k,\ell} |B_{k\ell}|\,|C_{k\ell}|k_0(\mu_k, \mu_\ell) \le \sqrt{q_\sigma(B)q_\sigma(C)} < \infty.$$

The inequality is Cauchy–Schwarz for these weighted sequences. By (B.8), the regularized pairing $\mathrm{Tr}(B^\dagger \mathcal{T}_{\sigma+\varepsilon I}(C))$ is the same sum with k_ε in place of k_0, and $k_\varepsilon \le k_0$ by (B.2), so the same Cauchy–Schwarz bound is a summable majorant for it. Thus monotone convergence applies to $q_\sigma(C)$ and dominated convergence applies to each displayed mixed sum as $\varepsilon \downarrow 0$.

In particular, the notation $[a_j, \log\sigma] = \mathcal{T}_\sigma([a_j, \sigma])$ means these scalar pairings, obtained by letting $\varepsilon \downarrow 0$ in (B.10). It does not assert that either unbounded product $a_j \log\sigma$ or $\log\sigma\, a_j$ is defined on every vector. All limiting logarithmic calculations below use this interpretation.

B.2 The commutator chain rule

Proof of Lemma 7. The logarithmic divided-difference and resolvent formulas are standard [19, Sec. 2.5, Example 1]. We record the domain and convergence bounds needed for the unbounded a_j. Writing $e_j = \mathrm{Tr}(\omega a_j^\dagger a_j)$, Hilbert–Schmidt factorization gives

$$\|a_j\omega\|_1 \le \sqrt{e_j}, \qquad \|\omega a_j\|_1 \le \sqrt{e_j + 1}.$$

Moreover, $\sum_k \mu_k(\|a_j|k\rangle\|^2 + \|a_j^\dagger|k\rangle\|^2) < \infty$ and $\mu_k > 0$, so every eigenvector belongs to both domains. Consequently,

$$(C_j)_{k\ell} = (\mu_\ell - \mu_k)(a_j)_{k\ell}, \qquad [a_j, \log\omega]_{k\ell} = (\log\mu_\ell - \log\mu_k)(a_j)_{k\ell}. \tag{B.17}$$

Multiplication by the divided difference in (2.42) proves the chain rule on the finite eigenvector span. With $\log\omega$ replaced by $\log(\omega + \varepsilon I)$, the same computation gives the entries $(\log(\mu_\ell + \varepsilon) - \log(\mu_k + \varepsilon))(a_j)_{k\ell} = k_\varepsilon(\mu_k, \mu_\ell)(C_j)_{k\ell}$, by (2.43) and (B.1); these are the entries of $\mathcal{T}_{\omega+\varepsilon I}(C_j)$ by (B.8), which proves (2.47). The right side is trace class by Lemma 22(a):

$$\|\mathcal{T}_{\omega+\varepsilon I}(C_j)\|_1 \le \varepsilon^{-1}\|C_j\|_1.$$

For the limiting pairing, the kernel $k_0(\mu_k, \mu_\ell) = \int_0^1 ((1-u)\mu_k + u\mu_\ell)^{-1}\, du$ of (B.1) satisfies

$$\sum_{k,\ell} |X_{k\ell}|\, |(C_j)_{k\ell}|\, k_0(\mu_k, \mu_\ell) \le \sqrt{q_\omega(X) q_\omega(C_j)} < \infty. \tag{B.18}$$

Fubini's theorem and (2.43) yield

$$\mathrm{Tr}[X^\dagger \mathcal{T}_\omega(C_j)] = \sum_{k,\ell} \overline{X_{k\ell}}(C_j)_{k\ell}\, k_0(\mu_k, \mu_\ell) = \int_0^1 \mathrm{Tr}[X^\dagger \mathcal{S}_{j,u}(\omega)]\, du.$$

By (B.8), the regularized pairing $\mathrm{Tr}[X^\dagger \mathcal{T}_{\omega+\varepsilon I}(C_j)]$ is the same sum with the coefficient $k_\varepsilon(\mu_k, \mu_\ell)$, which increases to $k_0(\mu_k, \mu_\ell)$ by (B.2). The summable majorant in (B.18) therefore also proves convergence of the regularized pairing. □

B.3 The integrated heat-flow identity

This subsection proves Lemma 8.

Proof of Lemma 8. Step 1: smoothing, the generator, and faithfulness. Write $\widetilde{D}_x = D_{z(x)}$ and $q_t(x) = (2\pi t)^{-m}\exp(-|x|^2/(2t))$. Then

$$\rho_t = \int q_t(x)\widetilde{D}_x \rho \widetilde{D}_x^\dagger\, dx, \qquad \widetilde{D}_h \rho_t \widetilde{D}_h^\dagger = \int q_t(x-h)\widetilde{D}_x \rho \widetilde{D}_x^\dagger\, dx.$$

For every multi-index α and integer $r \geq 0$,

$$\left\| \int (\partial_t^r \partial_x^\alpha q_t)(x) \widetilde{D}_x \rho \widetilde{D}_x^\dagger \, dx \right\|_1 \leq \|\partial_t^r \partial_x^\alpha q_t\|_{L^1} < \infty,$$

locally uniformly for $t > 0$. Differentiating these trace-class integrals and using $\partial_t q_t = \frac{1}{2} \sum_k \partial_k^2 q_t$ gives

$$\delta_k \rho_t = -\int \partial_k q_t(x) \widetilde{D}_x \rho \widetilde{D}_x^\dagger \, dx, \qquad \dot{\rho}_t = \frac{1}{2} \sum_{k=1}^{2m} \delta_k^2 \rho_t. \tag{B.19}$$

For a finite-energy state σ, each quadrature satisfies

$$\|R_k \sigma\|_1 \leq \|R_k \sigma^{1/2}\|_2 \|\sigma^{1/2}\|_2 = (\operatorname{Tr} \sigma R_k^2)^{1/2} < \infty.$$

Thus $[R_k, \sigma]$ has a trace-class extension, and

$$e^{-ihR_k} \sigma e^{ihR_k} - \sigma = -i \int_0^h e^{-irR_k} [R_k, \sigma] e^{irR_k} \, dr$$

identifies the trace-norm orbit derivative with $-i[R_k, \sigma]$. For $\sigma = \rho_t$, the canonical commutation relations on $\mathcal{D}_{\mathrm{fin}}$ give

$$\frac{1}{2}(\delta_{2j-1}^2 + \delta_{2j}^2)\rho_t = -\frac{1}{2}([a_j, [a_j^\dagger, \rho_t]] + [a_j^\dagger, [a_j, \rho_t]]) = -[a_j^\dagger, [a_j, \rho_t]],$$

since the difference of the two double commutators is $[[a_j, a_j^\dagger], \rho_t] = [I, \rho_t] = 0$. The left side supplies the trace-class extension of the weak double commutator. Hence our generator is $-\sum_j [a_j^\dagger, [a_j, \cdot]]$, precisely the normalization of [6, Sec. 2.2].

For faithfulness, positivity of q_t and continuity of the integrand give

$$\langle v, \rho_t v \rangle = \int q_t(x) \|\rho^{1/2} \widetilde{D}_x^\dagger v\|^2 \, dx = 0 \quad \Longrightarrow \quad v \perp \widetilde{D}_x \operatorname{supp} \rho \quad \text{for every } x.$$

The closed span of these Weyl translates is nonzero and invariant under every Weyl unitary. Irreducibility therefore makes it the whole Hilbert space [20, Proposition 14.7]; consequently $v = 0$.

Step 2: differentiate the regularized entropy. Fix $t > 0$, set $\sigma = \rho_t$, and let F_ε be the regularized entropy of Lemma 21. For each k, put $\sigma_h = \widetilde{D}_{he_k} \sigma \widetilde{D}_{he_k}^\dagger$. The scalar function $h \mapsto \operatorname{Tr}\big((\delta_k \sigma_h) \log(\sigma_h + \varepsilon I)\big)$ is constant, by unitary invariance of the trace and $\delta_k \sigma_h = \widetilde{D}_{he_k} (\delta_k \sigma) \widetilde{D}_{he_k}^\dagger$. Its first factor is differentiable in trace norm, by (B.19), and its second in operator norm, by Lemma 22(b); differentiating at $h = 0$ and using the chain rule (B.9) gives

$$\operatorname{Tr}\big((\delta_k^2 \sigma) \log(\sigma + \varepsilon I)\big) = -\operatorname{Tr}\big((\delta_k \sigma) \mathcal{T}_{\sigma + \varepsilon I}(\delta_k \sigma)\big). \tag{B.20}$$

By (B.4), (B.19), and (B.20),

$$\begin{aligned} \frac{d}{dt} F_\varepsilon(\rho_t) &= -\operatorname{Tr}\big(\dot{\rho}_t \log(\rho_t + \varepsilon I)\big) = -\frac{1}{2} \sum_{k=1}^{2m} \operatorname{Tr}\big((\delta_k^2 \rho_t) \log(\rho_t + \varepsilon I)\big) \\ &= \frac{1}{2} \sum_{k=1}^{2m} \operatorname{Tr}\big((\delta_k \rho_t) \mathcal{T}_{\rho_t + \varepsilon I}(\delta_k \rho_t)\big) = \sum_{j=1}^{m} \operatorname{Tr}\big(C_j(t)^\dagger \mathcal{T}_{\rho_t + \varepsilon I}(C_j(t))\big) =: J_\varepsilon(\rho_t) \geq 0, \end{aligned} \tag{B.21}$$

where $C_j(t) = [a_j, \rho_t] = -(\delta_{2j-1}\rho_t + i\delta_{2j}\rho_t)/\sqrt{2}$ by (2.1) and $\delta_{2j-1} = -i[P_j, \cdot]$, $\delta_{2j} = i[Q_j, \cdot]$ (see (3.22)); the last equality holds because $\mathcal{T}_{\rho_t + \varepsilon I}$ preserves adjoints and is self-adjoint for the

Hilbert–Schmidt inner product (Lemma 22(a)), so the mixed terms $\pm\frac{i}{2}\operatorname{Tr}[(\delta_{2j-1}\rho_t)\mathcal{T}(\delta_{2j}\rho_t)] \mp \frac{i}{2}\operatorname{Tr}[(\delta_{2j}\rho_t)\mathcal{T}(\delta_{2j-1}\rho_t)]$ cancel. The map $t \mapsto J_\varepsilon(\rho_t)$ is continuous: ρ_t and $C_j(t)$ are continuous in trace norm by (B.19), and the Lipschitz bound in (B.7) gives $\|\mathcal{T}_{\sigma+\varepsilon I}(C) - \mathcal{T}_{\sigma'+\varepsilon I}(C')\|_1 \le \varepsilon^{-1}\|C - C'\|_1 + \varepsilon^{-2}\|\sigma - \sigma'\|_1\|C\|_1$.

Step 3: remove the regularization after integrating in time. Integrating (B.21) on $[t_0, t_1]$ gives

$$F_\varepsilon(\rho_{t_1}) - F_\varepsilon(\rho_{t_0}) = \int_{t_0}^{t_1} J_\varepsilon(\rho_t)\, dt.$$

In an eigenbasis of ρ_t, the entrywise form (B.8) gives $J_\varepsilon(\rho_t) = \sum_{j,k,\ell} |(C_j(t))_{k\ell}|^2 k_\varepsilon(\mu_k, \mu_\ell)$ with the kernel k_ε of (B.1), which increases to $J_{\mathrm{KMB}}(\rho_t)$ as $\varepsilon \downarrow 0$, by (B.2) and (2.46). The endpoints have finite energy and hence finite entropy, and $F_\varepsilon \uparrow S + 1$ by Lemma 21(a). Monotone convergence on both sides therefore proves (2.49); in particular $\int_{t_0}^{t_1} J_{\mathrm{KMB}}(\rho_t)\, dt < \infty$. Section 4 supplies continuity of the integrand and hence upgrades (2.49) to a pointwise derivative. □

B.4 Analytic justification of the thermal proof

The algebraic proof of Theorem 1 is given in Section 3. The next lemma justifies trace-norm differentiation, removal of the entropy regularizer, and passage to zero SNR. In particular, the bounds below are obtained before differentiating the entropy.

Lemma 23 (Thermal differentiation under a second-moment bound)**.** *Let $N > 0$, put $\nu = N + \frac{1}{2}$, and suppose $V_Z = \mathbb{E}\|Z\|^2 < \infty$. For the thermal displacement model* (3.1)*, put $\Gamma_j(s) = \mathbb{E}[Z_j \rho_{s,Z}]$ and $C_j(s) = [a_j, \rho_s]$. Then the following statements hold.*

(a) The state path is locally C^1 in trace norm on $(0, \infty)$, and the Weyl orbits of ρ_s and $\Gamma_j(s)$ are continuously differentiable in trace norm. Consequently,

$$\frac{d}{ds} F_\varepsilon(\rho_s) = -\frac{1}{\sqrt{s}} \operatorname{Re} \sum_j \operatorname{Tr}(\Gamma_j(s)^\dagger \mathcal{T}_{\rho_s+\varepsilon I}(C_j(s))) =: h_\varepsilon(s). \tag{B.22}$$

(b) The spectral pairing

$$h(s) = -\frac{1}{\sqrt{s}} \operatorname{Re} \sum_j \operatorname{Tr}(\Gamma_j(s)^\dagger \mathcal{T}_{\rho_s}(C_j(s)))$$

is absolutely convergent at every $s > 0$. Moreover, $h_\varepsilon(s) \to h(s)$ with an integrable bound on each $(0, s_1]$, and

$$S(\rho_{s_1}) - S(\rho_{s_0}) = \int_{s_0}^{s_1} h(s)\, ds, \qquad 0 \le s_0 < s_1 < \infty. \tag{B.23}$$

(c) Entropy is continuous at $s = 0$, and

$$R(s) = \int_0^1 \frac{\mathsf{M}_u(Z \mid A_s)}{N + 1 - u}\, du$$

is continuous on $[0, \infty)$, with

$$R(0) = \log \frac{N+1}{N}\, \mathbb{E}\|Z - \mathbb{E}Z\|^2.$$

The annihilator pairings in (3.12) *are continuous pairings in the estimation completion.*

Proof. Step 1: trace-norm differentiation. Put $L_z = \sum_j (z_j a_j^\dagger - \overline{z}_j a_j)$. The thermal second moments give $\operatorname{Tr}(\tau_\nu L_z^\dagger L_z) = (2N+1)\|z\|^2$. Since $L_z^\dagger = -L_z$, Hilbert–Schmidt factorization yields

$$\|[L_z, \tau_\nu]\|_1 \le \|L_z\tau_\nu\|_1 + \|\tau_\nu L_z\|_1 \le 2\|L_z\tau_\nu^{1/2}\|_2\|\tau_\nu^{1/2}\|_2 = 2\sqrt{2N+1}\,\|z\|.$$

Set $C_{N,m} = 2\sqrt{2N+1}$. Differentiating the unitary orbit gives

$$\frac{d}{ds}\rho_{s,z} = \frac{1}{2\sqrt{s}} D_{\sqrt{s}z}[L_z, \tau_\nu] D_{\sqrt{s}z}^\dagger, \qquad \left\|\frac{d}{ds}\rho_{s,z}\right\|_1 \le \frac{C_{N,m}\|z\|}{2\sqrt{s}}. \tag{B.24}$$

For $0 < s_0 \le s \le s_1$, the expected bound is at most $C_{N,m}\sqrt{V_Z}/(2\sqrt{s_0})$. Dominated convergence therefore makes ρ_s locally C^1 in trace norm and gives the averaged derivative (3.6). Moreover,

$$\operatorname{Tr}(\rho_s H_0) = \mathbb{E}\operatorname{Tr}(\rho_{s,Z} H_0) = mN + sV_Z < \infty.$$

Every displaced thermal state is faithful, so their average is faithful.

For a real displacement derivative δ_k, put $B_k = (\Omega R)_k$, with the quadrature vector R from Section 2.1. The Weyl relation (2.3) gives $D_{\sqrt{s}z}^\dagger B_k D_{\sqrt{s}z} = B_k + cI$ for a real scalar c; hence

$$\begin{aligned}
\delta_k\rho_{s,z} &= -iD_{\sqrt{s}z}[D_{\sqrt{s}z}^\dagger B_k D_{\sqrt{s}z}, \tau_\nu] D_{\sqrt{s}z}^\dagger \\
&= -iD_{\sqrt{s}z}[B_k + cI, \tau_\nu] D_{\sqrt{s}z}^\dagger = D_{\sqrt{s}z}(\delta_k\tau_\nu) D_{\sqrt{s}z}^\dagger, \\
\|\delta_k\rho_{s,z}\|_1 &= \|\delta_k\tau_\nu\|_1.
\end{aligned}$$

Here $[cI, \tau_\nu] = 0$, which is the scalar cancellation being used.

The corresponding bound for the integrand defining $\delta_k\Gamma_j(s)$ is $|z_j|\|\delta_k\tau_\nu\|_1$, whose expectation is finite. Thus the Weyl orbits of both ρ_s and $\Gamma_j(s)$ satisfy the hypotheses of Lemma 22(b).

Write $L_\varepsilon = \log(\rho_s + \varepsilon I)$ and $A_{j,\varepsilon} = \mathcal{T}_{\rho_s+\varepsilon I}(C_j(s))$. Equation (B.11) and the adjoint relation $[a_j^\dagger, \Gamma_j] = -[a_j, \Gamma_j^\dagger]^\dagger$ give

$$\operatorname{Tr}([a_j, \Gamma_j^\dagger] L_\varepsilon) = -\operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon}), \qquad \operatorname{Tr}([a_j^\dagger, \Gamma_j] L_\varepsilon) = \overline{\operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon})}.$$

Substitution into (B.4) and (3.6) therefore gives

$$\begin{aligned}
\frac{d}{ds}F_\varepsilon(\rho_s) &= -\frac{1}{2\sqrt{s}}\sum_j \operatorname{Tr}(([a_j^\dagger, \Gamma_j] - [a_j, \Gamma_j^\dagger]) L_\varepsilon) \\
&= -\frac{1}{\sqrt{s}}\operatorname{Re}\sum_j \operatorname{Tr}(\Gamma_j^\dagger A_{j,\varepsilon}),
\end{aligned}$$

which proves part (a). Every logarithm in this calculation is bounded.

Step 2: a bound for the unregularized pairing. In an eigenbasis of ρ_s, define

$$q_u(C) = \sum_{k,\ell} \frac{|C_{k\ell}|^2}{(1-u)\mu_k + u\mu_\ell}, \qquad q_{\mathrm{KMB}}(C) = \int_0^1 q_u(C)\,du.$$

The normal equation and (2.31) imply

$$\sum_j q_u(\Gamma_j) = \sum_j \|\widehat{Z}_{j,u}\|_{\rho_s,u}^2 = V_Z - \mathsf{M}_u(Z \mid A_s) \le V_Z, \qquad \sum_j q_{\mathrm{KMB}}(\Gamma_j) \le V_Z.$$

To justify the annihilator in (3.4), let $P_L = 1_{H_0 \le L}$ and $T_L = a_j(I - P_L)$, with integer L. Then

$$\begin{aligned}\|a_j - a_j P_L\|_{\rho_s,u}^2 &= (1-u)\,\mathrm{Tr}(\rho_s a_j a_j^\dagger 1_{H_0 \ge L}) + u\,\mathrm{Tr}(\rho_s a_j^\dagger a_j 1_{H_0 > L}) \\ &\longrightarrow 0 \qquad (L \to \infty).\end{aligned}$$

The two products are computed on the number basis defined in Section 2.1: a_j lowers total photon number by one, so

$$T_L^\dagger T_L = a_j^\dagger a_j 1_{H_0 > L}, \qquad T_L T_L^\dagger = a_j a_j^\dagger 1_{H_0 \ge L}.$$

Since $a_j^\dagger a_j \le H_0$ and $a_j a_j^\dagger \le H_0 + I$,

$$\mathrm{Tr}(\rho_s a_j^\dagger a_j 1_{H_0>L}) \le \mathrm{Tr}(\rho_s H_0 1_{H_0>L}) \longrightarrow 0, \qquad \mathrm{Tr}(\rho_s a_j a_j^\dagger 1_{H_0 \ge L}) \le \mathrm{Tr}(\rho_s (H_0+I) 1_{H_0 \ge L}) \longrightarrow 0.$$

The convergence follows from the finite sum $\sum_{\boldsymbol{n}} (|\boldsymbol{n}|+1)\langle \boldsymbol{n}|\rho_s|\boldsymbol{n}\rangle < \infty$. The operators $a_j P_L$ are bounded, so Lemma 4 extends (3.3) to the completion and yields (3.4) there. This step uses only the thermal relation, not entropy differentiation.

Writing $n_u = N + 1 - u \ge N$, that relation gives

$$\mathcal{S}_{j,u}(\rho_s) = \frac{\sqrt{s}\,\widehat{Z}_{j,u} - a_j}{n_u}, \qquad \sum_j \|a_j\|_{\rho_s,u}^2 = sV_Z + mn_u.$$

Consequently,

$$\begin{aligned}J_u(\rho_s) &= \frac{1}{n_u^2}\sum_j \|\sqrt{s}\,\widehat{Z}_{j,u} - a_j\|_{\rho_s,u}^2 \\ &\le \frac{2}{n_u^2}\left(s\sum_j \|\widehat{Z}_{j,u}\|_{\rho_s,u}^2 + \sum_j \|a_j\|_{\rho_s,u}^2\right) \\ &\le \frac{2}{N^2}\big(2sV_Z + m(N+1)\big). \end{aligned} \tag{B.25}$$

The first inequality is $\|x-y\|^2 \le 2\|x\|^2 + 2\|y\|^2$. Tonelli gives the same upper bound for $J_{\mathrm{KMB}}(\rho_s) = \int_0^1 J_u(\rho_s)\,du$. Cauchy–Schwarz for the spectral sum now yields

$$\begin{aligned}&\sum_{j,k,\ell} |(\Gamma_j)_{k\ell}|\,|C_j(s)_{k\ell}| k_0(\mu_k,\mu_\ell) \\ &\quad\le \left(\sum_j q_{\mathrm{KMB}}(\Gamma_j)\right)^{1/2}\left(\sum_j q_{\mathrm{KMB}}(C_j(s))\right)^{1/2} \le \sqrt{V_Z J_{\mathrm{KMB}}(\rho_s)}. \end{aligned} \tag{B.26}$$

This proves absolute convergence of $h(s)$.

By the entrywise form (B.8), $h_\varepsilon(s)$ is the spectral sum defining $h(s)$ with k_ε in place of k_0. Since $k_\varepsilon \le k_0$ and $k_\varepsilon \uparrow k_0$ by (B.2), dominated convergence with the majorant (B.26) gives $h_\varepsilon(s) \to h(s)$. In addition, for $0 < s \le s_1$,

$$|h_\varepsilon(s)|,\ |h(s)| \le s^{-1/2}\sqrt{\frac{2V_Z}{N^2}\big(2sV_Z + m(N+1)\big)} \le K_{s_1} s^{-1/2},$$

where $K_{s_1} = \sqrt{2V_Z(2s_1V_Z + m(N+1))/N^2}$ and $\int_0^{s_1} K_{s_1} s^{-1/2}\,ds = 2K_{s_1}\sqrt{s_1} < \infty$. For $s_0 > 0$, integrate part (a) before removing ε:

$$\begin{aligned} S(\rho_{s_1}) - S(\rho_{s_0}) &= \lim_{\varepsilon\downarrow 0}[F_\varepsilon(\rho_{s_1}) - F_\varepsilon(\rho_{s_0})] \\ &= \lim_{\varepsilon\downarrow 0}\int_{s_0}^{s_1} h_\varepsilon(s)\,ds = \int_{s_0}^{s_1} h(s)\,ds. \end{aligned}$$

The first equality uses finite endpoint entropy and Lemma 21(a); the last uses the displayed integrable bound.

Step 3: continuity at zero and the first-moment pairing. Trace-norm continuity of Weyl conjugation and dominated convergence give

$$\|\rho_s - \tau_\nu\|_1 \le \mathbb{E}\|D_{\sqrt{s}Z}\tau_\nu D_{\sqrt{s}Z}^\dagger - \tau_\nu\|_1 \longrightarrow 0.$$

Since $\sup_{0\le s\le s_1}\operatorname{Tr}(\rho_s H_0) = mN + s_1V_Z < \infty$, energy-constrained entropy continuity [21, Lemma 15] yields $S(\rho_s) \to S(\tau_\nu)$. Therefore

$$S(\rho_{s_1}) - S(\tau_\nu) = \lim_{s_0\downarrow 0}\int_{s_0}^{s_1} h(s)\,ds = \int_0^{s_1} h(s)\,ds,$$

which extends part (b) to $s_0 = 0$.

Put $V_c = \mathbb{E}\|Z - \mathbb{E}Z\|^2$. Constant estimation gives

$$0 \le \frac{\mathsf{M}_u(Z \mid A_s)}{N+1-u} \le \frac{V_c}{N}, \qquad 0 < u < 1.$$

Lemma 18 and Corollary 19 supply pointwise continuity of the numerator, including its value V_c at zero. Dominated convergence proves continuity of R, with

$$R(0) = V_c\int_0^1 \frac{du}{N+1-u} = V_c\log\frac{N+1}{N}.$$

Finally, Lemma 4 and Step 2 imply

$$|\operatorname{Tr}(\Gamma_j^\dagger(a_j - a_jP_L))| \le \|\widehat{Z}_{j,u}\|_{\rho_s,u}\|a_j - a_jP_L\|_{\rho_s,u} \longrightarrow 0.$$

The bounded pairings can also be passed through the ensemble average. Indeed, Cauchy–Schwarz gives an integrable bound,

$$\mathbb{E}[|Z_j|\sqrt{N + s\|Z\|^2 + 1}] \le \sqrt{\mathbb{E}|Z_j|^2}\sqrt{N + sV_Z + 1} < \infty.$$

Using $\operatorname{Tr}(\rho_{s,z}a_j) = \sqrt{s}\,z_j$, we obtain

$$\begin{aligned} \operatorname{Tr}(\Gamma_j^\dagger a_j) &= \lim_{L\to\infty}\mathbb{E}[\overline{Z}_j\operatorname{Tr}(\rho_{s,Z}a_jP_L)] \\ &= \mathbb{E}[\overline{Z}_j\operatorname{Tr}(\rho_{s,Z}a_j)] = \sqrt{s}\,\mathbb{E}|Z_j|^2. \end{aligned}$$

This proves the last assertion of part (c) and (3.12). □

The algebra in the main proof identifies $h(s) = R(s)$. Consequently,

$$I(s) = S(\rho_s) - S(\tau_\nu) = \int_0^s R(r)\,dr, \qquad R \in C([0,\infty)).$$

The fundamental theorem of calculus now gives $I'(s) = R(s)$ for every $s > 0$ and $I'(0+) = R(0)$. This order does not require trace-norm differentiability of the state at zero; for a nonzero input mean, $\rho_s - \rho_0$ may have a term of order $\sqrt{s}$.

B.5 Real-coordinate scores and the displacement derivative

We record the real-coordinate form of the entropy calculation in the proof of Theorem 1. This also explains the scope of the corresponding calculation for a general initial state. Use the derivations and scores defined in (3.22).

For $X \in \mathbb{R}^{2m}$ and $Z = z(X)$, write $\Gamma_k^X(s) = \mathbb{E}[X_k \rho_{s,X}]$, where $\rho_{s,x} = D_{\sqrt{s}z(x)} \rho_0 D^\dagger_{\sqrt{s}z(x)}$ for an initial state ρ_0 ($\rho_0 = \tau_\nu$ in the thermal model). The state derivative, when justified in trace norm, is

$$\rho_s' = \frac{1}{2\sqrt{s}} \sum_{k=1}^{2m} \delta_k \Gamma_k^X. \tag{B.27}$$

Here is a precise sufficient condition for the following entropy calculation on an open interval $J \subset (0, \infty)$. Assume that ρ_s is faithful and C^1 in trace norm, that the Weyl orbits of ρ_s and $\Gamma_k^X(s)$ are continuously differentiable in trace norm, that (B.27) holds, and that $S(\rho_0), S(\rho_s) < \infty$. Assume also, for every k and $s \in J$,

$$q_{\rho_s}(\Gamma_k^X(s)) < \infty, \qquad q_{\rho_s}(\delta_k \rho_s) < \infty.$$

Then all mixed logarithmic pairings below are absolutely convergent by Cauchy–Schwarz. Define

$$H_\varepsilon(s) = \frac{1}{2\sqrt{s}} \sum_k \operatorname{Tr}\big(\Gamma_k^X(s) \mathcal{T}_{\rho_s + \varepsilon I}(\delta_k \rho_s)\big).$$

The finite quadratic forms give convergence to the corresponding spectral sum $H(s)$. Finally, assume that H is continuous and that on every compact subinterval $K \subset J$ there is $b_K \in L^1(K)$ with $|H_\varepsilon(s)| \le b_K(s)$ for $0 < \varepsilon \le 1$. For example, the latter follows if $(2\sqrt{s})^{-1} \sum_k \sqrt{q_{\rho_s}(\Gamma_k^X) q_{\rho_s}(\delta_k \rho_s)}$ is integrable on K.

For $a < b$ in J, Lemma 21, followed by dominated convergence, gives

$$S(\rho_b) - S(\rho_a) = \lim_{\varepsilon \downarrow 0} \int_a^b H_\varepsilon(s)\, ds = \int_a^b H(s)\, ds.$$

The fundamental theorem of calculus and unitary invariance now give

$$\begin{aligned}
\frac{d}{ds} I(X; A_s) &= \lim_{\varepsilon \downarrow 0} \big[- \operatorname{Tr}(\rho_s' \log(\rho_s + \varepsilon I)) \big] \\
&= \lim_{\varepsilon \downarrow 0} \left[- \frac{1}{2\sqrt{s}} \sum_k \operatorname{Tr}\big((\delta_k \Gamma_k^X) \log(\rho_s + \varepsilon I)\big) \right] \\
&= \lim_{\varepsilon \downarrow 0} \frac{1}{2\sqrt{s}} \sum_k \operatorname{Tr}\big(\Gamma_k^X \mathcal{T}_{\rho_s + \varepsilon I}(\delta_k \rho_s)\big) \\
&= \frac{1}{2\sqrt{s}} \sum_k \operatorname{Tr}\big(\Gamma_k^X \mathcal{T}_{\rho_s}(\delta_k \rho_s)\big).
\end{aligned}$$

The second equality uses (B.27); the third is the second identity in (B.9), applied with $B = \Gamma_k^X$ and the real derivative δ_k; and the last is the assumed convergence of the spectral sums. This argument states the required limiting hypotheses without presuming an unregularized operator product with $\log \rho_s$.

For the change to complex coordinates, use

$$\Gamma_j = \frac{\Gamma_{2j-1}^X + i \Gamma_{2j}^X}{\sqrt{2}}, \qquad [a_j, \rho_s] = - \frac{\delta_{2j-1} \rho_s + i \delta_{2j} \rho_s}{\sqrt{2}}. \tag{B.28}$$

The operators Γ_k^X and the trace-norm derivatives $\delta_k\rho_s$ are self-adjoint, and $\mathcal{T}_{\rho_s}$ preserves adjoints. Their trace pairings are therefore real, so expansion gives

$$\begin{aligned}
&\operatorname{Re}\operatorname{Tr}[\Gamma_j^\dagger \mathcal{T}_{\rho_s}([a_j,\rho_s])] \\
&\quad = -\frac{1}{2}\operatorname{Re}\operatorname{Tr}\Big[\Gamma_{2j-1}^X \mathcal{T}_{\rho_s}(\delta_{2j-1}\rho_s) + \Gamma_{2j}^X \mathcal{T}_{\rho_s}(\delta_{2j}\rho_s) \\
&\qquad\qquad + i\Gamma_{2j-1}^X \mathcal{T}_{\rho_s}(\delta_{2j}\rho_s) - i\Gamma_{2j}^X \mathcal{T}_{\rho_s}(\delta_{2j-1}\rho_s)\Big] \\
&\quad = -\frac{1}{2}\Big[\operatorname{Tr}\big(\Gamma_{2j-1}^X L_{2j-1}^{\mathrm{KMB}}(\rho_s)\big) + \operatorname{Tr}\big(\Gamma_{2j}^X L_{2j}^{\mathrm{KMB}}(\rho_s)\big)\Big].
\end{aligned}$$

The two terms multiplied by i have zero real part. Summing over j proves

$$\begin{aligned}
\frac{d}{ds} I(X;A_s) &= \frac{1}{2\sqrt{s}} \sum_{k=1}^{2m} \operatorname{Tr}[\Gamma_k^X L_k^{\mathrm{KMB}}(\rho_s)] \\
&= -\frac{1}{\sqrt{s}} \operatorname{Re} \sum_{j=1}^{m} \operatorname{Tr}[\Gamma_j^\dagger \mathcal{T}_{\rho_s}([a_j,\rho_s])].
\end{aligned} \tag{B.29}$$

Equation (B.29) identifies the entropy score before invoking any thermal property. For a general initial state, this identity is conditional on the regularity, spectral-convergence, domination, and continuity conditions just stated. Finite entropy alone does not imply them. The thermal model with $N > 0$ and $\mathbb{E}\|Z\|^2 < \infty$, and the Gaussian-noise model with a finite-energy initial state, satisfy these conditions by the preceding lemmas and Section 4.

C The known Gaussian measurement optimum in our normalization

The one-mode Gaussian Bayesian measurement problem is solved in [18, Sec. 7.5], in the unnumbered calculation between Eqs. (371) and (372). For a displaced thermal state with mean photon number $N \geq 0$ and a centered circular Gaussian displacement θ with $\mathbb{E}|\theta|^2 = N_1 > 0$, its minimum complex squared error over all POVMs and estimators is

$$R_{\mathbb{C}} = \frac{N_1 N}{N_1 + N} + \frac{N_1^2}{(N_1 + N + 1)(N_1 + N)} = \frac{N_1(N+1)}{N_1 + N + 1}. \tag{C.1}$$

It is attained by the heterodyne POVM (2.23) followed by the estimate $\widehat{\theta} = N_1\alpha/(N_1 + N + 1)$. We record the change of normalization and the reduction from arbitrary joint measurements to this one-mode result.

Let the real signal $X \sim \mathsf{N}(0, \sigma^2 I_{2m})$ be encoded by the thermal experiment (3.1), that is, by $\rho_{s,x} = D_{\sqrt{s}z(x)} \tau_\nu D_{\sqrt{s}z(x)}^\dagger$, and define the measurement-based error

$$\mathrm{MMSE}_{\mathrm{meas}}(X \mid A_s) = \inf_{\Pi, \widehat{x}} \mathbb{E}\|X - \widehat{x}(Y)\|^2, \qquad P_{Y|X=x}(dy) = \operatorname{Tr}(\rho_{s,x}\Pi(dy)),$$

where Π ranges over all POVMs on the m modes and $\widehat{x}$ over all measurable estimators of the outcome Y. In contrast to the operator error of Subsection 3.3, a measurement must produce all coordinates in a single classical outcome. First assume $s > 0$ and $\sigma^2 > 0$. For one mode put

$$\theta = \sqrt{\frac{s}{2}}(X_1 + iX_2), \qquad N_1 = s\sigma^2, \qquad N = \nu - \frac{1}{2}.$$

For corresponding real and complex estimates,

$$(X_1 - \widehat{X}_1)^2 + (X_2 - \widehat{X}_2)^2 = \frac{2}{s}|\theta - \widehat{\theta}|^2.$$

Consequently (C.1) gives the one-mode minimum

$$r_s = \frac{2}{s} R_{\mathbb{C}} = \frac{2\sigma^2(\nu + 1/2)}{\nu + s\sigma^2 + 1/2}. \tag{C.2}$$

For m modes write $x = (x_1, \dots, x_m)$ with $x_j \in \mathbb{R}^2$, let $\mu = \mathsf{N}(0, \sigma^2 I_2)$, and write the conditional state as $\rho_{s,x} = \bigotimes_{j=1}^m \rho_{s,x_j}^{(j)}$. Any joint POVM and measurable estimator can be combined into a POVM $M(d\widehat{x})$ on the estimate space $\mathbb{R}^{2m}$. For each j, average all other modes:

$$\overline{\rho}_{-j} = \bigotimes_{k \neq j} \int_{\mathbb{R}^2} \rho_{s,x_k}^{(k)} \, \mu(dx_k).$$

For a Borel set $B \subseteq \mathbb{R}^2$, put $B^{\uparrow j} = \{\widehat{x} : \widehat{x}_j \in B\}$ and define

$$M_j(B) = \mathrm{Tr}_{-j}\Big[(I_j \otimes \overline{\rho}_{-j}^{1/2}) M(B^{\uparrow j}) (I_j \otimes \overline{\rho}_{-j}^{1/2})\Big]. \tag{C.3}$$

The partial trace is understood weakly: for every one-mode state τ_j,

$$\mathrm{Tr}(\tau_j M_j(B)) = \mathrm{Tr}\Big[(\tau_j \otimes \overline{\rho}_{-j}) M(B^{\uparrow j})\Big],$$

where tensor factors are ordered with mode j first. Trace-class duality gives a unique bounded effect $M_j(B)$. Positivity, countable additivity in the weak operator topology, and $M_j(\mathbb{R}^2) = I$ follow from the same properties of M. Thus M_j is a POVM to which (C.2) applies. Independence and Tonelli's theorem give

$$\begin{aligned} &\int_{\mathbb{R}^{2m}} \int_{\mathbb{R}^{2m}} \|x - \widehat{x}\|^2 \, \mathrm{Tr}(\rho_{s,x} M(d\widehat{x})) \, \mu^{\otimes m}(dx) \\ &\quad = \sum_{j=1}^m \int_{\mathbb{R}^2} \int_{\mathbb{R}^2} \|x_j - \widehat{x}_j\|^2 \, \mathrm{Tr}(\rho_{s,x_j}^{(j)} M_j(d\widehat{x}_j)) \, \mu(dx_j) \geq m r_s. \end{aligned} \tag{C.4}$$

Product heterodyne measurement (2.23) attains each one-mode bound: by (2.9), its real outcome is $Y = \sqrt{s}X + Z_0$ with $Z_0 \sim \mathsf{N}(0, (\nu + 1/2) I_{2m})$ independent of X, and the posterior-mean estimator $\mathbb{E}[X \mid Y] = \sqrt{s}\,\sigma^2 Y / (\nu + s\sigma^2 + 1/2)$ has risk $2m\sigma^2(\nu + 1/2)/(\nu + s\sigma^2 + 1/2) = m r_s$. Therefore

$$\mathrm{MMSE}_{\mathrm{meas}}(X \mid A_s) = \frac{2m\sigma^2(\nu + 1/2)}{\nu + s\sigma^2 + 1/2}. \tag{C.5}$$

At $s = 0$, the state is independent of X, so the prior mean is optimal and the risk is $2m\sigma^2$. If $\sigma^2 = 0$, the input is deterministic and the risk is zero. These cases agree with (C.5) without division by s.

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